KU LEUVEN

FACULTY OF SCIENCE

Maximising Edge Density in Graphs Subject to Connectivity Constraints

Erdős Problem 915, from Proof to Search

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Foreword

This thesis asks how many edges a network can have when no two vertices may be joined by too many independent routes. The starting point is Erdős Problem 915, which asks this for edge-disjoint paths in simple graphs. Edges may also be directed, may be repeated, or may contain more than two vertices, and independent routes may share no edge or share no vertex. These choices give sixteen problems, and the thesis treats all of them.

Chapter 1 states the problem and its variants, proves the main results, and lists the open problems. The definitions are in its first three sections, and the notation is also listed under Abbreviations and Symbols. A reader who wants only the results can read these sections and then go to Section 1.5, where a table and a figure give what is known about each of the sixteen problems, followed by the open problems. The shorter proofs are in Chapter 1, and the longer ones are in Appendix A. That appendix has three parts, on undirected graphs, directed graphs and hypergraphs.

Chapter 2 explains how a Python program can measure connectivity and search for dense examples, using pruning and metaheuristics to make the search faster. No proof uses the program, so a reader interested only in the mathematics can skip this chapter. It needs only the definitions of Chapter 1. Appendix B prints the parts of the program that Chapter 2 leaves out, and the output of one full run.

I thank my girlfriend Sara for her unwavering support and love. I also thank my friends and family for their help throughout this journey.

Contribution Statement

I used artificial intelligence to write code, to suggest proofs, to help check them, to format LaTeX and to check spelling. Results from the literature are cited where they are used. The code is available in the repository [1].

Area	Main results
Undirected graphs and multigraphs	A proof of Mader's bound with the Gomory–Hu tree, graphs that attain it for every $n \geq m$, and the exact multigraph edge value.
Multigraphs, vertex version	The exact value at $m = 2$ and $m = 3$.
Hypergraphs, edge version	An upper bound for every r-uniform hypergraph, attained on stated ranges.
Hypergraphs, vertex version	The exact value at $m = 2$ and $m = 3$, for simple hypergraphs at $m = 3$ when $r < n$.
Simple digraphs, $m = 2$	The exact value, for both separations.
Directed multigraphs	The exact value for every n and m, and the extremal multigraphs for $n \geq \max(8, 2m)$.
Simple digraphs and directed hypergraphs	Upper bounds with the right leading term, $n^2/4$ for simple digraphs and $(m-1)n^2/(4(r-1))$ for directed hypergraphs in the three orientation models, valid for every n.

Short Summary

Erdős Problem 915 asks how many edges a simple graph on n vertices can have if no pair is joined by m edge-disjoint paths [2, 3]. Writing the maximum as $\ell_m(n)$, Mader's bound is

$$\ell_m(n) \le \left\lfloor \frac{m(n-1)}{2} \right\rfloor .$$

We prove this bound with the Gomory–Hu tree and give graphs that reach it for every $n \ge m \ge 2$.

The thesis also studies the problem in eight models. Edges may be directed or not, may be repeated or not, and may join two vertices or more (hypergraphs). In each model we can ask for edge-disjoint routes or for vertex-disjoint routes, which gives sixteen problems.

The proved results include:

- Multigraphs: exactly $(m-1)(n-1)$ edges. With vertex-disjoint routes, exactly $n-1$ at $m=2$ and $2(n-1)$ at $m=3$.
- r-uniform hypergraphs: at most $\lfloor (m-1)(n-1)/(r-1) \rfloor$ hyperedges, attained on stated ranges.
- Hypergraphs with vertex-disjoint routes: exactly $\lfloor (n-1)/(r-1) \rfloor$ at $m=2$, and exactly $\lfloor 2(n-1)/(r-1) \rfloor$ at $m=3$, for simple hypergraphs when $r < n$.
- Simple digraphs at $m=2$: exactly $\max\bigl(2(n-1), \lfloor n^2/4 \rfloor\bigr)$ arcs, for both separations.
- Directed multigraphs: exactly $(m-1)\max\bigl(2(n-1), \lfloor n^2/4 \rfloor\bigr)$ arcs.
- Simple digraphs: $n^2/4 + \Theta_m(n)$ arcs for fixed $m \ge 3$, for both separations.
- Directed hypergraphs: $(1+o(1))(m-1)n^2/(4(r-1))$ hyperedges for fixed m and r, in the three orientation models.

The exact value for simple digraphs is open for $m \ge 3$.

Chapter 2 explains how a program measures connectivity with maximum flow and searches for dense examples. No result depends on this program.

Abbreviations and Symbols

Symbol	Meaning
$G = (V, E)$, $D = (V, A)$	vertex set V, with edge multiset E or arc multiset A
n, m, r	vertex count, forbidden-connectivity threshold, hyperedge size
$\lambda_G(u, v)$	maximum number of edge-disjoint u–v paths
$\kappa_G(u, v)$	maximum number of incidence-separated u–v paths
$\lambda_G^{\max}$, $\kappa_G^{\max}$	maximum local connectivity over distinct vertex pairs, ordered in directed models
$\ell_m(n)$, $L_m(n)$	simple and multigraph undirected edge maxima
$k_m(n)$, $K_m(n)$	simple and multigraph undirected vertex maxima
$\ell_m^{\mathrm{dir}}(n)$, $L_m^{\mathrm{dir}}(n)$	simple and multigraph directed arc maxima
$k_m^{\mathrm{dir}}(n)$, $K_m^{\mathrm{dir}}(n)$	simple and multigraph directed vertex maxima
$\ell_m^{(r)}(n)$, $L_m^{(r)}(n)$	simple and multi r-uniform hypergraph edge maxima
$k_m^{(r)}(n)$, $K_m^{(r)}(n)$	simple and multi r-uniform hypergraph vertex maxima
$M(n)$	$\max(2(n-1), \lfloor n^2/4 \rfloor)$, the directed arc maximum at $m = 2$
$\mu(u, v)$	edge or arc multiplicity
$O(f)$, $\Theta(f)$	at most, or between two positive constant multiples of, f for large n. A subscript such as Θ_m means the constants depend on m
$o(1)$	a quantity that tends to 0

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Chapter 1

The Problem and Its Results

This chapter defines the problem and its variants, and states the results that are proved.

1.1 Graphs and routes

A *multigraph* G consists of a finite set $V(G)$ of *vertices* and a multiset $E(G)$ of *edges*. Each edge joins two different vertices, and edges can have identical copies. The number of such identical edges between u and v is the multiplicity $\mu(u, v)$. If $\mu(u, v) \leq 1$ for every pair, then G is a *simple graph*. The *size* of G is its number of edges, counted with multiplicity: $|E(G)| = \sum_{\{u,v\}} \mu(u, v)$. This thesis tries to maximize the size. Drawing vertices as dots and edges as lines is the standard way to visualize graphs.

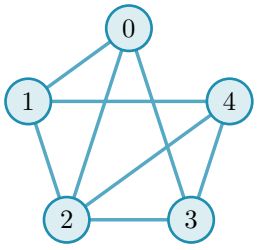


Figure 1.1. The simple graph $K_5 - \{04, 13\}$ has five vertices and eight edges, where K_n is the complete graph on n vertices.

We write $n := |V(G)|$. The *degree* of a vertex is the number of edges at it. From now on, graphs are simple unless we say otherwise. A *path* from u to v is a sequence $v_0, e_1, v_1, \ldots, e_q, v_q$ of distinct vertices v_i and edges e_i, where $v_0 = u$, $v_q = v$ and e_i joins v_{i-1} and v_i. In a multigraph, two parallel edges give two different paths of length one. In a simple graph we can write a path as a list of vertices. A graph is *connected* if every two vertices are joined by a path. A *tree* is a connected graph in which every two vertices are joined by exactly one path. A tree on n vertices has $n - 1$ edges [4]. A *forest* is a graph without cycles. A *spanning tree* of a graph G is a tree on all vertices of G that uses only edges of G. We also call a path a *route*. Two u–v paths are *edge-disjoint* if they have no edge in common. The *local edge-connectivity* $\lambda_G(u, v)$ is the largest number of pairwise edge-disjoint u–v paths. Its largest value over all pairs is

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v).$$

If G has fewer than two vertices, we set $\lambda_G^{\max} = 0$.

Theorem 1.1 (Menger [5, 6])**.** *Let u and v be distinct vertices of a multigraph G. The largest*

number of pairwise edge-disjoint u–v paths equals the smallest number of edges whose removal leaves no u–v path.

Figure 1.2 shows an example.

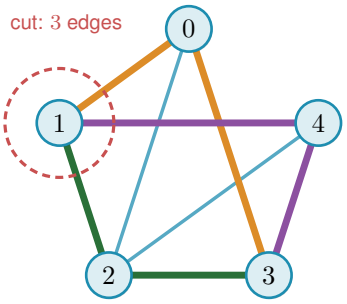


Figure 1.2. Menger's theorem on Figure 1.1: three edge-disjoint 1–3 routes (coloured) meet the minimum three-edge cut around vertex 1, so $\lambda(1,3) = 3$.

1.2 Erdős Problem 915

Fix an integer $m \geq 2$. The condition $\lambda_G^{\max} \leq m-1$ means that no two vertices are joined by m edge-disjoint paths. By Menger's theorem, it also means that any two vertices can be separated by removing at most $m-1$ edges. We ask how many edges a graph can have under this condition.

Question 1.2 (Erdős Problem 915 [3]). How many edges can a graph on n vertices have if no two of its vertices are joined by m disjoint paths?

The database [3] notes that the original statement leaves open whether "disjoint" means that the paths share no edge, or that they share no vertex except their two ends. This thesis takes edge-disjoint paths as the main question and treats vertex-disjoint paths as a variant (Section 1.3).

We write

$$\ell_m(n) = \max\{\, |E(G)| : |V(G)| = n,\ \lambda_G^{\max} \leq m-1 \,\}$$

for the answer. A graph with $\lambda_G^{\max} \leq m-1$ is called *feasible*. Mader proved the following upper bound. Here $\lfloor x \rfloor$ is x rounded down and $\lceil x \rceil$ is x rounded up.

Theorem 1.3 (Mader [7])**.** *For all $n \geq 2$ and $m \geq 2$,*

$$\ell_m(n) \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor .$$

Bollobás and Erdős [2] found graphs that attain this bound when $m-1$ divides $n-1$: copies of K_m that share one vertex. For $n \geq m$ the bound is always attained (Theorem A.3). For $2 \leq n < m$ the complete graph K_n satisfies the condition, so $\ell_m(n) = \binom{n}{2}$. This is smaller than $\lfloor m(n-1)/2 \rfloor$, except when $(n,m) = (2,3)$.

We prove Theorem 1.3 in a different way from Mader, with the Gomory–Hu tree. It stores the $\binom{n}{2}$ values $\lambda_G(u,v)$ in only $n-1$ weights.

Theorem 1.4 (Gomory–Hu [8])**.** *Let G be a multigraph on the vertex set V. There is a tree T on V, with a weight $c(t)$ on each tree edge t, such that:*

(1) Removing t from T splits V into two parts S and $V \setminus S$, and $c(t)$ is the number of edges of G with one end in S and one end in $V \setminus S$.

(2) For distinct u, v, the value $\lambda_G(u, v)$ is the smallest weight on the u–v path of T.

Such a tree is a *Gomory–Hu tree* of G. Its edges need not be edges of G. If $t = xy$ is a tree edge, then the x–y path of T is t itself, so $c(t) = \lambda_G(x, y)$ by (2). This thesis uses the theorem only for undirected graphs.

1.2.1 Proof of Mader's bound

Let T be a Gomory–Hu tree of G. For an edge e of G, let $\mathrm{dist}_T(e)$ be the number of tree edges on the path in T between the ends of e (Figure 1.3). An edge $e = uv$ of G *crosses* a tree edge t if u and v lie in different parts when t is removed from T. This happens exactly when t lies on the u–v path of T. So e crosses $\mathrm{dist}_T(e)$ tree edges.

The proof has three steps.

(1) The tree weights add up to at most $(m-1)(n-1)$.

(2) The same sum counts every edge of G once for each tree edge it crosses (Lemma 1.5).

(3) In a simple graph, at most $n-1$ edges cross only one tree edge, and every other edge crosses at least two (Lemma 1.6).

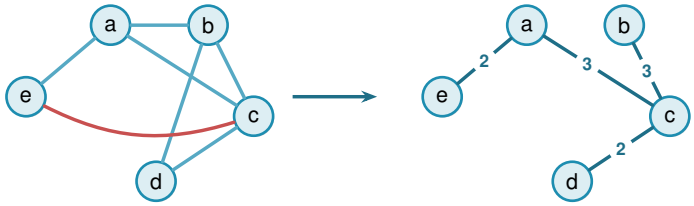


Figure 1.3. A graph G and a Gomory–Hu tree T on the same vertices. Each edge e of G crosses $\mathrm{dist}_T(e)$ tree edges. In a simple graph, at most $n-1$ edges have $\mathrm{dist}_T(e) = 1$.

Lemma 1.5 (Double-counting identity)**.** *Let G be a multigraph and T a Gomory–Hu tree of G with weights c. Then*

$$\sum_{t \in E(T)} c(t) \;=\; \sum_{e \in E(G)} \mathrm{dist}_T(e).$$

Parallel edges are counted separately in the right-hand sum.

Proof. Count the pairs (t, e) where t is a tree edge, e is an edge of G, and e crosses t. For a fixed t, exactly $c(t)$ edges cross t, by property (1) of the Gomory–Hu theorem. For a fixed e, exactly $\mathrm{dist}_T(e)$ tree edges are crossed by e. So both sums count the same pairs.

□

Lemma 1.6 (Sum of tree distances)**.** *For a simple graph* G, $\sum_{e \in E(G)} \operatorname{dist}_T(e) \geq 2|E(G)| - (n-1)$.

Proof. Let E_1 be the set of edges with $\operatorname{dist}_T(e) = 1$. Such an edge joins two vertices that are adjacent in T. The tree T has $n-1$ edges, and since G is simple, each of them gives at most one edge of E_1. So $|E_1| \leq n-1$. Every other edge has $\operatorname{dist}_T(e) \geq 2$. So

$$\sum_{e \in E(G)} \operatorname{dist}_T(e) \geq |E_1| + 2\big(|E(G)| - |E_1|\big) = 2|E(G)| - |E_1| \geq 2|E(G)| - (n-1). \qquad \square$$

Proof of Theorem 1.3. Let G be a simple graph on n vertices with $\lambda_G^{\max} \leq m-1$, and let T be a Gomory–Hu tree of G. Each tree edge $t = xy$ has weight $c(t) = \lambda_G(x,y) \leq m-1$. There are $n-1$ tree edges, so $\sum_t c(t) \leq (m-1)(n-1)$. With Lemmas 1.5 and 1.6 this gives

$$2|E(G)| - (n-1) \leq \sum_{e \in E(G)} \operatorname{dist}_T(e) = \sum_{t \in E(T)} c(t) \leq (m-1)(n-1).$$

So $2|E(G)| \leq m(n-1)$. Since $|E(G)|$ is an integer, $|E(G)| \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor$. $\square$

The factor $\frac{1}{2}$ comes from step (3). In a multigraph step (3) fails, and the bound is $(m-1)(n-1)$ (Theorem 1.8).

1.3 Variants of the problem

We change four assumptions. The first three give eight graph models. In each model we can ask for edge-disjoint or for vertex-disjoint routes (Section 1.3.1).

The first assumption is the size of an edge. A *hyperedge* is a set of $r \geq 2$ vertices. In an r*-uniform hypergraph* every hyperedge has exactly r vertices. A *Berge path* is a sequence of distinct vertices and distinct hyperedges, alternating, in which each hyperedge contains the vertex before it and the vertex after it. For $r = 2$ it is an ordinary path. The size of a hypergraph is its number of hyperedges, counted with multiplicity.

The second assumption is repetition. A *simple* graph or hypergraph has each edge at most once. A *multigraph* or *multihypergraph* may repeat edges.

The third assumption is direction. A directed edge, or *arc*, is an ordered pair of vertices. A directed hyperedge splits its r vertices into a non-empty set of *tails* and a non-empty set of *heads*. A directed Berge route enters a hyperedge at a tail and leaves it at a head. The allowed numbers of tails are a parameter of the directed hypergraph models. In the *forward* model every hyperedge has one tail and $r-1$ heads, in the *backward* model it has $r-1$ tails and one head, and the *general* model allows every split. For $r = 2$ the three coincide, and a hyperedge is an arc. Unless a statement names one of them, the directed hypergraph models are forward. In the figures, one colour is one hyperedge, not a path of ordinary edges. Figure 1.4 shows the eight models.

Digraphs have no Gomory–Hu tree, but Menger's theorem (Theorem 1.1) also holds for directed multigraphs, with directed paths and arcs. In a digraph, the *out-degree* $d^+(v)$ is the

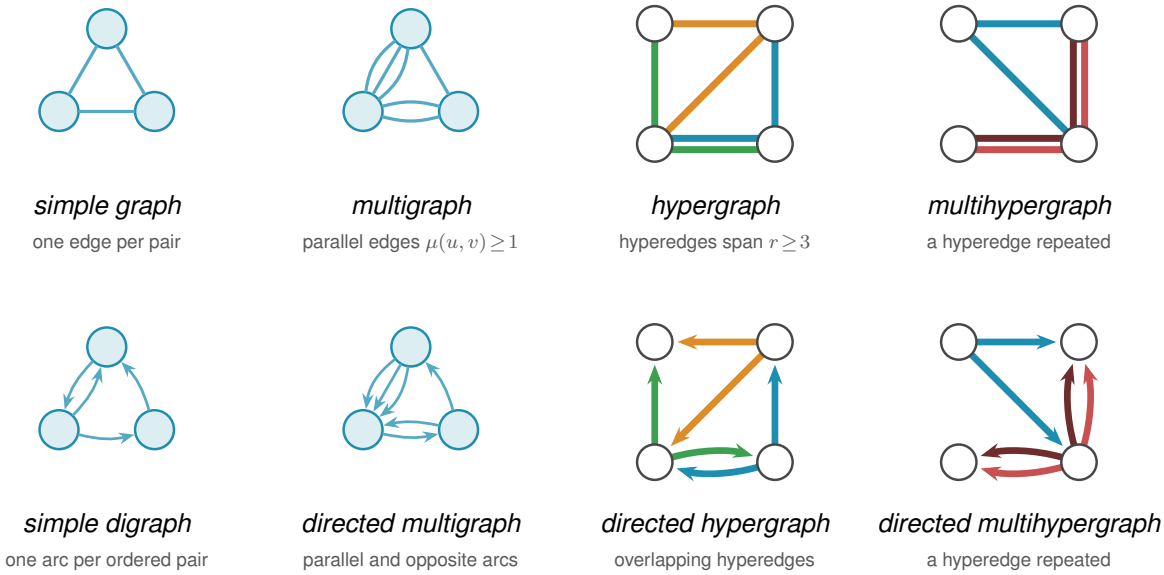


Figure 1.4. The eight models side by side, with the four undirected on top and the four directed below. The simple graph is a triangle, the multigraph has parallel edges, the hypergraph has three hyperedges, and the multihypergraph has one repeated hyperedge. The directed versions have arcs instead of edges, with the same pattern of repetition and hyperedges. In the two multihypergraph panels the two shades of red are the two copies of one hyperedge. Each directed hyperedge drawn has one tail.

number of arcs that leave v. The vertices these arcs point to are the *out-neighbours* of v. The *in-degree* $d^-(v)$ is the number of arcs that enter v, and they come from the *in-neighbours* of v. The *total degree* is $d(v) = d^+(v) + d^-(v)$. Figure 1.5 shows three arc-disjoint routes from u to v. Removing three arcs cuts all of them, as Menger's theorem predicts. The out-degree of u and the in-degree of v also bound the number of routes:

Lemma 1.7 (Degree bound)**.** *Every pair u, v of distinct vertices of a graph G satisfies $\lambda_G(u,v) \leq \min(\deg u, \deg v)$, and every ordered pair of a digraph D satisfies $\lambda_D(u,v) \leq \min(d^+(u), d^-(v))$.*

Proof. Every edge-disjoint u–v path uses a distinct edge at u, so there are at most $\deg u$ of them, and likewise at most $\deg v$. In the directed case each arc-disjoint path leaves u by a distinct out-arc and enters v by a distinct in-arc. □

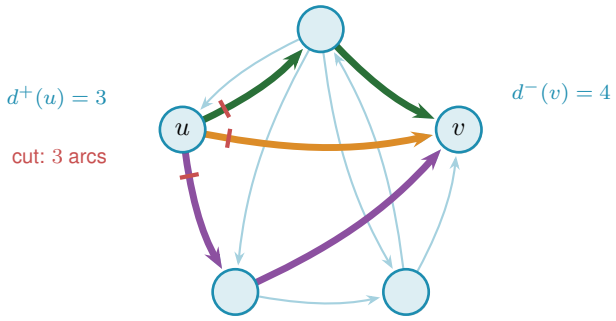


Figure 1.5. Three arc-disjoint $u \to v$ routes (coloured) and the corresponding three-arc directed cut (red) in a *simple* digraph. Thus $\lambda(u,v) = 3$, meeting the degree bound $\min(d^+(u), d^-(v)) = 3$.

The fourth assumption is *separation*. Two u–v paths are *internally vertex-disjoint* if they have no vertex in common except u and v. The largest number of such paths is the *local vertex-connectivity* $\kappa_G(u, v)$. Its largest value over all pairs is $\kappa_G^{\max}$. Internally vertex-disjoint paths are also edge-disjoint, so $\kappa_G(u, v) \leq \lambda_G(u, v)$. Section 1.3.1 extends this to multigraphs and hypergraphs. Figure 1.6 shows that the vertex and edge-disjointness can differ.

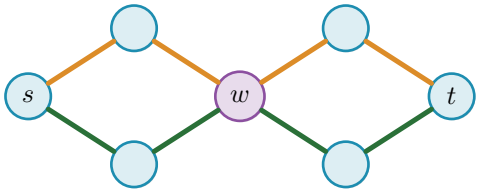


Figure 1.6. The separation axis: two edge-disjoint s–t routes both pass through w, so $\lambda(s, t) = 2$ but $\kappa(s, t) = 1$.

1.3.1 The incidence convention

For multigraphs and hypergraphs we must say when two routes count as vertex-disjoint. This thesis uses the following convention. For simple graphs it gives the usual notion.

The *incidence graph* $I(\mathcal{H})$ has a node for each vertex and a node for each edge copy. An edge copy is joined to the vertices it contains. A route from u to v is a path in $I(\mathcal{H})$. Two routes are *separated* if they have no inner node in common. In the directed case we use the directed incidence graph: each tail has an arc to the edge copy, and the edge copy has an arc to each head. For example, an arc $u \to v$ becomes $u \to e \to v$.

The inner nodes are vertices and edge copies, so separated routes share no inner vertex and no edge copy. Sharing no vertex is not enough for hypergraphs, because two Berge routes can use the same hyperedge without meeting in a vertex. Since separated routes also share no edge copy, $\kappa \leq \lambda$ holds in every model.

In a multigraph, q parallel copies of uv give q separated routes. So the vertex problems for multigraphs, $K_m(n)$ and $K_m^{\mathrm{dir}}(n)$, differ from the simple ones. In a feasible multigraph no multiplicity is larger than $m - 1$, since m parallel copies already give m separated routes.

In every model, a graph is *feasible* if no two vertices are joined by m disjoint routes of the kind considered. Table 1.1 lists the notation for all sixteen problems. Figure 1.10 shows what is known about each of them.

Menger's theorem, the Gomory–Hu tree and Mader's bound are classical. So are the results of Bártfai, Bollobás, Leonard, Sørensen and Thomassen. The main new results are:

- ★ a proof of Mader's bound with the Gomory–Hu tree, graphs that attain it for every $n \geq m$, and the multigraph edge value (Section 1.2.1, Theorems A.3 and 1.8),
- ★ the multigraph vertex value at $m = 2$ and $m = 3$ (Theorem A.33 and Corollary A.35),
- ★ the hypergraph edge bound and the ranges where it is attained (Theorem 1.13),

Table 1.1. Notation for the extremal functions. The third column uses the incidence convention defined above, in every model.

Model	edge-disjoint routes	vertex-disjoint routes
Simple graph	$\ell_m(n)$	$k_m(n)$
Multigraph	$L_m(n)$	$K_m(n)$
r-uniform simple hypergraph	$\ell_m^{(r)}(n)$	$k_m^{(r)}(n)$
r-uniform multihypergraph	$L_m^{(r)}(n)$	$K_m^{(r)}(n)$
Simple digraph	$\ell_m^{\mathrm{dir}}(n)$	$k_m^{\mathrm{dir}}(n)$
Directed multigraph	$L_m^{\mathrm{dir}}(n)$	$K_m^{\mathrm{dir}}(n)$
Directed r-uniform simple hypergraph	$\ell_m^{\mathrm{dir}\,(r)}(n)$	$k_m^{\mathrm{dir}\,(r)}(n)$
Directed r-uniform multihypergraph	$L_m^{\mathrm{dir}\,(r)}(n)$	$K_m^{\mathrm{dir}\,(r)}(n)$

- the hypergraph vertex value at $m = 2$ and $m = 3$, for simple hypergraphs at $m = 3$ when $r < n$ (Theorems 1.15 and 1.16),
- the simple digraph values at $m = 2$, for both separations (Theorems 1.11 and 1.12),
- the directed multigraph value for every n and m, and its extremal multigraphs for $n \geq \max(8, 2m)$ (Theorems 1.18 and A.22),
- the quadratic bounds for simple digraphs and for directed hypergraphs in the three orientation models (Theorems 1.17, 1.22 and A.50).

1.4 Results and proofs

1.4.1 Undirected graphs

Theorem 1.8. *For $n \geq 1$ and $m \geq 2$, undirected multigraphs satisfy $L_m(n) = (m-1)(n-1)$.*

Proof of Theorem 1.8. For the lower bound, take any tree on the n vertices and give every edge multiplicity $m-1$. This multigraph has $(m-1)(n-1)$ edges. Two vertices u, v are joined by exactly one path in the tree. Removing the $m-1$ copies of one edge of this path separates u from v. So $\lambda(u,v) \leq m-1$ for every pair.

For the upper bound, let $\lambda_G^{\max} \leq m-1$ and let T be a Gomory–Hu tree of G. Every edge of G crosses at least one tree edge. So by Lemma 1.5, $|E(G)| \leq \sum_e \mathrm{dist}_T(e) = \sum_t c(t) \leq (m-1)(n-1)$. □

In simple graphs, the edge and vertex problems have the same answer for $m \leq 4$.

Theorem 1.9 (Bártfai [9], Bollobás [10], Leonard [11])**.** *For all $n \geq m$ and $2 \leq m \leq 4$,*

$$k_m(n) = \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor .$$

Since $\kappa \leq \lambda$, every graph with $\lambda^{\max} \leq m-1$ also has $\kappa^{\max} \leq m-1$. So $k_m(n) \geq \ell_m(n)$ (Figure 1.7). Sørensen and Thomassen showed that the two differ for $m = 5$.

Theorem 1.10 (Sørensen–Thomassen [12])**.** *For simple graphs,*

$$k_5(n) = \left\lfloor \frac{5(n-1)}{2} \right\rfloor = \ell_5(n) \quad (5 \le n \le 13), \qquad k_5(n) = \left\lfloor \frac{8n}{3} \right\rfloor - 4 \quad (n \ge 14).$$

So $k_5(n) > \ell_5(n)$ for every $n \ge 14$. Sørensen and Thomassen count the least number of edges that forces m disjoint paths. This is one more than our maximum (Remark A.5). They give one formula with exceptions at $n = 7$ and $n = 12$. Splitting the statement at $n = 14$, as above, removes these exceptions.

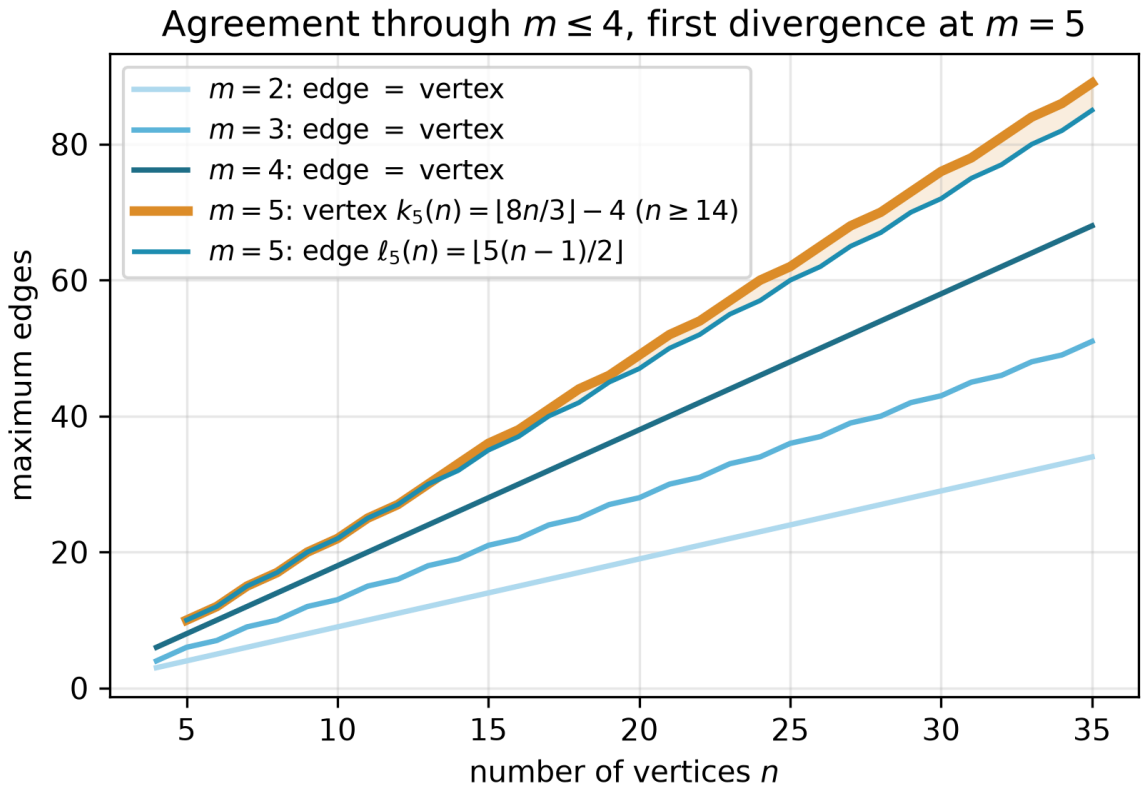


Figure 1.7. The edge and vertex extremal functions for $m = 2, 3, 4$ (blue shades, single curve per m since they agree) and $m = 5$ (thin blue for edge, thick orange for vertex). The vertex function diverges from the edge function at $n = 14$ and grows faster thereafter.

1.4.2 Directed graphs at the first threshold

Theorem 1.11 (Directed arc value at $m = 2$)**.** *For simple digraphs,*

$$\ell_2^{\mathrm{dir}}(n) = \max\left(2(n-1),\ \left\lfloor \frac{n^2}{4} \right\rfloor\right).$$

Two constructions attain the two terms (Figure 1.8). The *bidirected star* joins one vertex to every other vertex by an arc in each direction, so it has $2(n-1)$ arcs. The one-way complete bipartite digraph has all arcs from one half of the vertices to the other half, so it has $\lfloor n^2/4 \rfloor$ arcs. In both, every ordered pair is joined by at most one directed path. Section A.2 gives the details.

Theorem 1.12 (Directed vertex value at $m = 2$)**.** *For simple digraphs,* $k_2^{\mathrm{dir}}(n) = \ell_2^{\mathrm{dir}}(n)$.

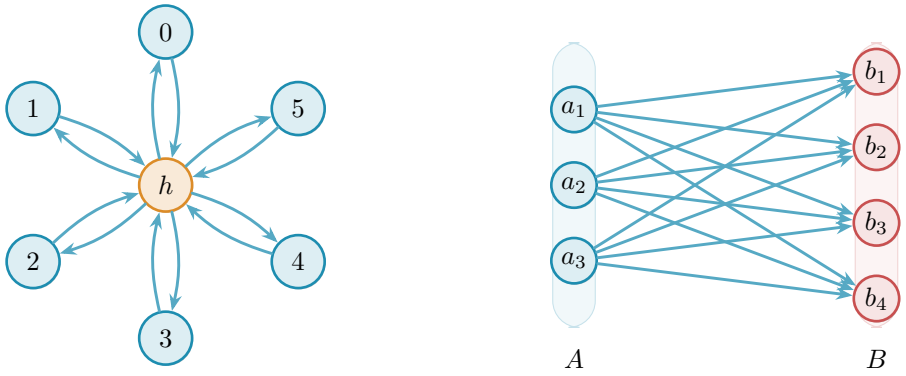


Figure 1.8. The two branches tie at $n = 7$, where each holds 12 arcs at $\lambda^{\max} = 1$. Left: the bidirected star, a hub joined to six spokes both ways, $2(n-1) = 12$. Right: the one-way complete bipartite digraph $B_{3,4}$, every arc from A to B, $\lfloor n^2/4 \rfloor = 12$.

The star has more arcs for $n < 7$, and the bipartite digraph has more for $n > 7$. So the directed value grows quadratically in n, while the undirected value grows linearly. In the bipartite digraph all arcs point the same way, so a route crosses between the parts only once and never comes back. An undirected graph with $\lambda^{\max} \leq 1$ has no cycle, so it has at most $n - 1$ edges.

For the upper bound we use a *reachability skeleton*: a subdigraph, with at most one copy of each arc, in which u can reach v exactly when u can reach v in the whole digraph. Lemma A.13 in the appendix shows that there is always a skeleton with at most $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$ arcs.

Proof of Theorem 1.12. Write $M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$.

Lower bound. The bidirected star (Construction A.8 at $m = 2$) and the one-way bipartite digraph (Construction A.9) have $\lambda^{\max} = 1$. Since $\kappa \leq \lambda$, they also have $\kappa^{\max} \leq 1$. So $\ell_2^{\mathrm{dir}}(n) \geq M(n)$ and $k_2^{\mathrm{dir}}(n) \geq M(n)$.

Upper bound. Let D be a simple digraph on $n \geq 2$ vertices with $\kappa_D^{\max} \leq 1$. By Lemma A.13, D has a reachability skeleton R with $|A(R)| \leq M(n)$.

We claim that R contains every arc of D. Suppose an arc (u, v) of D is not in R. Since u reaches v in D, it also reaches v in R, along a directed path P. The path P is not the arc (u, v), so it has length at least two. The arc $u \to v$ has no inner vertex, so P and the arc are two internally vertex-disjoint routes. So $\kappa_D(u, v) \geq 2$, a contradiction.

So $|A(D)| = |A(R)| \leq M(n)$. With the lower bound, $k_2^{\mathrm{dir}}(n) = M(n)$. □

Proof of Theorem 1.11. The two constructions above give $\ell_2^{\mathrm{dir}}(n) \geq M(n)$. Since $\kappa \leq \lambda$, every arc-feasible digraph is vertex-feasible, so Theorem 1.12 bounds it. So

$$M(n) \;\leq\; \ell_2^{\mathrm{dir}}(n) \;\leq\; k_2^{\mathrm{dir}}(n) \;\leq\; M(n),$$

so all three are equal. □

1.4.3 Undirected hypergraphs

Theorem 1.13 (Hypergraph edge bound)**.** *For $n \geq r \geq 2$ and $m \geq 2$, an r-uniform multihypergraph on n vertices with $\lambda^{\max} \leq m-1$ has size at most $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$. There are simple hypergraphs attaining the bound when $m-1 \leq \binom{n-2}{r-2}$, and multihypergraphs attaining it when $m-1 \leq \binom{n-2}{r-2}$ or $(r-1) \mid n-1$.*

Lemma 1.14 (Cut induction for multihypergraphs)**.** *Let $\mathcal{H}$ be a multihypergraph on $n \geq 1$ vertices, not assumed uniform, with $\lambda^{\max}_{\mathcal{H}} \leq k$. Then, the sum running over hyperedge copies,*

$$\sum_{e} \big(|e|-1\big) \;\leq\; k\,(n-1).$$

Proof. Induction on n. At $n = 1$ there is no hyperedge, since a hyperedge has at least two vertices. So both sides are 0.

Let $n \geq 2$ and fix two vertices $u \neq v$. Since $\lambda_{\mathcal{H}}(u,v) \leq k$, Theorem A.23 gives a set F of at most k hyperedge copies whose removal separates u from v. Let S be the set of vertices that u can reach in $\mathcal{H} - F$, and let $\bar{S} = V \setminus S$. Then $u \in S$ and $v \in \bar{S}$, so both sets are non-empty. Every copy that meets both S and $\bar{S}$ lies in F. Indeed, a copy outside F with a vertex $x \in S$ and a vertex $y \in \bar{S}$ would let u reach x and then y in $\mathcal{H} - F$, so y would lie in S. So at most k copies meet both sides.

Form two *trace* multihypergraphs $\mathcal{H}[S]$ and $\mathcal{H}[\bar{S}]$: replace every copy e by $e \cap S$, or by $e \cap \bar{S}$, and drop the sets with fewer than two vertices. A Berge route in a trace is also a Berge route in $\mathcal{H}$, if we replace each trace copy by the copy it came from. Different trace copies come from different copies of $\mathcal{H}$. So a family of hyperedge-disjoint routes in a trace gives one of the same size in $\mathcal{H}$. So both traces have $\lambda^{\max} \leq k$. Both have fewer than n vertices, so the induction hypothesis applies to both.

Now count. By induction, the copies of $\mathcal{H}[S]$ contribute at most $k(|S|-1)$, and those of $\mathcal{H}[\bar{S}]$ at most $k(|\bar{S}|-1)$. A copy inside one side contributes $|e|-1$ to that side. A copy that meets both sides satisfies

$$|e| - 1 \;=\; \big(|e \cap S| - 1\big) \;+\; \big(|e \cap \bar{S}| - 1\big) \;+\; 1,$$

where each bracket is the contribution to one trace (zero if that part was dropped). Summing over all copies gives

$$\sum_{e} \big(|e|-1\big) \;\leq\; k\big(|S|-1\big) + k\big(|\bar{S}|-1\big) + \big|\{\text{crossing copies}\}\big| \;\leq\; k(n-2)+k \;=\; k(n-1). \;\square$$

A *hypertree* is a hypergraph whose incidence graph (Section 1.3.1) is a tree.

Proof of Theorem 1.13. Upper bound. Every hyperedge has $|e| = r$, so Lemma 1.14 with $k = m-1$ gives $(r-1)|E(\mathcal{H})| \leq (m-1)(n-1)$. Since $|E(\mathcal{H})|$ is an integer, we can round down.

Lower bound. Suppose $r-1$ divides $n-1$. Take a hub h and split the other $n-1$ vertices into groups $G_1, G_2, \ldots$ of size $r-1$. The hyperedges are the sets $\{h\} \cup G_i$. The incidence graph

of this hypergraph is a tree, so we call it a *star hypertree*. Give each hyperedge multiplicity $m-1$. Every vertex other than h lies in exactly $m-1$ hyperedge copies. Every pair contains such a vertex, and every route from or to it uses one of these copies. So removing these $m-1$ copies separates the pair, and every pair has $\lambda \le m-1$. There are $(m-1)(n-1)/(r-1)$ hyperedges.

If $m-1 \le \binom{n-2}{r-2}$ and $r \ge 3$, Theorem A.24 below attains the bound with a simple hypergraph. If $r=2$, this condition means $m=2$. Then the bound is $n-1$, and a tree attains it. A simple hypergraph is also a multihypergraph, so in this range multihypergraphs attain the bound too. □

The star hypertree of Figure 1.9 attains the bound at $m=2$. It has a hub joined to disjoint groups of $r-1$ vertices. For $r \ge 3$, simple hypergraphs attain the bound whenever $m-1 \le \binom{n-2}{r-2}$ (Theorem A.24). So for $r \ge 3$, unlike for graphs, repeated edges often do not help. The condition says that there are enough different hyperedges. When it fails, the simple value can be smaller. For example, at $n=r=m=3$ there is only one possible hyperedge. A simple hypergraph has one hyperedge, while a multihypergraph can have two copies of it (Remark A.25).

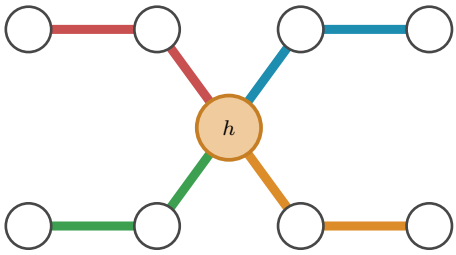


Figure 1.9. The $r=3$, $n=9$ star hypertree: four hyperedges meet only at the hub h, attaining $\lfloor (n-1)/(r-1) \rfloor = 4$ with $\lambda^{\max}=1$.

Theorem 1.15 (Hypergraph vertex value at $m=2$)**.** *For $r \ge 2$ and $n \ge 1$, the maximum number of hyperedges of an r-uniform multihypergraph on n vertices with $\kappa^{\max} \le 1$ is*

$$K_2^{(r)}(n) \;=\; \left\lfloor \frac{n-1}{r-1} \right\rfloor ,$$

attained by the star hypertree of Figure 1.9, which is simple. In particular $K_2^{(r)}(n) = k_2^{(r)}(n) = \ell_2^{(r)}(n)$: the vertex and edge problems agree at $m=2$ for every uniformity, and repeated hyperedges never help.

Proof of Theorem 1.15. Let q be the number of hyperedges. We claim that $\kappa_{\mathcal{H}}^{\max} \le 1$ exactly when $I(\mathcal{H})$ is a forest. The graph $I(\mathcal{H})$ is bipartite, with vertex nodes on one side and hyperedge nodes on the other. So a cycle in it alternates between the two kinds of nodes. Suppose $I(\mathcal{H})$ has a cycle $v_1, e_1, v_2, e_2, \dots, v_\ell, e_\ell, v_1$, with $\ell \ge 2$ and all v_i and e_i distinct. Then v_1, e_1, v_2 and $v_1, e_\ell, v_\ell, \dots, e_2, v_2$ are two Berge routes from v_1 to v_2 that share no hyperedge and no inner vertex. So $\kappa(v_1, v_2) \ge 2$. Conversely, two such routes between u and v are two paths in $I(\mathcal{H})$ with the same ends and no common inner node. Together they form a cycle in $I(\mathcal{H})$. This proves the claim.

Upper bound. Each hyperedge copy contains r vertices, so $I(\mathcal{H})$ has rq edges and $n+q$ nodes. A forest on $n+q$ nodes has at most $n+q-1$ edges. So $rq \le n+q-1$, that is, $q \le (n-1)/(r-1)$. Since q is an integer, $q \le \lfloor (n-1)/(r-1) \rfloor$.

Lower bound. Take the star hypertree with $\lfloor (n-1)/(r-1) \rfloor$ hyperedges, and leave the remaining vertices in no hyperedge. Its hyperedges meet only at the hub, and every other vertex lies in at most one hyperedge, so its incidence graph is a forest. So it has $\kappa^{\max} \le 1$, and it is simple.

The three values agree. The star hypertree also has $\lambda^{\max} \le 1$, since removing the one hyperedge at a non-hub vertex separates it from every other vertex. So it is feasible in all three problems, and $\ell_2^{(r)}(n),\ k_2^{(r)}(n),\ K_2^{(r)}(n) \ge \lfloor (n-1)/(r-1) \rfloor$. A simple hypergraph is a multihypergraph, and $\kappa \le \lambda$, so $\ell_2^{(r)}(n) \le k_2^{(r)}(n) \le K_2^{(r)}(n)$. The upper bound above gives $K_2^{(r)}(n) \le \lfloor (n-1)/(r-1) \rfloor$, so all three are equal. □

Theorem 1.16 (Hypergraph vertex value at $m = 3$)**.** *For $r \ge 2$ and $n \ge 1$, every r-uniform multihypergraph on n vertices with $\kappa^{\max} \le 2$ has at most $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ hyperedges. For $2 < r < n$, which is exactly the range where $2 \le \binom{n-2}{r-2}$, the bound is attained by a simple hypergraph, so*

$$k_3^{(r)}(n) \;=\; \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor \;=\; \ell_3^{(r)}(n).$$

For multihypergraphs the bound is attained for every $n \ge r$, so $K_3^{(r)}(n) = \lfloor 2(n-1)/(r-1) \rfloor$ for all $n \ge r \ge 2$.

Both proofs use the incidence graph (Section A.7). At $m = 3$ the proof bounds the cycle rank of the incidence graph, its number of edges minus its number of nodes plus one. We have no such bound for $m \ge 4$.

For $r = 2$ these results are about multigraphs. They give $K_2(n) = n-1$, attained by trees, and $K_3(n) = 2(n-1)$, attained by a tree with every edge doubled (Corollary A.35). For $m = 5$ the vertex and edge values differ. On four vertices, join two of them by 2 parallel edges, and join each of the other two to both of them by 3 parallel edges. This multigraph has 14 edges and $\kappa^{\max} = 4$ (Section A.8), while $L_5(4) = 12$. For $m = 4$ this is open. Section A.8 discusses larger m.

1.4.4 Directed bounds beyond the first threshold

Theorem 1.17 (The quadratic bound with a linear error)**.** *For all $m \ge 2$ and $n \ge 2$, every simple digraph D on n vertices with $\lambda_D^{\max} \le m-1$, or with $\kappa_D^{\max} \le m-1$, satisfies*

$$|A(D)| \;\le\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; (m-1)(n-1),$$

so for each fixed m,

$$\ell_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; \Theta_m(n) \quad \textit{and} \quad k_m^{\mathrm{dir}}(n) \;=\; \frac{n^2}{4} \;+\; \Theta_m(n) \qquad (m \ge 3).$$

Appendix A gives the proof. It counts routes of length two. For an ordered pair (u, v), the routes $u \to x \to v$ with different midpoints x, together with the arc $u \to v$, are disjoint. So a feasible digraph has at most $m - 1$ of them. Take a vertex v of largest out-degree Δ. Summing this over the pairs (v, w) shows that the out-neighbours of v send at most $(m-1)(n-1)$ arcs in total. The other $n - 1 - \Delta$ vertices send at most Δ arcs each. Together with the Δ arcs out of v this gives at most $\Delta(n - \Delta) + (m - 1)(n - 1)$ arcs, and $\Delta(n - \Delta) \le \lfloor n^2/4 \rfloor$. The routes of length two are also internally vertex-disjoint, so the same proof works for vertex separation.

Theorem 1.18 (Directed multigraph value, every n and m)**.** *For all $n \ge 2$ and $m \ge 2$,*

$$L_m^{\mathrm{dir}}(n) \;=\; (m-1)\, M(n), \qquad M(n) := \max\Big(2(n-1),\; \Big\lfloor \frac{n^2}{4} \Big\rfloor\Big).$$

The lower bound uses thickened trees and thickened one-way bipartite digraphs.

If we multiply every multiplicity by the same integer, every cut and so every connectivity is multiplied by that integer.

Lemma 1.19 (Thickening)**.** *Let D_0 be a directed multigraph and $k \ge 1$ an integer. Write $k \cdot D_0$ for the multigraph obtained by multiplying every multiplicity $\mu(u, v)$ by k. Then $\lambda_{k \cdot D_0}(u, v) = k \cdot \lambda_{D_0}(u, v)$ for every ordered pair $u \ne v$, and $|A(k \cdot D_0)| = k\,|A(D_0)|$.*

Proof. By Menger's theorem, $\lambda_{D_0}(u, v)$ is the smallest number of arcs, counted with multiplicity, in a u–v cut. Multiplying every multiplicity by k multiplies the size of every cut by k. So it multiplies the smallest cut by k too. The number of arcs is also multiplied by k. □

Construction 1.20 (Thickened tree and thickened bipartite digraph)**.** Let $m \ge 2$. (a) Take any tree on the n vertices and put $m - 1$ arcs in each direction on every tree edge. With one arc in each direction this is the bidirected tree, with $\lambda^{\max} = 1$. So by Lemma 1.19 the thickened tree has $\lambda^{\max} = m - 1$ and $2(m - 1)(n - 1)$ arcs. (b) Take the one-way bipartite digraph of Construction A.9 and repeat every arc $m - 1$ times. By the same lemma it has $\lambda^{\max} = m - 1$ and $(m - 1)\lfloor n^2/4 \rfloor$ arcs.

The larger of the two gives $L_m^{\mathrm{dir}}(n) \ge (m-1)M(n)$.

For the upper bound we want to delete a part of D that lowers every positive connectivity, and whose size we can bound. Deleting one vertex does not work: it need not lower every connectivity. Instead we delete a reachability skeleton. By Lemma A.13 it has at most $M(n)$ arcs, and the next lemma shows that deleting it lowers every positive connectivity.

Lemma 1.21 (Deleting a skeleton lowers every positive connectivity)**.** *Let D be a directed multigraph with $\lambda^{\max}(D) \le k$ for some $k \ge 1$, and let R be a reachability skeleton of D. Let $D - R$ be D with one copy of each arc of R removed. Then for every ordered pair $u \ne v$,*

$$\lambda_{D-R}(u, v) \;\le\; \max\big(\lambda_D(u, v) - 1, 0\big),$$

and consequently $\lambda^{\max}(D - R) \le k - 1$.

The skeleton keeps reachability. So if routes from u to v remain after the deletion, one more route runs inside the deleted arcs.

Proof. Fix $u \neq v$ and let $t = \lambda_{D-R}(u, v)$. If $t = 0$ there is nothing to prove. Otherwise $D - R$ has $t \geq 1$ arc-disjoint u–v paths, so u reaches v in D. Since R is a reachability skeleton, there is a u–v path P in R. The arcs of P are the removed copies, so P shares no arc with the t paths in $D - R$. So D has $t + 1$ arc-disjoint u–v paths, and $t + 1 \leq \lambda_D(u, v)$. Taking the maximum over all pairs gives $\lambda^{\max}(D - R) \leq k - 1$. □

Proof of Theorem 1.18. The lower bound is Construction 1.20. For the upper bound, fix n. We show by induction on $k \geq 0$ that every directed multigraph D on n vertices with $\lambda^{\max}(D) \leq k$ has $|A(D)| \leq k\, M(n)$. For $k = 0$ there are no arcs, since one arc is already one route.

Let $k \geq 1$, and let R be a reachability skeleton of D with $|A(R)| \leq M(n)$ (Lemma A.13). By Lemma 1.21, $\lambda^{\max}(D - R) \leq k - 1$, so by induction $|A(D - R)| \leq (k - 1)M(n)$. Every arc of D is either in R or in $D - R$. So

$$|A(D)| \;=\; |A(R)| \;+\; |A(D - R)| \;\leq\; M(n) \;+\; (k - 1)M(n) \;=\; k\, M(n).$$

With $k = m - 1$ this gives $L_m^{\mathrm{dir}}(n) \leq (m - 1)M(n)$. □

For $n \geq \max(8, 2m)$ the bipartite construction is the only extremal multigraph, up to reversing all arcs (Theorem A.22).

Theorem 1.22 (The directed hypergraph constant)**.** *Let $r \geq 2$ and $m \geq 2$. Every forward directed r-uniform multihypergraph $\mathcal{H}$ on $n \geq 2$ vertices with $\kappa_{\mathcal{H}}^{\max} \leq m - 1$ satisfies*

$$|E(\mathcal{H})| \;\leq\; \frac{m - 1}{r - 1} \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; (m - 1)^2(n - 1).$$

Since $\kappa \leq \lambda$, the same bound holds when $\lambda_{\mathcal{H}}^{\max} \leq m-1$. The construction of Proposition A.45 has $(1 + o(1))(m - 1)n^2/(4(r - 1))$ hyperedges, so the leading term is right.

Appendix A gives the proof. It uses the *shadow* digraph, which has an arc $u \to v$ when some hyperedge has tail u and head v. Each hyperedge gives $r - 1$ arcs of the shadow, and each arc comes from at most $m - 1$ hyperedges. The route condition on the hypergraph limits the routes of length two in the shadow. Lemma A.18 then bounds the number of arcs of the shadow.

The backward model is the forward model with every hyperedge reversed, so it has the same extremal values (Lemma A.47). The general model has the same leading term, with a larger linear term (Theorem A.50).

At $r = 2$ this also covers directed multigraphs with vertex separation. At $m = 2$ no arc can be repeated, so $K_2^{\mathrm{dir}}(n) = \ell_2^{\mathrm{dir}}(n)$. For fixed m, the extremal multigraphs of Theorem 1.18 are feasible here too, since $\kappa \leq \lambda$. With Theorem 1.22 at $r = 2$ this gives $K_m^{\mathrm{dir}}(n) = (m - 1)n^2/4 + O_m(n)$ (Corollary A.42).

Section A.4.1 compares Theorem 1.17 with a theorem of Huang and Lyu [13].

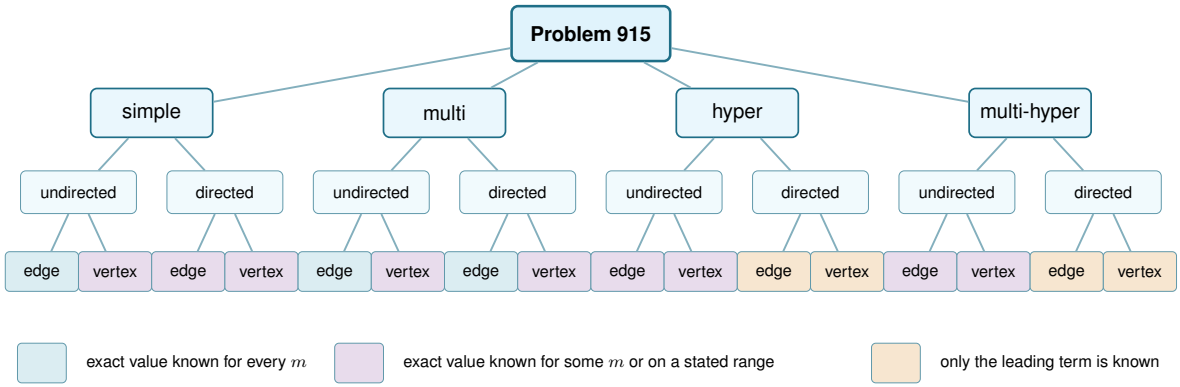


Figure 1.10. Overview of the sixteen variants. Blue: the exact value is known for every m. Purple: the exact value is known for some m or on a stated range. Amber: only the leading term is known.

1.5 Overview and open problems

Table 1.2 and Figure 1.10 collect what is known about the sixteen problems.

Model	Edge or arc separation	Vertex separation
Simple graph	$\lfloor m(n-1)/2 \rfloor$ for $n \geq m$	Exact for $m \leq 5$. Open for $m \geq 6$
Multigraph	$(m-1)(n-1)$	$n-1$ at $m=2$, $2(n-1)$ at $m=3$. Open for $m \geq 4$
Simple hypergraph	At most $\lfloor (m-1)(n-1)/(r-1) \rfloor$, equal when $m-1 \leq \binom{n-2}{r-2}$	$\lfloor (n-1)/(r-1) \rfloor$ at $m=2$, $\lfloor 2(n-1)/(r-1) \rfloor$ at $m=3$ for $r<n$. Open for $m \geq 4$
Multihypergraph	At most $\lfloor (m-1)(n-1)/(r-1) \rfloor$, equal when $m-1 \leq \binom{n-2}{r-2}$ or $(r-1) \mid (n-1)$	$\lfloor (n-1)/(r-1) \rfloor$ at $m=2$, $\lfloor 2(n-1)/(r-1) \rfloor$ at $m=3$. Open for $m \geq 4$
Simple digraph	$M(n)$ at $m=2$. $n^2/4+\Theta_m(n)$ for $m \geq 3$	Same
Directed multigraph	$(m-1)M(n)$	$M(n)$ at $m=2$. $(m-1)n^2/4+O_m(n)$
Directed simple hypergraph	$(1+o(1))(m-1)n^2/(4(r-1))$	Same
Directed multihypergraph	$(1+o(1))(m-1)n^2/(4(r-1))$	Same

Table 1.2. Status of the sixteen problems. Here $m \geq 2$, hypergraphs have $r \geq 3$, and asymptotic statements are for fixed m and r as $n \to \infty$. The directed hypergraph rows hold in the forward, backward and general models.

The main open problems are:

- Find $\ell_m^{\mathrm{dir}}(n)$ for $m \geq 3$, or at least the coefficient of the linear term.
- Is $k_m^{\mathrm{dir}}(n) = \ell_m^{\mathrm{dir}}(n)$ for $m \geq 3$?
- Find the simple vertex value $k_m(n)$ for $m \geq 6$, and the multigraph vertex value $K_m(n)$ for $m \geq 4$. Is $K_4(n) = L_4(n) = 3(n-1)$ for every n?
- Find the hypergraph edge value outside the range where it is attained, and a vertex bound for hypergraphs at $m \geq 4$.

- Find exact values for directed hypergraphs, and compare the different ways to orient a hyperedge (Section A.9) beyond the leading term.

Chapter 2

Measuring and Searching by Machine

Chapter 1 stated the problem and the proved results. This chapter explains how a program can measure connectivity in every variant, check small graphs one by one, and search for dense examples. No result in this thesis depends on the output of this program. Its output for $n, m \leq 7$ is printed in Appendix B. Section 2.5 shows how an output can be checked without reading the code.

The program is one Python file of about 520 lines, `program/erdos915.py`. It uses the library `networkx` for maximum flows and the program `geng` of `nauty` for one search. This chapter prints the main parts of the program, each next to the idea it implements. The remaining parts, such as the imports and small helper functions, are in Appendix B, so every line of the program is printed once. The names in the code follow the words of the text. Figure 2.1 shows the structure of the program.

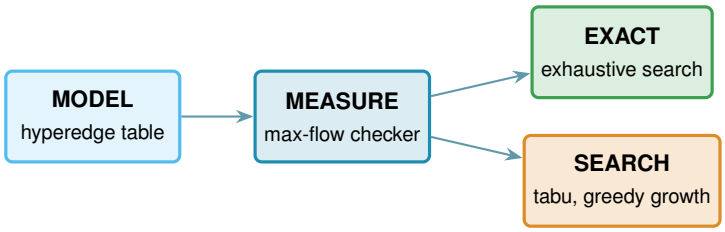


Figure 2.1. One model feeds the max-flow checker. Exhaustive search uses it to find exact values, and heuristic search uses it to find dense examples.

2.1 Modelling

The program treats all eight models in one way. A model is described by four facts: whether it is directed, whether edges may repeat, the edge size r, and, for directed hypergraphs, the allowed numbers of tails. A graph is the case $r = 2$. The program runs each directed hypergraph model in three versions, forward, backward and general (Section 1.3), so `ALL_MODELS` lists twelve entries for the eight models. The separation, edge or vertex, is a separate choice, so the eight models give the sixteen problems. A `Problem` bundles a model, a separation, the number of vertices n and the threshold m, and every function below takes one (Listing 2.1).

```
24 @dataclass(frozen=True)
25 class Model:
26     NAME: str
27     IS_DIRECTED: bool
28     ALLOWS_REPEATED_EDGES: bool
```

```
    EDGE_SIZE: int               # r: 2 for graphs, at least 3 otherwise
    TAIL_COUNTS: tuple = (1,)    # directed: the allowed numbers of tails


def ALL_MODELS(HYPEREDGE_SIZE=3):
    r = HYPEREDGE_SIZE
    splits = [("one tail", (1,)), ("one head", (r - 1,)),
              ("any split", tuple(range(1, r)))]
    return [
        Model("simple graph",             False, False, 2),
        Model("multigraph",               False, True,  2),
        Model("simple hypergraph",        False, False, r),
        Model("multihypergraph",          False, True,  r),
        Model("simple digraph",           True,  False, 2),
        Model("directed multigraph",      True,  True,  2),
    ] + [
        Model(f"{name}, {split}", True, repeated, r, tail_counts)
        for name, repeated in [("directed hypergraph", False),
                               ("directed multihypergraph", True)]
        for split, tail_counts in splits
    ]


class Separation(Enum):
    EDGE_DISJOINT = "edge"
    VERTEX_DISJOINT = "vertex"


@dataclass(frozen=True)
class Problem:
    MODEL: Model
    SEPARATION: Separation
    VERTEX_COUNT: int            # n
    THRESHOLD_M: int             # feasible: fewer than m disjoint routes
```

Listing 2.1. The models, with three versions of each directed hypergraph model, and the two separations.

Every edge is stored as a hyperedge: its vertices, and its tails if the model is directed. An ordinary edge uv is the hyperedge $\{u, v\}$ without tails, and an arc $u \to v$ is the hyperedge $\{u, v\}$ with tail u. A route may enter a hyperedge at an `entry_vertices` vertex and leave it at an `exit_vertices` vertex. In an undirected hyperedge every vertex is both. In a directed one only the tails are entries and only the heads are exits. A candidate graph is a table that gives every possible hyperedge its multiplicity. In a simple model every multiplicity is 0 or 1. In a multi model it is at most $m - 1$, since m copies of one edge already give m disjoint routes (Listing 2.2). Figure 2.2 shows two such tables.

```
@dataclass(frozen=True, order=True)
class Hyperedge:
    TAILS: tuple                 # empty for an undirected hyperedge
```

```
    MEMBERS: tuple               # all its vertices, sorted

    def entry_vertices(self):
        return self.TAILS or self.MEMBERS

    def exit_vertices(self):
        return tuple(v for v in self.MEMBERS if v not in self.TAILS)


def possible_hyperedges(problem):
    model, n = problem.MODEL, problem.VERTEX_COUNT
    hyperedges = []
    for members in itertools.combinations(range(n), model.EDGE_SIZE):
        if not model.IS_DIRECTED:
            hyperedges.append(Hyperedge((), members))
            continue
        for tail_count in model.TAIL_COUNTS:
            hyperedges += [Hyperedge(tails, members) for tails
                           in itertools.combinations(members, tail_count)]
    return hyperedges


def largest_multiplicity(problem):
    if problem.MODEL.ALLOWS_REPEATED_EDGES:
        return problem.THRESHOLD_M - 1
    return 1
```

Listing 2.2. Hyperedges, the list of all possible hyperedges, and the largest useful multiplicity.

2.2 Measuring

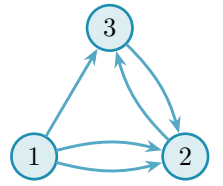
Every search below needs $\lambda^{\max}$ or $\kappa^{\max}$ of a candidate graph. This section explains how the program computes them.

To compute $\lambda(u, v)$ we use a maximum flow. In a network every link has a capacity, and a *maximum flow* is the largest amount that can be sent from u to v. We give each pair of vertices a link with capacity $\mu(u, v)$. In a digraph the link has a direction. In an undirected graph it can be used in both directions. The max-flow min-cut theorem of Ford and Fulkerson [6] says that the maximum flow equals the smallest capacity of a cut. By Menger's theorem, this is $\lambda(u, v)$. So computing a maximum flow computes $\lambda(u, v)$. We call this routine the *checker*.

2.2.1 Transformations

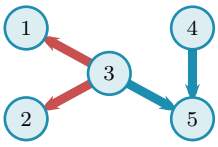
The checker works on one kind of network for all sixteen problems.

Every hyperedge, also an ordinary edge, becomes a *gate* (Figure 2.4b). The gate has an entry node and an exit node, joined by one arc whose capacity is the multiplicity of the hyperedge. Each vertex of the hyperedge gets an arc of unlimited capacity to the entry node and one from the exit node. So the arc inside the gate is the only bottleneck. For a directed hyperedge, only



directed multigraph
(a)

TAILS	MEMBERS	copies
$\{1\}$	$\{1,2\}$	**2**
$\{1\}$	$\{1,3\}$	1
$\{2\}$	$\{2,3\}$	1
$\{3\}$	$\{2,3\}$	1



directed hypergraph
(b)

TAILS	MEMBERS	copies
$\{\mathbf{3}\}$	$\{1,2,3\}$	1
$\{\mathbf{3},\mathbf{4}\}$	$\{3,4,5\}$	1

Figure 2.2. The program stores every model in one way: a table that gives each hyperedge its number of copies. (a) A directed multigraph. The arc $u \to v$ is the hyperedge with members $\{u, v\}$ and tail u, and the two parallel arcs from 1 to 2 are one row with 2 copies. (b) A directed 3-uniform hypergraph in the general model. Each hyperedge records its tails and its three members: the red one has tail 3, the blue one has tails 3 and 4 and head 5. In an undirected model there are no tails. A hyperedge that does not occur has 0 copies and is left out.

the tails have an arc to the entry, and only the heads have an arc from the exit (Figure 2.4c). So a route must go from a tail to a head. Theorem A.23 shows that the maximum flow in this network is the number of hyperedge-disjoint Berge routes. Figure 2.3 shows an example.

Every vertex v is split into two nodes v^{in} and v^{out}, joined by one arc (Figure 2.4a). Routes arrive at v^{in} and leave from v^{out}. For vertex separation this arc has capacity one, so at most one route can pass through v. For edge separation its capacity is unlimited, and it changes nothing. The gates keep their capacity, so parallel copies still give separate routes, as in the incidence convention of Section 1.3.1. Listing 2.3 builds this network.

```
113 def build_flow_network(problem, multiplicity):
114     network = networkx.DiGraph()
115     if problem.SEPARATION is Separation.VERTEX_DISJOINT:
116         through_vertex = 1
117     else:
118         through_vertex = math.inf
119     for v in range(problem.VERTEX_COUNT):
120         network.add_edge(("in", v), ("out", v), capacity=through_vertex)
121     for hyperedge, copies in multiplicity.items():
122         if copies == 0:
123             continue
124         enter, leave = ("enter", hyperedge), ("exit", hyperedge)
125         network.add_edge(enter, leave, capacity=copies)
126         for v in hyperedge.entry_vertices():
127             network.add_edge(("out", v), enter, capacity=math.inf)
128         for v in hyperedge.exit_vertices():
129             network.add_edge(leave, ("in", v), capacity=math.inf)
```

```
    return network
```

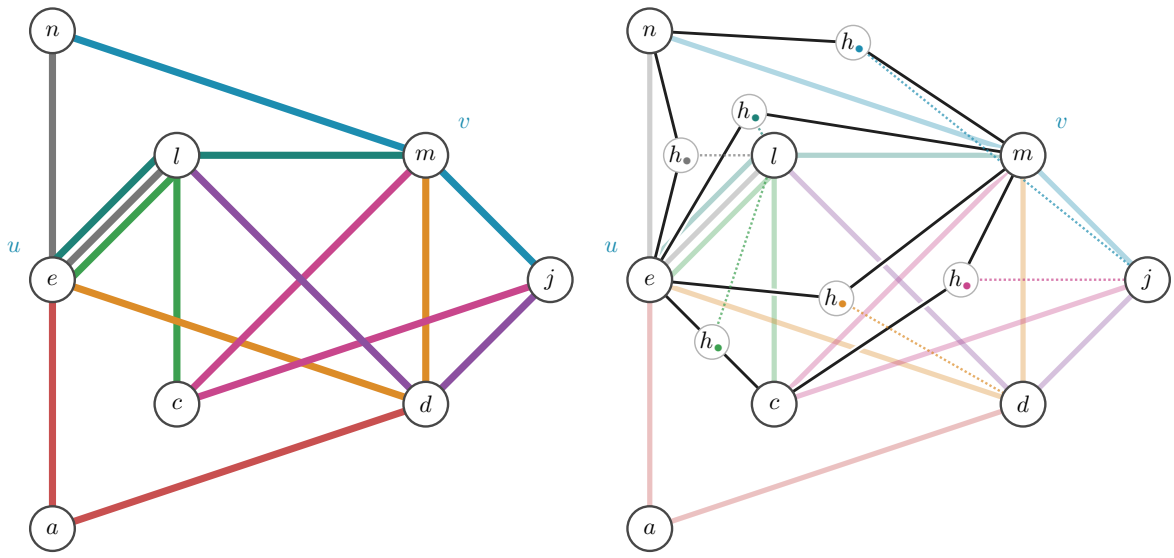
Listing 2.3. The flow network: split vertices and one gate per hyperedge.

The number of disjoint routes from u to v is the maximum flow from u^{out} to v^{in} (Listing 2.4). A graph is feasible when this number is below m for every pair, ordered in directed models. The flow may stop as soon as it reaches m, since a larger value changes nothing. This is the `stop_at` argument.

```
def disjoint_routes(network, source, target, stop_at=None):
    return networkx.maximum_flow_value(
        network, ("out", source), ("in", target),
        flow_func=networkx.algorithms.flow.edmonds_karp, cutoff=stop_at)


def is_feasible(problem, multiplicity):
    m = problem.THRESHOLD_M
    network = build_flow_network(problem, multiplicity)
    return all(disjoint_routes(network, u, v, stop_at=m) < m
               for u, v in vertex_pairs(problem))
```

Listing 2.4. Counting disjoint routes, and the feasibility test.



(a) (b)

Figure 2.3. (a) An undirected 3-uniform hypergraph. (b) The relevant part of its gate network for the pair u, v. The four black routes use pairwise disjoint sets of gates, so $\lambda(u, v) \geq 4$. The vertex v lies in only four hyperedges, and removing those four hyperedges separates u from v, so $\lambda(u, v) \leq 4$. Hence $\lambda(u, v) = 4$. Each gate is drawn as one point. Dotted lines indicate additional incidences of the displayed hyperedges.

2.3 Checking every graph of a fixed order

To find an exact value, the program must check every graph, or exclude it by a correct pruning rule. A simple graph on n vertices is a choice of $\mu(u, v) \in \{0, 1\}$ for each pair, and a multigraph

is a choice in $\{0, \ldots, m-1\}$. A digraph makes this choice for each ordered pair. A hypergraph is a choice of some of the $\binom{n}{r}$ possible hyperedges. A multihypergraph also chooses a multiplicity for each. A forward directed hypergraph has r possible hyperedges on each r-set, one for each choice of tail, and chooses some of these $r\binom{n}{r}$ hyperedges. A backward one also has r, and a general one has $2^r - 2$, one for each non-empty proper subset of tails. The program goes through all choices, checks each one, and keeps the densest feasible graph. If the search is stopped early, it gives only a lower bound.

The number of vertices matters most. The number of candidates is (values per entry) to the power (number of entries). For graphs the number of entries grows like n^2, and for r-uniform hypergraphs like n^r. A larger m gives more values per entry. Direction gives more entries, since both directions of a pair are chosen separately. Directed hypergraphs have the most candidates, because each r-set carries several possible hyperedges. The separation does not change the candidates. It only changes the test for each candidate.

So there are $2^{\binom{n}{2}}$ simple graphs and $2^{n(n-1)}$ simple digraphs on n labelled vertices. With multiplicities at most $m-1$ there are $m^{\binom{n}{2}}$ multigraphs and $m^{n(n-1)}$ directed multigraphs. Figure 2.5 shows how fast this grows.

2.3.1 Pruned search

A pruned search skips whole families of graphs. Fix an order on the possible edges and decide them one at a time. A partly built graph stands for all graphs that agree with it so far. A pruning rule can discard this whole family.

The first rule uses that adding an edge never lowers connectivity. If a partly built graph already breaks the condition, so does every way to complete it, and the branch is dropped.

The second rule bounds the size of every completion: the edges already placed, plus the number of undecided edges times the largest allowed multiplicity. If this cannot beat the best graph found so far, the branch is dropped (Figure 2.6).

Both rules only drop graphs that are infeasible or cannot beat the best one. So a finished pruned search gives the same answer as the full search.

Listing 2.5 decides the possible hyperedges one at a time, in a fixed order. For each it tries $1, 2, \ldots$ copies, and stops at the first number of copies that makes the graph infeasible, since more copies stay infeasible. Then it also tries 0 copies. The first line of `decide` is the second rule: if even all undecided hyperedges at full multiplicity cannot beat the best graph so far, the branch is dropped. After the time limit the search stops and reports the best graph so far as a lower bound only.

```
162 def exhaustive_search(problem, deadline):
163     hyperedges = possible_hyperedges(problem)
164     cap = largest_multiplicity(problem)
165     best = {"size": -1, "graph": {}}
166     multiplicity = {}
167 
168     def decide(index, size):
```

```
        if time.time() > deadline:
            raise OutOfTime
        undecided = len(hyperedges) - index
        if size + undecided * cap <= best["size"]:
            return                          # cannot beat the best
        if index == len(hyperedges):
            best["size"], best["graph"] = size, dict(multiplicity)
            return
        hyperedge = hyperedges[index]
        for copies in range(1, cap + 1):
            multiplicity[hyperedge] = copies
            if not is_feasible(problem, multiplicity):
                break                       # more copies stay infeasible
            decide(index + 1, size + copies)
        multiplicity.pop(hyperedge, None)
        decide(index + 1, size)

    try:
        decide(0, 0)
        return best["size"], best["graph"], True
    except OutOfTime:
        return best["size"], best["graph"], False
```

Listing 2.5. The pruned exhaustive search.

2.3.2 Generating search

The full search meets the same graph many times, with the vertices renamed. Another method avoids this. Prof. Jan Goedgebeur pointed us to the `nauty` toolkit of McKay and Piperno [14]. Its program `geng` lists the undirected graphs on n vertices, each only once up to renaming the vertices. For directed multigraphs, our program uses each such graph as a *support*. Each support edge $\{u, v\}$ gets multiplicities $\mu(u, v)$ and $\mu(v, u)$ between 0 and $m-1$, not both zero. Renaming the vertices of a digraph renames its support in the same way. So digraphs on different supports from `geng` stay different, even after renaming vertices. After each choice the checker tests the partial digraph, and a branch that breaks the condition is dropped. Digraphs of the largest size are reported once each, up to renaming the vertices. For this the program compares digraphs after trying every renaming of the vertices. It does so only when the search finishes, which happens only for small n. The supports are read from `geng` one at a time (Listing B.4), so the search can stop at the time limit before all of them are listed. We wrote our own step for choosing the multiplicities, because it can prune after each choice. Listing 2.6 shows the whole search. For each support edge $\{a, b\}$, `choices` lists the pairs $(\mu(a, b), \mu(b, a))$ that are not both zero.

```
def best_digraphs_on_support(problem, support_edges, deadline):
    m = problem.THRESHOLD_M
    choices = [(forward, backward) for forward in range(m)
               for backward in range(m) if forward + backward > 0]
    best = {"size": -1, "graphs": []}
    multiplicity = {}
```

```
201
202     def choose(index, size):
203         if time.time() > deadline:
204             raise OutOfTime
205         undecided = len(support_edges) - index
206         if size + undecided * 2 * (m - 1) < best["size"]:
207             return
208         if index == len(support_edges):
209             if size > best["size"]:
210                 best["size"], best["graphs"] = size, []
211             best["graphs"].append(dict(multiplicity))
212             return
213         a, b = support_edges[index]
214         a_to_b, b_to_a = Hyperedge((a,), (a, b)), Hyperedge((b,), (a, b))
215         for forward, backward in choices:
216             multiplicity[a_to_b], multiplicity[b_to_a] = forward, backward
217             if is_feasible(problem, multiplicity):
218                 choose(index + 1, size + forward + backward)
219         del multiplicity[a_to_b], multiplicity[b_to_a]
220
221     try:
222         choose(0, 0)
223         return best["size"], best["graphs"], True
224     except OutOfTime:
225         return best["size"], best["graphs"], False
```

Listing 2.6. The generating search for directed multigraphs, with `geng` from `nauty`.

2.4 The heuristic search

For larger n the full search is impossible. Instead we search for dense graphs in a guided way. A *heuristic* is a rule that moves the search towards better graphs. The methods below also allow moves that make the graph worse, so that the search does not get stuck.

For graphs and digraphs, each candidate gets an *energy*

$$\Phi(G) = -|E(G)| + \Lambda \cdot \max\bigl(0,\ c^{\max}(G) - (m-1)\bigr).$$

Here $c^{\max}(G)$ is $\lambda^{\max}(G)$ or $\kappa^{\max}(G)$, depending on the separation, and $\Lambda > 0$ is a fixed penalty. Low energy means many edges and no excess connectivity. In the program Λ is `PENALTY_PER_EXCESS_ROUTE` (Listing 2.7).

```
275 PENALTY_PER_EXCESS_ROUTE = 10
276
277
278 def energy(size, highest_connectivity, threshold_m):
279     excess = max(0, highest_connectivity - (threshold_m - 1))
280     return -size + PENALTY_PER_EXCESS_ROUTE * excess
```

Listing 2.7. The energy of a candidate graph.

The search starts from the empty graph and changes one multiplicity by one at each step. We use *tabu search* [15]. At each step it makes the best allowed move, even if every move raises the energy. After a move, the changed pair is *tabu* for a fixed number of steps, so the search cannot undo the move at once. A move that gives a better graph than any seen so far is always allowed. When the search stalls, it makes a small change to escape: it removes all copies of three random hyperedges. Listing 2.8 lists the moves, and Listing 2.9 is the search itself. The search keeps the exact connectivity of every pair of the current graph in `connectivity`, and `highest_after` uses it to judge a move cheaply (Section 2.4.2).

```
def moves(problem, multiplicity):
    cap = largest_multiplicity(problem)
    for hyperedge in possible_hyperedges(problem):
        copies = multiplicity.get(hyperedge, 0)
        if copies < cap:
            yield hyperedge, +1
        if copies > 0:
            yield hyperedge, -1


def highest_after(problem, multiplicity, move, connectivity):
    """Measures only near the cap: one copy more raises a value by at
    most one, so only pairs already at m-1 need a check, and one copy
    less never raises a value."""
    cap = problem.THRESHOLD_M - 1
    current_highest = max(connectivity.values(), default=0)
    _, change = move
    if change < 0 and current_highest <= cap:
        return current_highest
    if change > 0 and current_highest < cap:
        return current_highest + 1
    network = build_flow_network(problem, apply(multiplicity, move))
    at_cap = [pair for pair, value in connectivity.items()
              if value >= cap]
    return max((disjoint_routes(network, u, v) for u, v in at_cap),
               default=0)
```

Listing 2.8. The moves: one copy more or one copy less of one hyperedge.

```
TABU_TENURE = 7
STEPS_WITHOUT_PROGRESS_BEFORE_KICK = 30


def tabu_search(problem, deadline, seed):
    generator = random.Random(seed)
    m = problem.THRESHOLD_M
    multiplicity, tabu_until, step = {}, {}, 0
    connectivity = all_connectivities(problem, multiplicity)
    best_size, best_graph, last_progress = 0, {}, 0
    while time.time() < deadline:
        step += 1
```

```
        scored = []
        for move in moves(problem, multiplicity):
            highest = highest_after(problem, multiplicity, move,
                                    connectivity)
            size = edge_count(multiplicity) + move[1]
            new_best = highest <= m - 1 and size > best_size
            if tabu_until.get(move[0], 0) > step and not new_best:
                continue                    # tabu, unless a new best
            scored.append((energy(size, highest, m),
                           generator.random(), move))
        if not scored:
            break
        _, _, move = min(scored)
        multiplicity = apply(multiplicity, move)
        tabu_until[move[0]] = step + TABU_TENURE
        connectivity = all_connectivities(problem, multiplicity)
        size = edge_count(multiplicity)
        if max(connectivity.values(), default=0) <= m - 1 \
                and size > best_size:
            best_size, best_graph = size, multiplicity
            last_progress = step
        if step - last_progress > STEPS_WITHOUT_PROGRESS_BEFORE_KICK:
            kicked = generator.sample(list(multiplicity),
                                      min(3, len(multiplicity)))
            multiplicity = dict(multiplicity)
            for hyperedge in kicked:
                multiplicity[hyperedge] = 0
            connectivity = all_connectivities(problem, multiplicity)
            last_progress = step
    return best_size, best_graph
```

Listing 2.9. Tabu search.

The program also implements simulated annealing on the same energy. It accepts a worse move with a probability that decreases over time. At temperature T a move that raises the energy by Δ is accepted with probability $e^{-\Delta/T}$, and T shrinks by a fixed factor at every step. The driver in Listing 2.13 does not use it, so its code is in Appendix B (Listing B.6).

Every feasible graph can be built from the empty graph by adding edges one at a time, staying feasible. Still, it can help to pass through infeasible graphs: exchanging two edges may need an extra edge first. Tabu search allows this, but it only keeps feasible results.

Each search keeps the densest feasible graph it has seen.

2.4.1 Random greedy growth for hypergraphs

For hypergraphs we use random greedy growth. Each run starts with the empty hypergraph and a shuffled list of all possible hyperedges: each hyperedge once in the simple model, and $m - 1$ times in the multi model. The procedure adds the hyperedges one by one, and keeps each one only if the hypergraph stays feasible. This is repeated with new random orders, and the densest

result is kept.

At the end of a run no single hyperedge can be added, because a rejected hyperedge stays infeasible when more hyperedges are added. This does not mean the result is as large as possible: an early choice can block several later ones. We use this method instead of tabu search because there are many possible hyperedges, for example $\binom{n}{r}$ in the undirected simple case.

```
def random_greedy_growth(problem, deadline, seed):
    generator = random.Random(seed)
    cap = largest_multiplicity(problem)
    best_size, best_graph = 0, {}
    while time.time() < deadline:
        copies = [hyperedge for hyperedge in possible_hyperedges(problem)
                  for _ in range(cap)]
        generator.shuffle(copies)
        multiplicity = {}
        for hyperedge in copies:
            if time.time() > deadline:
                break                          # the partial graph is feasible
            grown = apply(multiplicity, (hyperedge, +1))
            if is_feasible(problem, grown):
                multiplicity = grown
        if edge_count(multiplicity) > best_size:
            best_size, best_graph = edge_count(multiplicity), multiplicity
    return best_size, best_graph
```

Listing 2.10. Random greedy growth.

2.4.2 Measuring only near the connectivity cap

Removing an edge never raises $\lambda^{\max}$, since it creates no new route. Adding an edge raises $\lambda(u, v)$ by at most one for every pair. Indeed, take a largest family of disjoint u–v routes after the addition. At most one of them uses the new edge, and the others are disjoint routes in the old graph. The same holds for $\kappa^{\max}$, since separated routes also share no edge. So the search does not need to run the checker at every step. Removing an edge from a feasible graph keeps it feasible. Adding an edge to a graph with $\lambda^{\max} \leq m - 2$ also keeps it feasible. Also, an addition can only push a pair above $m - 1$ if that pair is already at $m - 1$. So `highest_after` in Listing 2.8 answers without the checker when the graph stays feasible for sure. Otherwise it measures only the pairs at the cap. After a move is made, the search measures all pairs once, so the next step again starts from exact values.

2.5 Checking the output without trusting the program

The program is too long to check by hand, so a value it prints is not a proof. We solve this by separating *finding* from *checking*. The program may find a graph in any way, correct or not. Together with the graph it writes a short *certificate*: a list that a reader can check by hand, or with a few lines of independent code, without trusting the program.

Edge separation. For every pair u, v, the certificate gives a set of at most $m - 1$ edges (or arcs, or hyperedge copies) whose removal leaves no route from u to v. To check it, remove the set and follow the remaining edges from u: v must not be reached. Then every route from u to v uses an element of the set. Disjoint routes use different elements, so there are at most $m - 1$ of them, and no pair has m disjoint routes. This check uses only that a route through a removed set must use one of its elements. It needs neither Menger's theorem nor a maximum flow. For undirected graphs one tree is enough: any tree on the vertices such that, for each tree edge, at most $m - 1$ edges of G cross it. Take a pair u, v and a tree edge on the tree path between them. Removing this tree edge splits the vertices into two parts, with u and v on different sides. Every u–v route uses an edge of G that crosses between the parts, and there are at most $m - 1$ of these. So $n - 1$ counts replace $\binom{n}{2}$ checks. A Gomory–Hu tree (Theorem 1.4) of a feasible graph is such a tree.

Vertex separation. For every pair u, v, the certificate gives a set of at most $m - 1$ inner nodes of the incidence graph, vertices or edge copies, whose removal leaves no route from u to v. It is checked in the same way. Separated routes share no inner node, so again each of them uses a different element of the set.

Size of the graph. The number of edges is counted directly.

The program writes the certificate from a smallest cut of the flow network (Listing 2.11). A cut uses only arcs of limited capacity: the arc inside a gate, which stands for a hyperedge, and, for vertex separation, the arc inside a split vertex. So the cut translates directly into removed hyperedges and removed vertices.

```
def write_certificate(problem, multiplicity):
    network = build_flow_network(problem, multiplicity)
    certificate = {}
    for u, v in vertex_pairs(problem):
        _, (source_side, _) = networkx.minimum_cut(
            network, ("out", u), ("in", v))
        removed_vertices, removed_hyperedges = [], []
        for node in source_side:
            for neighbour in network.successors(node):
                if neighbour in source_side:
                    continue
                if node[0] == "in":
                    removed_vertices.append(node[1])
                else:
                    removed_hyperedges.append(node[1])
        certificate[(u, v)] = (removed_vertices, removed_hyperedges)
    return certificate
```

Listing 2.11. Writing a certificate from smallest cuts.

The checker in Listing 2.12 computes no maximum flow and uses none of the search code. It uses only the list of vertex pairs and the entry and exit vertices of a hyperedge, from Listings 2.2 and B.2. It counts the removed elements, with every copy of a removed hyperedge, and then follows the remaining hyperedges from u, entering each only at an entry vertex. If v is never

reached, the certificate is correct for this pair. Its correctness can be read off from its 28 lines and these two short helpers.

```
def check_certificate(problem, multiplicity, certificate):
    """No flows and no search code, only following edges."""
    for u, v in vertex_pairs(problem):
        removed_vertices, removed_hyperedges = certificate[(u, v)]
        if problem.SEPARATION is Separation.EDGE_DISJOINT \
                and removed_vertices:
            return False
        if u in removed_vertices or v in removed_vertices:
            return False
        size = len(set(removed_vertices)) + sum(
            multiplicity[e] for e in set(removed_hyperedges))
        if size > problem.THRESHOLD_M - 1:
            return False
        reached, frontier = {u}, [u]
        while frontier:
            vertex = frontier.pop()
            for hyperedge, copies in multiplicity.items():
                if copies == 0 or hyperedge in removed_hyperedges:
                    continue
                if vertex not in hyperedge.entry_vertices():
                    continue
                for nxt in hyperedge.exit_vertices():
                    if nxt not in reached and nxt not in removed_vertices:
                        reached.add(nxt)
                        frontier.append(nxt)
        if v in reached:
            return False
    return True
```

Listing 2.12. The independent certificate checker.

So every lower bound the program finds can be checked: the graph is feasible by its certificate, and its size is counted. An upper bound is different. A certificate that no graph on n vertices has more edges would be the whole search, which is too long to check. So the thesis takes no upper bound or exact value from the program. Every upper bound is proved by hand, and every construction in the thesis is checked by hand in Chapter 1 or Appendix A. The exhaustive and heuristic searches of this chapter show which bounds are worth trying to prove, and which constructions to try.

2.6 Running the program

Listing 2.13 puts the parts together. For each case, the driver allocates a nominal thirty-second search budget. The first fifteen seconds are allocated to an exact search: the generating search for directed multigraphs, and the pruned search for the other models. If that search does not finish, a further fifteen seconds are allocated to a heuristic search: tabu search for graphs and digraphs, and random greedy growth for hypergraphs. Deadline checks occur between operations, so

individual operations can overrun the budget. Certification and, for directed multigraphs, counting the graphs up to renaming the vertices add further overhead. Every result is then certified and checked. The program saves, for each case, the model, the separation, n and m, the hyperedges of the graph with their multiplicities, its size, the certificate for every vertex pair, and whether the search finished and the certificate passed. It writes these to the file `witnesses.json`, so a reader can recount the edges and check every certificate without rerunning the search. The cases are independent, so the program runs them in parallel, one per processor core. For each problem, and each version of a directed hypergraph problem, it prints a table with one row per n and one column per m (Listing B.7). Each entry is the number of edges of the densest feasible graph found, marked when the search finished, and marked differently if the certificate failed. A finished search gives an exact value, but only the lower bound is certified (Section 2.5).

The true maximum never decreases when n or m grows: an isolated vertex can be added, and a graph with no m disjoint routes also has no $m+1$. The printed entries need not follow this. A larger n or m also gives more candidates, often many more, and every case gets the same thirty-second budget. So an unfinished search covers a smaller part of a larger search space, and its entry can be smaller than the entry for a smaller n or m.

```
SECONDS_PER_CASE = 30

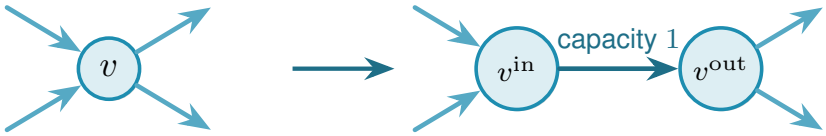

def solve(problem):
    model = problem.MODEL
    deadline = time.time() + SECONDS_PER_CASE / 2
    if model.IS_DIRECTED and model.ALLOWS_REPEATED_EDGES \
            and model.EDGE_SIZE == 2:
        size, graph, _, exact = generating_search(problem, deadline)
    else:
        size, graph, exact = exhaustive_search(problem, deadline)
    if not exact:
        deadline = time.time() + SECONDS_PER_CASE / 2
        if model.EDGE_SIZE == 2:
            search = tabu_search
        else:
            search = random_greedy_growth
        found_size, found_graph = search(problem, deadline, seed=1)
        if found_size > size:
            size, graph = found_size, found_graph
    certificate = write_certificate(problem, graph)
    checked = check_certificate(problem, graph, certificate)
    return size, exact, checked, witness(problem, graph, certificate)


def witness(problem, graph, certificate):
    """The graph and its certificate, in a form a reader can check."""
    def as_list(hyperedge):
        return [list(hyperedge.TAILS), list(hyperedge.MEMBERS)]
    return {
```

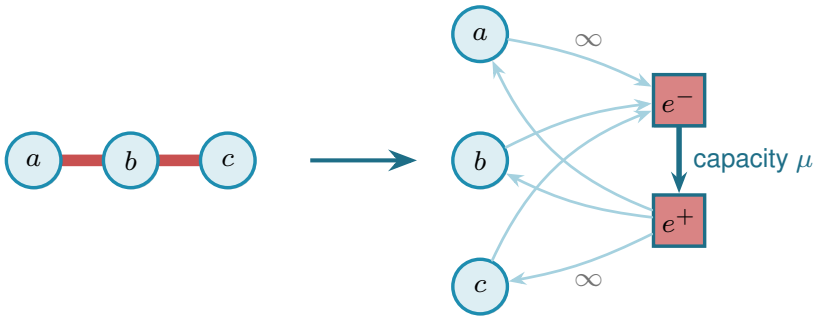
```
        "hyperedges": [as_list(e) + [copies]
                       for e, copies in sorted(graph.items()) if copies],
        "certificate": [[u, v, removed_vertices,
                         [as_list(e) for e in removed_hyperedges]]
                        for (u, v), (removed_vertices, removed_hyperedges)
                        in certificate.items()]}
```

Listing 2.13. The driver: a thirty-second budget per case, then a certificate.

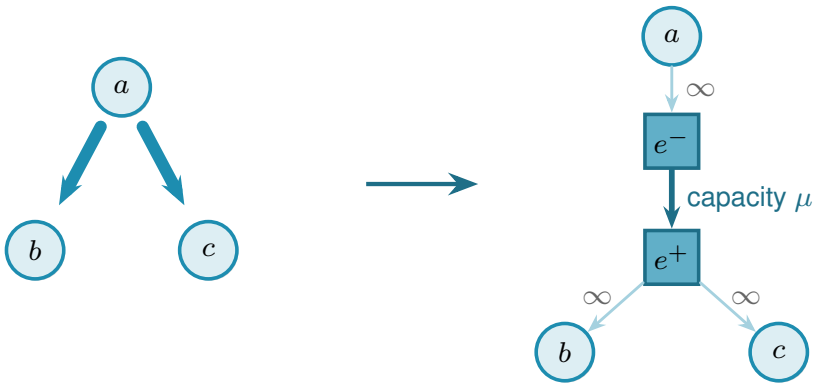
Section B.2 prints the full output for $2 \le m \le 7$ and $n \le 7$.



(a) Splitting each vertex v into v^{in} and v^{out}, joined by an arc of capacity one, turns vertex-disjoint routes into arc-disjoint routes. Only this inner arc has capacity one.



(b) Transform a hyperedge into a gate with capacity equal to the multiplicity of the hyperedge.



(c) A directed hyperedge with one tail and two heads is transformed into a gate. This gate has capacity equal to the multiplicity of the hyperedge and every route must enter at the tail and leave at a head. With several tails, each tail has an arc to e^-.

Figure 2.4. The building blocks of the connectivity checker: vertex splitting, hyperedge gates and directed hyperedge gates.

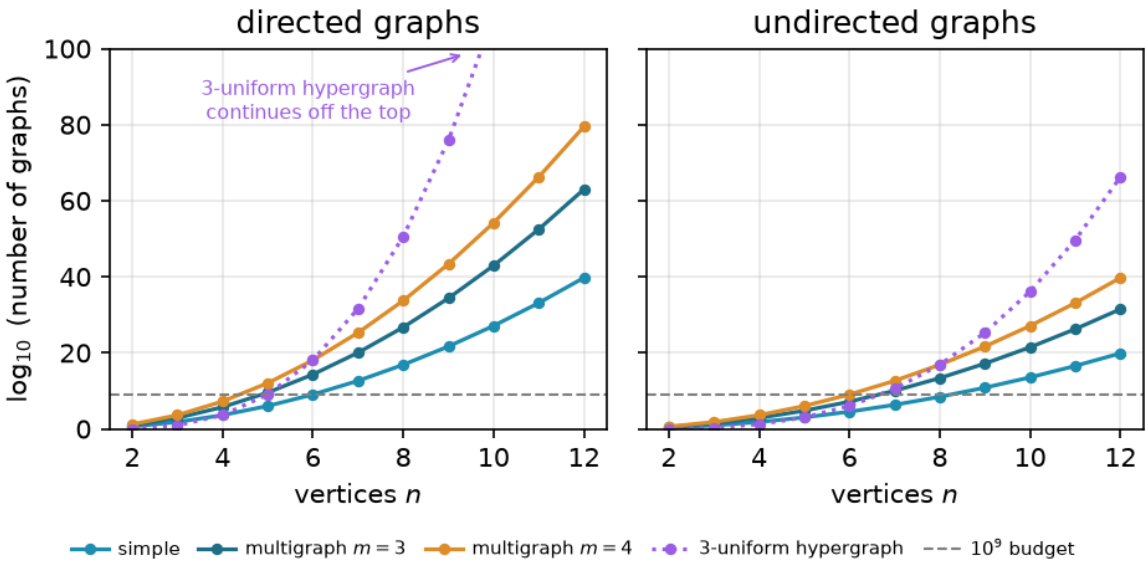


Figure 2.5. The number of candidates for blind enumeration, on a logarithmic scale. Every model passes 10^9 candidates before $n = 10$. Directed models grow fastest, because they have more entries to choose. The directed hypergraph curve is the forward model.

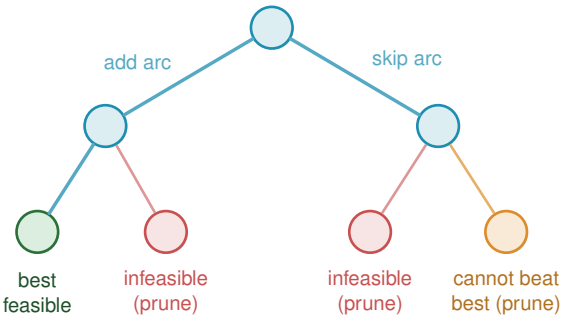


Figure 2.6. Each level of the partially built graph fixes one adjacency. Infeasible partial graphs (red) and branches unable to beat the incumbent (orange) are discarded.

Appendix A

Proofs

This appendix contains the proofs that are not in Chapter 1, and the lemmas they need. The definitions are in Chapter 1. New terms are defined where they are first used.

The appendix has three parts. PART I is about undirected graphs, PART II about directed graphs, and PART III about hypergraphs. The three main proof ideas are the reachability skeleton (directed multigraphs), the incidence-rank lemma (hypergraphs at $m = 3$), and the two-step budget (directed graphs and hypergraphs).

Part I. Undirected graphs

Graphs that attain Mader's bound, and the vertex problem as a problem about blocks.

A.1 Graphs that attain Mader's bound

The constructions below need hypergraphs in which no vertex lies in too many hyperedges. Recall that a graph is a 2-uniform hypergraph, and the *degree* of a vertex is the number of hyperedges that contain it.

Lemma A.1 (Sparse uniform hypergraphs exist)**.** *Let r, n and d be integers with $r \geq 1$, $n \geq r$ and $0 \leq d \leq \binom{n-1}{r-1}$. There is a simple r-uniform hypergraph on n vertices with exactly $\lfloor \frac{dn}{r} \rfloor$ hyperedges and maximum degree at most d.*

Proof. Let $N := \lfloor \frac{dn}{r} \rfloor$. Since $n\binom{n-1}{r-1} = r\binom{n}{r}$ (both count the pairs of an r-set and one of its vertices), we get $N \leq \frac{n}{r}\binom{n-1}{r-1} = \binom{n}{r}$. So there are at least N different r-sets. There are finitely many families of N different r-sets. Choose one with the smallest value of $\sum_x \deg(x)^2$. We show that any two degrees in this family differ by at most one.

Suppose instead that $\deg(x) \geq \deg(y) + 2$ for two vertices x, y. Let a be the number of sets in the family that contain x but not y, let b be the number that contain y but not x, and let c be the number that contain both. Then $\deg(x) = a+c$ and $\deg(y) = b+c$, so $a-b = \deg(x)-\deg(y) \geq 2$. For a set A that contains x but not y, the set $A-x+y$ is an r-set that contains y but not x. Different sets A give different sets $A - x + y$. So there are a such new sets, but the family has only $b < a$ sets that contain y but not x. So for some A, the set $A - x + y$ is not in the family. Replace A by $A - x + y$. The family still has N different sets. The degree of x drops by one, the degree of y rises by one, and the other degrees stay the same. So the sum changes by

$$(\deg(x) - 1)^2 + (\deg(y) + 1)^2 - \deg(x)^2 - \deg(y)^2 = 2\big(\deg(y) - \deg(x)\big) + 2 \leq -2.$$

This contradicts the choice of the family. So any two degrees differ by at most one.

Each of the N sets has r vertices, so the degrees add up to rN. Their average is rN/n. Since any two degrees differ by at most one, the largest degree is $\lceil rN/n \rceil$. Since $N \leq dn/r$, we have $rN/n \leq d$. As d is an integer, $\lceil rN/n \rceil \leq d$. □

Construction A.2 (Hub-and-spokes graph)**.** Let $n \geq m \geq 2$. The graph $H_{m,n}$ has a vertex h and further vertices $v_1, \ldots, v_{n-1}$. Join h to every v_i. On $v_1, \ldots, v_{n-1}$, add a graph with $\lfloor (m-2)(n-1)/2 \rfloor$ edges in which every vertex has degree at most $m-2$.

Such a graph exists. For $n = 2$ we have $m = 2$, and no edges are needed. For $n \geq 3$, apply Lemma A.1 with $r = 2$, with $n-1$ vertices and with $d = m-2$. The lemma needs $m-2 \leq n-2$, and this holds since $m \leq n$. If $m-1$ divides $n-1$, one can instead take disjoint copies of K_{m-1}. In that case $H_{m,n}$ consists of copies of K_m that share the vertex h.

Theorem A.3 (Extremal graphs exist)**.** *For all $n \geq m \geq 2$ the graph $H_{m,n}$ has $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ edges and $\lambda^{\max} \leq m-1$.*

Proof. The hub h has $n-1$ edges, and the other edges number $\lfloor (m-2)(n-1)/2 \rfloor$. Since $n-1$ is an integer, the total is

$$(n-1) + \left\lfloor \frac{(m-2)(n-1)}{2} \right\rfloor = \left\lfloor \frac{2(n-1)+(m-2)(n-1)}{2} \right\rfloor = \left\lfloor \frac{m(n-1)}{2} \right\rfloor.$$

Now take two vertices u, v. Only one vertex is not of the form v_i, so at least one of u, v is some v_i. This v_i has one edge to h and at most $m-2$ other edges, so its degree is at most $m-1$. Lemma 1.7 says that $\lambda(u,v)$ is at most the degree of u and at most the degree of v. So $\lambda(u,v) \leq m-1$. □

So for $n \geq m$ the bound of Theorem 1.3 is attained. The next theorem describes what equality means for the Gomory–Hu tree.

Theorem A.4 (Equality in the Gomory–Hu bound)**.** *Let G be simple with $\lambda_G^{\max} \leq m-1$ and $|E(G)| = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$, and let T be a Gomory–Hu tree of G.*

(a) If $m(n-1)$ is even, then every tree edge has weight $m-1$, every tree edge is an edge of G, and every edge of G has $\operatorname{dist}_T(e) \leq 2$.

(b) If $m(n-1)$ is odd, then exactly one of these three statements fails, and it fails by one: one tree edge has weight $m-2$, or one tree edge is not an edge of G, or one edge of G has $\operatorname{dist}_T(e) = 3$.

Proof. We first recall the proof of Theorem 1.3 in Chapter 1. The tree T has $n-1$ edges, and a tree edge t has weight $c(t)$. An edge e of G crosses $\operatorname{dist}_T(e)$ tree edges: the edges of the path in T between the ends of e. Let E_1 be the set of edges e of G with $\operatorname{dist}_T(e) = 1$. The proof uses one identity and three inequalities.

- ★ The identity is $\sum_e \operatorname{dist}_T(e) = \sum_t c(t)$ (Lemma 1.5).
- ★ Every weight is at most $m-1$, so $\sum_t c(t) \leq (m-1)(n-1)$.
- ★ Every edge outside E_1 has $\operatorname{dist}_T(e) \geq 2$, so $\sum_e \operatorname{dist}_T(e) \geq |E_1| + 2\big(|E| - |E_1|\big) = 2|E| - |E_1|$.
- ★ An edge in E_1 joins the two ends of a tree edge. Since G is simple, each tree edge gives at most one such edge, so $|E_1| \leq n-1$.

Each inequality has a gap, which is a non-negative integer:

$$S_1 = (m-1)(n-1) - \sum_t c(t), \qquad S_2 = \sum_{e:\, \operatorname{dist}_T(e) \geq 2} \big(\operatorname{dist}_T(e) - 2\big), \qquad S_3 = (n-1) - |E_1|.$$

So S_1 is the sum of $(m-1) - c(t)$ over the tree edges, S_2 adds up how far edges go beyond tree distance two, and S_3 is the number of tree edges whose ends are not joined by an edge of

G. The edges in E_1 add $|E_1|$ to $\sum_e \mathrm{dist}_T(e)$. The other edges add $2(|E| - |E_1|) + S_2$. With the identity this gives two expressions for the same sum:

$$\sum_e \mathrm{dist}_T(e) \;=\; 2|E| - |E_1| + S_2 \qquad \text{and} \qquad \sum_e \mathrm{dist}_T(e) \;=\; \sum_t c(t) \;=\; (m-1)(n-1) - S_1.$$

We substitute $|E_1| = (n-1) - S_3$ into the first and compare with the second:

$$\begin{aligned} 2|E| - (n-1) + S_2 + S_3 \;&=\; (m-1)(n-1) - S_1, \\ \text{so} \qquad 2|E| \;&=\; m(n-1) - (S_1 + S_2 + S_3). \end{aligned}$$

If $m(n-1)$ is even, then $|E| = m(n-1)/2$, so $S_1+S_2+S_3 = 0$. The three gaps are non-negative, so all three are zero. $S_1 = 0$ says that every tree edge has weight $m-1$. $S_3 = 0$ says that every tree edge is an edge of G. $S_2 = 0$ says that every edge of G has $\mathrm{dist}_T(e) \le 2$. This is (a).

If $m(n-1)$ is odd, then $|E| = (m(n-1)-1)/2$, so $S_1 + S_2 + S_3 = 1$. So exactly one gap is one, and the other two are zero. $S_1 = 1$ says that one tree edge has weight $m-2$. $S_3 = 1$ says that one tree edge is not an edge of G. $S_2 = 1$ says that one edge of G has $\mathrm{dist}_T(e) = 3$. This is (b). □

For $m = 2$ this shows that the extremal graphs are exactly the trees on the n vertices. Indeed, $m(n-1) = 2(n-1)$ is even, so by (a) every tree edge is an edge of G. So G contains T. Both have $n-1$ edges, so $G = T$ is a tree. Conversely, a tree has $n-1$ edges and $\lambda^{\max} = 1$.

Remark A.5 (Forcing and avoiding counts). This thesis maximises the number of edges with no m disjoint routes between any pair. Sørensen and Thomassen, and the Erdős Problems database, give instead the least number of edges that forces such a pair [3, 12]. Their number is one more than ours.

For example, Theorem 4 of [12] gives the forcing number $\lfloor 8n/3 \rfloor - 3$ for $n \ge 6$, except 16 at $n = 7$ and 28 at $n = 12$. Subtracting one gives our values. For $6 \le n \le 13$ these numbers, the two exceptions included, equal $\lfloor 5(n-1)/2 \rfloor + 1$. Subtracting one gives the two cases of Theorem 1.10. At $n = 13$ both maxima are 30, and the edge and vertex values first differ at $n = 14$.

The same care is needed for the Bollobás–Erdős conjecture of Section A.1.1. Its forcing number $1 + \binom{m}{2}p$ is for graphs on $1 + (m-1)p$ vertices. Each copy of K_m adds $\binom{m}{2}$ edges but only $m-1$ vertices, so in our notation the rate is $m/2$ edges per vertex.

A.1.1 The vertex problem is a block problem

The vertex problem $k_m(n)$ is open for $m \ge 6$. This section reduces it to 2-connected graphs (Theorem A.6).

A graph on more than k vertices is *k-connected* if removing any $k-1$ vertices leaves it connected. A *cut vertex* is a vertex whose removal disconnects the graph. A *bridge* is an edge whose removal disconnects its component. A *block* is a maximal connected subgraph that has an edge and no cut vertex of its own. In a simple graph, a block is either a bridge or a maximal

2-connected subgraph.

The next theorem says that the best graph is built from blocks: choose block sizes b_i, where a block of size b_i costs $b_i - 1$ vertices and gives at most $h_m(b_i)$ edges, with total cost at most $n - 1$.

Theorem A.6 (The vertex problem reduces to blocks)**.** *For $b \geq 2$ let*

$$h_m(b) \;:=\; \max\Big\{\, |E(B)| \;:\; B \text{ 2-connected or a single edge},\ |V(B)| = b,\ \kappa_B^{\max} \leq m-1 \,\Big\},$$

with $h_m(b) := -\infty$ when no such B exists. Then for all $m \geq 2$ and $n \geq 2$,

$$k_m(n) \;=\; \max\Big\{ \sum_i h_m(b_i) \;:\; b_i \geq 2,\ \sum_i (b_i - 1) \leq n-1 \Big\}.$$

Consequently $k_m(n) \geq k_m(n_1) + k_m(n_2)$ whenever $n = n_1 + n_2 - 1$, and

$$c_m \;:=\; \lim_{n\to\infty} \frac{k_m(n)}{n-1} \;=\; \sup_{b\geq 2} \frac{h_m(b)}{b-1}$$

exists, the limit being approached from below. The supremum is finite: Remark A.7 gives $c_m \leq m - 1$.

Proof. We use two facts about blocks. First, every edge lies in exactly one block. Second, a connected graph can be built by adding its blocks one at a time, so that each new block shares exactly one vertex with the blocks before it. So a block B adds $|V(B)| - 1$ new vertices, and in a component with c vertices, $\sum_B (|V(B)| - 1) = c - 1$.

Upper bound. Let G be feasible, and let u, v lie in a common block B. Suppose a u–v path leaves B at some vertex c of B. Since B is a block, every way from outside B back into B passes through c again. A path visits c only once, so this is impossible. So every u–v path stays in B, and $\kappa_G(u, v) = \kappa_B(u, v) \leq m - 1$. So every block is feasible. A block is a single edge or 2-connected, so $|E(B)| \leq h_m(|V(B)|)$. Every edge lies in exactly one block, so $|E(G)| = \sum_B |E(B)| \leq \sum_B h_m(|V(B)|)$. Adding the second fact over the components gives $\sum_B (|V(B)| - 1) \leq n - 1$. So the block sizes of G are one of the choices in the maximum, and $|E(G)|$ is at most that maximum.

Lower bound. Take blocks $B_1, \ldots, B_t$ with sizes b_i, with $\sum(b_i - 1) \leq n - 1$, each feasible with $h_m(b_i)$ edges. Glue them at one common vertex, and keep them disjoint otherwise. This graph has $1 + \sum(b_i - 1) \leq n$ vertices. Add isolated vertices to reach n. Its blocks are the B_i, so by the argument above a pair inside one block keeps its value of κ, which is at most $m - 1$. Two vertices in different blocks are separated by removing the common vertex, so $\kappa \leq 1 \leq m - 1$. So the graph is feasible, with $\sum_i h_m(b_i)$ edges.

Gluing. Take an extremal graph on n_1 vertices and one on n_2 vertices, and glue them at one vertex. Pairs inside one part keep their value of κ, as above, and pairs in different parts have $\kappa \leq 1$. The result is feasible on $n_1 + n_2 - 1$ vertices, so $k_m(n) \geq k_m(n_1) + k_m(n_2)$ for

$n = n_1 + n_2 - 1$.

The limit. Write $a_n = k_m(n)/(n-1)$ and $s = \sup_{b \geq 2} h_m(b)/(b-1)$. In the formula for $k_m(n)$, every block of size b_i gives at most $s\,(b_i - 1)$ edges. So $k_m(n) \leq s \sum_i (b_i - 1) \leq s\,(n-1)$, that is, $a_n \leq s$ for every n.

A lemma of Fekete [16] says: if numbers $x_1, x_2, \ldots$ satisfy $x_{i+j} \geq x_i + x_j$ for all i, j, then x_t/t tends to $\sup_t x_t/t$. Take $x_t = k_m(t+1)$. Gluing with $n_1 = i+1$ and $n_2 = j+1$ gives $x_{i+j} \geq x_i + x_j$. Also $x_t/t = a_{t+1}$. So a_n tends to $\sup_n a_n$. This supremum is at most s, since every $a_n \leq s$. It is also at least s: one block of size b is a feasible graph on b vertices, so $a_b \geq h_m(b)/(b-1)$ for every b. So a_n tends to s. Since $a_n \leq s$ for every n, it approaches s from below. □

We note three consequences.

First, at $m = 3$ the reduction gives Theorem 1.9 again. A block with $\kappa^{\max} \leq 2$ is an edge or a cycle. To see this, let B be a 2-connected block that is not a cycle. Since B is 2-connected, it contains a cycle C. Since B is connected and is not C, it has an edge wz that is not on C, with z on C. If w is on C, let P be this edge. Otherwise, $B - z$ is still connected, so it has a path from w to C. Let y be the first vertex of C on it, and let P be the path $z, w, \ldots, y$. In both cases P joins two different vertices x, y of C and has no inner vertex on C. The cycle C gives two internally disjoint x–y paths, and P is a third. So $\kappa(x, y) \geq 3$, which breaks the condition. So $h_3(2) = 1$ and $h_3(b) = b$ for $b \geq 3$, where b is the length of the cycle. The rate $h_3(b)/(b-1)$ is 1 at $b = 2$, $3/2$ at $b = 3$ and $b/(b-1) < 3/2$ for $b \geq 4$. So the best choice uses $\lfloor (n-1)/2 \rfloor$ triangles, and one extra edge when $n - 1$ is odd: $k_3(n) = 3\lfloor (n-1)/2 \rfloor + ((n-1) \bmod 2) = \left\lfloor \frac{3(n-1)}{2} \right\rfloor$.

Second, at $m = 4$, Theorem 1.9 gives $h_4(b) = 2(b-1)$ for every $b \geq 4$. So the rate $h_4(b)/(b-1)$ is 2 for $b \geq 4$, against 1 at $b = 2$ and $3/2$ at $b = 3$. The best choice uses blocks of size at least four, and $k_4(n) = 2(n-1)$ for $n \geq 4$. This uses Theorem 1.9, so it is not a new proof. It does show the extremal block: the hub-and-spokes graph $H_{4,b}$ of Construction A.2, with the graph on $v_1, \ldots, v_{b-1}$ chosen to be one cycle, so that $H_{4,b}$ is a hub joined to all vertices of a cycle. It is a single block with $2(b-1)$ edges.

Third, the Bollobás–Erdős construction, copies of K_m sharing one vertex, is the case $b = m$: $h_m(m) = \binom{m}{2}$, with rate $m/2$. Their conjecture says that $c_m = m/2$, so that no block beats K_m. Call $|E(G)|/(|V(G)| - 1)$ the *rate* of a graph G. A *bouquet* is a set of blocks that share one vertex and are otherwise disjoint. A bouquet with blocks B_i of sizes b_i has rate

$$\frac{\sum_i |E(B_i)|}{\sum_i (b_i - 1)} = \sum_i \frac{b_i - 1}{\sum_j (b_j - 1)} \cdot \frac{|E(B_i)|}{b_i - 1},$$

a weighted average of the rates of its blocks. For any graph G, the proof of Theorem A.6 gives $|E(G)| = \sum_B |E(B)|$ and $|V(G)| - 1 \geq \sum_B (|V(B)| - 1)$. So the rate of G is at most this weighted average for its own blocks, and some block has a rate at least the rate of G. So a counterexample must contain a block with rate above $m/2$. A single edge has rate $1 \leq m/2$, so this block is 2-connected.

The conjecture is false. Sørensen and Thomassen [12] showed that for each fixed m,

$$c_m \;\geq\; \frac{m(m-1)-2}{2m-3} \;=\; \frac{m}{2}+\frac{1}{4}+O(1/m).$$

This is larger than $m/2$ for every $m \geq 5$, because

$$2\big(m(m-1)-2\big)-m(2m-3) \;=\; m-4.$$

At $m = 5$, Theorem 1.10 gives $k_5(14) = \lfloor 112/3 \rfloor - 4 = 33$ edges on 14 vertices. Its rate is $33/13 > 5/2$, so by the bouquet argument above such a graph contains a block with rate above $5/2$. So $c_m = m/2$ for $m \leq 4$, $c_5 = 8/3$ by Theorem 1.10, and $m/2 < c_m \leq m-1$ for $m \geq 6$ by the next remark.

Remark A.7 (Feasible graphs are $(m-1)$-degenerate). A theorem of Mader [17] states that every simple graph of minimum degree at least $d \geq 2$ contains two adjacent vertices joined by d internally disjoint paths. Consequently, every graph feasible for the vertex problem has a vertex of degree at most $m-1$. Since taking a subgraph cannot increase local vertex-connectivity, every non-empty subgraph of a feasible graph has such a vertex. Thus every feasible graph is $(m-1)$-*degenerate*.

Deleting vertices successively and counting the edges in reverse order, the i-th vertex has at most $\min(i-1,\, m-1)$ neighbours among the earlier ones. This gives

$$\begin{aligned} k_m(n) \;\leq\; \sum_{i=1}^{n} \min(i-1,\, m-1) \;&=\; \binom{m-1}{2} + (n-m+1)(m-1) \\ &=\; (m-1)n - \binom{m}{2} \qquad (n \geq m-1). \end{aligned}$$

Hence $c_m \leq m-1$.

Part II. Directed graphs

The directed constructions, the reachability skeleton, the bounds for $m \geq 3$, and the extremal directed multigraphs.

A.2 The directed constructions

Three constructions give the directed lower bounds. Figure 1.8 shows the first two. Here we define them and check their connectivity.

Construction A.8 (Directed hub)**.** For $n \geq m \geq 2$, take a hub h and label the other $N = n-1$ vertices $0, \ldots, N-1$. Add the $2N$ hub arcs $h \to i$ and $i \to h$, and on the non-hub vertices the $(m-2)N$ ring arcs $i \to i+j \bmod N$ for $j = 1, \ldots, m-2$. The total is $m(n-1)$ arcs and every

non-hub vertex has $d^+ = d^- = m - 1$.

The ring arcs at i go to $m - 2$ different vertices, since $m - 2 < N$. With the hub arcs, every non-hub vertex has out-degree and in-degree $1 + (m - 2) = m - 1$. Every ordered pair (u, v) contains a vertex other than the hub. Lemma 1.7 says that $\lambda(u, v) \le d^+(u)$ and $\lambda(u, v) \le d^-(v)$. So $\lambda(u, v) \le m - 1$, that is, $\lambda^{\max} \le m - 1$. At $m = 2$ there are no ring arcs, and we get the *bidirected star*.

Construction A.9 (One-way bipartite digraph)**.** Split the vertices into parts A and B of sizes $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$, and include every arc from A to B and none the other way. The total is $\lfloor n^2/4 \rfloor$ arcs.

No arc leaves B and no arc enters A. So every route has length one, and $\lambda^{\max} = 1$.

Construction A.10 (Augmented bipartite digraph)**.** For $n \ge m \ge 2$, set $|B| = \lceil (n + m - 2)/2 \rceil$ and $|A| = n - |B|$. Number the vertices of B as $b_1, \ldots, b_{|B|}$, and read the indices modulo $|B|$. Include every arc from A to B, and inside B give each b_j the $m-2$ in-arcs from $b_{j-1}, \ldots, b_{j-(m-2)}$. The total is $\lfloor (n + m - 2)^2/4 \rfloor$ arcs.

Since $n \ge m$, we have $|B| \ge (2m - 2)/2 = m - 1$, so the vertices $b_{j-1}, \ldots, b_{j-(m-2)}$ are different from each other and from b_j.

Lemma A.11 (The augmented bipartite digraph is feasible)**.** *For $m \ge 2$ and $n \ge m$, the digraph of Construction A.10 satisfies $\lambda^{\max} = m - 1$ and has $\lfloor (n + m - 2)^2/4 \rfloor$ arcs. Hence $\ell_m^{\mathrm{dir}}(n) \ge \lfloor (n + m - 2)^2/4 \rfloor$.*

Proof. Counting. Each vertex of B receives one arc from every vertex of A and $m - 2$ arcs from inside B. No other arcs exist. So the number of arcs is $|B|\,(|A| + m - 2)$. Write $N = n + m - 2$. Then $|B| = \lceil N/2 \rceil$ and $|A| + m - 2 = N - |B| = \lfloor N/2 \rfloor$. For every integer N we have $\lfloor N/2 \rfloor \lceil N/2 \rceil = \lfloor N^2/4 \rfloor$ (check even and odd N).

Pairs from A to B. Take $a \in A$ and $b \in B$. The direct arc $a \to b$ is one route. For each of the $m - 2$ in-neighbours b' of b in B, the route $a \to b' \to b$ is another. These $m - 1$ routes use different arcs. Now take any a–b route. No arc enters A, so the route never comes back to A. So its last arc is $a \to b$ or one of the $m - 2$ arcs into b from B. Removing these $m - 1$ arcs therefore separates a from b. By Menger's theorem, $\lambda(a, b) = m - 1$.

Other pairs. For $b_i, b_j \in B$, a route from b_i stays in B, since no arc leaves B. So its last arc is one of the $m-2$ arcs into b_j from B, and $\lambda(b_i, b_j) \le m-2$. No arc leaves B, so there is no route from B to A. No arc goes from A to A, and a route from A never comes back to A, so there is no route between two vertices of A. So $\lambda^{\max} = m - 1$. □

For example, take $m = 3$ and $n = 10$. Then $|B| = 6$ and $|A| = 4$. There are 24 arcs from A to B, and 6 arcs forming a cycle inside B. Each pair a_i, b_j has the direct arc and one detour, so $\lambda^{\max} = 2$. This gives 30 arcs, more than the $\max(m(n - 1), \lfloor n^2/4 \rfloor) = 27$ of the first two

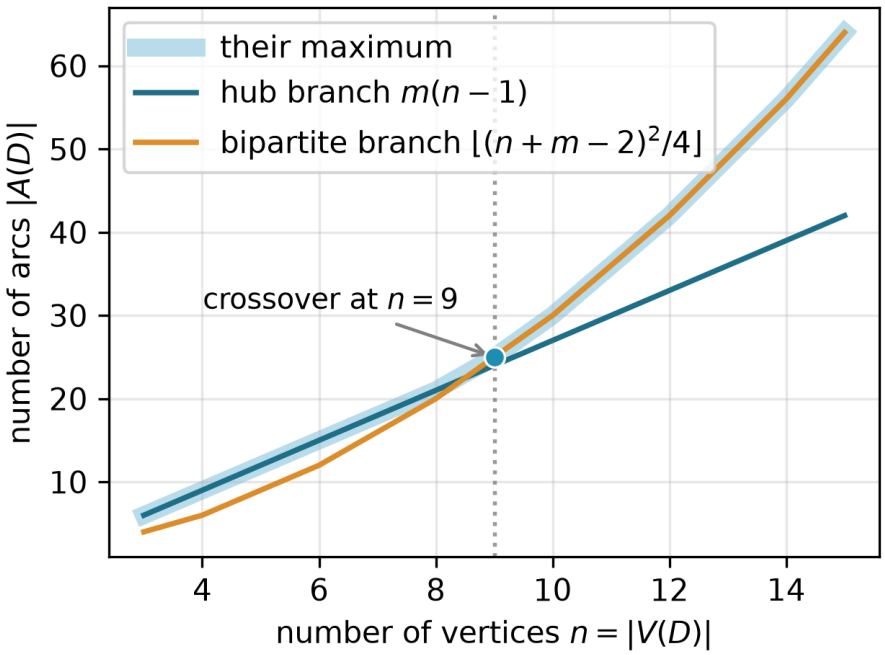


Figure A.1. The arc counts of the hub and of the augmented bipartite digraph at $m = 3$, and their maximum. These are lower bounds only.

constructions. The gain comes from making B larger than half of the vertices. So for $n \geq m$,

$$\ell_m^{\mathrm{dir}}(n) \;\geq\; \max\Big(m(n-1),\; \Big\lfloor \frac{(n+m-2)^2}{4} \Big\rfloor\Big).$$

For fixed m and large n the augmented bipartite digraph has more arcs (Figures A.1 and A.2).

A.3 The reachability skeleton

The proofs of Theorems 1.12 and 1.18 in Chapter 1 use the next lemma. Recall from Theorem 1.18 that

$$M(n) \;=\; \max\Big(2(n-1),\; \Big\lfloor \frac{n^2}{4} \Big\rfloor\Big).$$

Definition A.12 (Reachability skeleton)**.** Let D be a directed multigraph. A *reachability skeleton* of D is a subdigraph R on the same vertices, with at most one copy of each arc, such that for all $u \neq v$: u can reach v in R exactly when u can reach v in D.

A skeleton keeps the information about who can reach whom. It does not keep the number of disjoint paths.

Lemma A.13 (A sparse reachability skeleton always exists)**.** *Every loopless directed multigraph D on $n \geq 2$ vertices has a reachability skeleton R with $|A(R)| \leq M(n)$.*

Proof. We build R in three steps: first inside each strongly connected component, then between components, and then we count. A *strongly connected component* is a maximal set of vertices

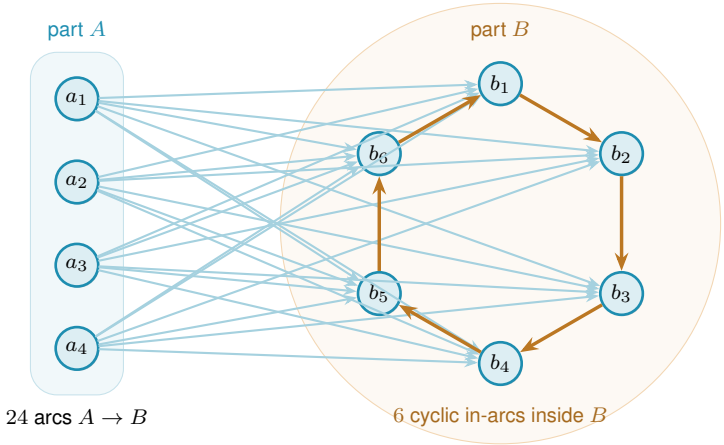


Figure A.2. The augmented bipartite digraph of Construction A.10 at $m = 3$, $n = 10$. All arcs between the parts go from A to B. Inside B, each vertex gets one arc from the previous vertex on a cycle. A pair a_i, b_j has the direct arc and one detour, so $\lambda^{\max} = 2$. The digraph has $24 + 6 = 30$ arcs, against 27 for the hub.

in which any two can reach each other. Every digraph splits into such components. Figure A.3 shows an example.

Step 1: inside each component. Take a component with n_i vertices and choose a vertex r in it. We choose $n_i - 1$ arcs along which r reaches every vertex of the component. Start with r alone. As long as some vertex of the component is not yet reached, there is an arc of D from a reached vertex to an unreached vertex of the component, since r reaches every vertex of the component along arcs inside it. Add this arc. Each arc reaches one new vertex, so this ends after $n_i - 1$ arcs. In the same way, with all directions reversed, there are $n_i - 1$ arcs along which every vertex of the component reaches r. Any two vertices u, v of the component are then joined by a walk from u to r and on to v along these arcs, and this walk contains a u–v path. So at most $2(n_i - 1)$ arcs keep all reachability inside the component.

Step 2: between components. Replace each component by one point. Put an arc from one point to another if D has an arc from the first component to the second. This gives a digraph Q_0 on q points, one per component. Q_0 has no directed cycle. Indeed, all vertices of the components on such a cycle could reach each other, so they would lie in one component. Now delete arcs from Q_0 one at a time, as long as this does not change which points of Q_0 reach which. Call the result Q. No arc of Q can be deleted without changing this.

The graph Q, with directions ignored, has no triangle. Indeed, suppose three points of Q are joined in pairs. Two points are never joined both ways, since that would be a directed cycle. Three arcs on three points that form no directed cycle always look like $x \to y$, $y \to z$ and $x \to z$, after renaming: x is the point with both arcs leaving, and z the point with both arcs entering. But then $x \to z$ could be deleted, since x still reaches z through y. This contradicts the choice of Q. By Mantel's theorem [18], a graph on q vertices with no triangle has at most $\lfloor q^2/4 \rfloor$ edges. So

$$|A(Q)| \;\le\; \left\lfloor \frac{q^2}{4} \right\rfloor .$$

Step 3: counting. Let R consist of the two trees of Step 1 in every component, and, for each arc of Q, one arc of D between the corresponding components. By Step 1, reachability inside each component is kept. Now suppose u reaches v in D, with u and v in different components. Then the component of u reaches the component of v in Q_0, and so also in Q. Follow the arcs of Q from the first component to the last. For each arc of Q, the set R has an arc of D between the two components. Inside each component, the arcs of Step 1 lead from the vertex where we enter to the vertex where we leave. So u reaches v in R. Since R is part of D, it creates no new reachability. So R is a reachability skeleton. If the components have sizes $n_1, \dots, n_q$, then

$$|A(R)| \;\le\; \sum_{i=1}^{q} 2(n_i - 1) \;+\; |A(Q)| \;\le\; 2(n-q) \;+\; \left\lfloor \frac{q^2}{4} \right\rfloor \;=:\; f(q).$$

It remains to bound $f(q)$ for $1 \le q \le n$. Since $\lfloor (q+1)^2/4 \rfloor - \lfloor q^2/4 \rfloor = \lfloor (q+1)/2 \rfloor$ (check even and odd q),

$$f(q+1) - f(q) \;=\; -2 \;+\; \left\lfloor \frac{q+1}{2} \right\rfloor .$$

These differences never decrease as q grows. So f first goes down (or stays level) and then goes up (or stays level). Its largest value on $\{1, \dots, n\}$ is therefore at $q = 1$ or at $q = n$. Since $f(1) = 2(n-1)$ and $f(n) = \lfloor n^2/4 \rfloor$, we get $f(q) \le M(n)$. □

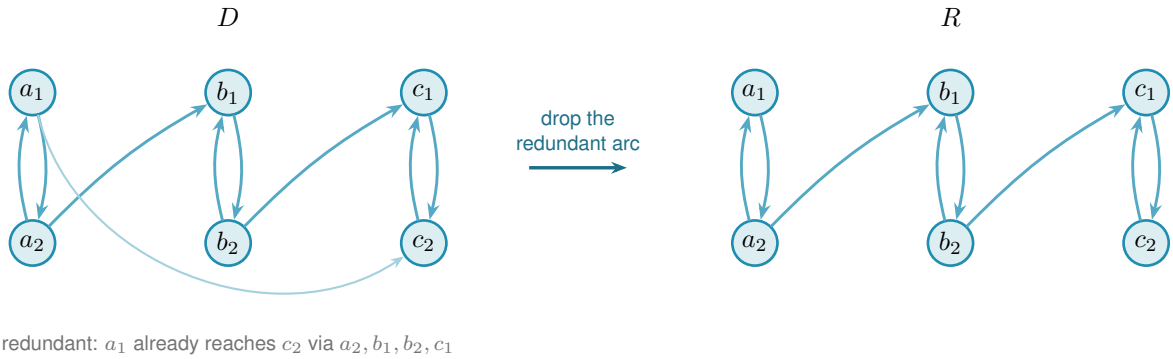


Figure A.3. The construction of Lemma A.13. Each bidirected pair A, B, C is a strongly connected component. Contracting the pairs gives $A \to B \to C$ and the redundant arc $A \to C$. Delete that arc to obtain the triangle-free digraph Q. The skeleton R keeps the internal arcs of each pair and the two remaining connections between pairs. Although $a_1 \to c_2$ is removed, a_1 still reaches c_2 through B.

A.4 Simple digraphs beyond the first threshold

A.4.1 The two-step budget

Definition A.14 (Two-step budget)**.** Let D be a simple digraph and $C \ge 1$. For $u \ne v$, let $p_2(u, v)$ be the number of vertices $x \notin \{u, v\}$ with arcs $u \to x$ and $x \to v$. Such an x is the *midpoint* of the route $u \to x \to v$. We call D *C-budgeted* if every ordered pair (u, v) of distinct vertices satisfies

$$\mathbf{1}_{(u,v) \in A(D)} + p_2(u, v) \;\le\; C.$$

Lemma A.15 (Feasible digraphs are budgeted)**.** *Let D be a simple digraph and $m \geq 2$. If $\lambda_D^{\max} \leq m-1$, or if $\kappa_D^{\max} \leq m-1$, then D is $(m-1)$-budgeted.*

Proof. Fix an ordered pair (u, v). Take the arc $u \to v$, if it exists, and the route $u \to x \to v$ for each of the $p_2(u, v)$ midpoints x. Two routes with different midpoints use different arcs, and none of them uses the arc $u \to v$. So these routes are arc-disjoint. Each two-step route has one inner vertex, its midpoint, and the arc has none. So the routes are also internally vertex-disjoint. So their number is at most $\lambda_D(u, v)$ and at most $\kappa_D(u, v)$, and either condition gives at most $m-1$. □

The next lemma turns the budget into a statement about degrees. A vertex with large in-degree and large out-degree is the midpoint of many two-step routes, and the budget limits these.

Lemma A.16 (The budget, read on degrees)**.** *Every C-budgeted simple digraph D on n vertices satisfies*

$$\sum_{x \in V} d^+(x)\, d^-(x) \;\leq\; C\, n(n-1).$$

Proof. Count the walks $u \to x \to v$ of length two. Choosing the middle vertex x, an in-neighbour u and an out-neighbour v gives $\sum_x d^-(x)\, d^+(x)$ walks. The walks with $u \neq v$ number $\sum_{u \neq v} p_2(u, v)$. A walk with $u = v$ is fixed by its first arc $u \to x$, so there are at most $|A(D)|$ of them. Summing the budget over all ordered pairs gives $|A(D)| + \sum_{u \neq v} p_2(u, v) \leq C\, n(n-1)$. So

$$\sum_{x} d^+(x)\, d^-(x) \;\leq\; \sum_{u \neq v} p_2(u, v) + |A(D)| \;\leq\; C\, n(n-1). \qquad \square$$

Lemma A.17 (Deleting the smaller half of every vertex)**.** *Every simple digraph D on n vertices can be made* one-way bipartite*, so that all remaining arcs go from one part to the other, by deleting at most $\sum_x \min(d^+(x), d^-(x))$ arcs. So*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; \sum_{x \in V} \min\bigl(d^+(x), d^-(x)\bigr).$$

Proof. Call x *source-like* if $d^+(x) \geq d^-(x)$, and *sink-like* otherwise. Let S and T be these two sets. Delete every arc into a source-like vertex and every arc out of a sink-like vertex. At a source-like vertex x this deletes its $d^-(x) = \min(d^+(x), d^-(x))$ in-arcs. At a sink-like vertex it deletes its $d^+(x) = \min(d^+(x), d^-(x))$ out-arcs. An arc may be deleted at both of its ends, so in total at most the sum above is deleted. An arc $a \to b$ survives only if a is not sink-like and b is not source-like, that is, $a \in S$ and $b \in T$. So the remaining arcs all go from S to T. Since $|S| + |T| = n$, there are at most $|S|\,|T| \leq \lfloor n^2/4 \rfloor$ of them. □

Lemma A.18 (The counting core)**.** *Every C-budgeted simple digraph D on $n \geq 2$ vertices satisfies*

$$|A(D)| \;\leq\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; C(n-1).$$

Proof. Let v be a vertex of largest out-degree, and write $\Delta = d^+(v)$. Let $N^+(v)$ be the set of the Δ vertices x with $v \to x$, and let $\delta(v)$ be the number of those x that also have $x \to v$. So $\delta(v) \le \Delta$.

First we count the routes of length two that start at v. We count the pairs (x, w) with $v \to x$, $x \to w$ and $w \neq v$ in two ways. Fixing $w \neq v$ first gives the midpoints of the routes $v \to x \to w$. The conditions $x \neq v$ and $x \neq w$ hold automatically, since $v \to x$ and $x \to w$. So this count is $\sum_{w \neq v} p_2(v, w)$. Fixing $x \in N^+(v)$ first gives the $d^+(x)$ out-neighbours of x, minus one exactly when $x \to v$. So this count is $\sum_{x \in N^+(v)} d^+(x) - \delta(v)$. Therefore

$$\sum_{w \neq v} p_2(v, w) \;=\; \sum_{x \in N^+(v)} d^+(x) \;-\; \delta(v).$$

Adding the budget over the $n-1$ pairs (v, w) with $w \neq v$ gives

$$\Delta \;+\; \sum_{x \in N^+(v)} d^+(x) \;-\; \delta(v) \;\le\; C(n-1),$$

since v has Δ out-arcs. With $\delta(v) \le \Delta$ this gives $\sum_{x \in N^+(v)} d^+(x) \le C(n-1)$.

Now we count the arcs by their tails, since every arc has one tail. Split the vertices into v, the set $N^+(v)$, and the remaining $n - 1 - \Delta$ vertices, each of out-degree at most Δ. So

$$|A(D)| \;\le\; \Delta \;+\; C(n-1) \;+\; (n-1-\Delta)\,\Delta \;=\; C(n-1) \;+\; \Delta(n-\Delta).$$

Finally $\Delta(n-\Delta) = n^2/4 - (\Delta - n/2)^2 \le n^2/4$, and $\Delta(n-\Delta)$ is an integer, so $\Delta(n-\Delta) \le \lfloor n^2/4 \rfloor$. □

We can now prove Theorem 1.17.

Theorem 1.17 (restated). *For all $m \ge 2$ and $n \ge 2$, every simple digraph D on n vertices with $\lambda_D^{\max} \le m-1$, or with $\kappa_D^{\max} \le m-1$, satisfies*

$$|A(D)| \;\le\; \left\lfloor \frac{n^2}{4} \right\rfloor \;+\; (m-1)(n-1),$$

so for each fixed $m \ge 3$, both $\ell_m^{\mathrm{dir}}(n)$ and $k_m^{\mathrm{dir}}(n)$ equal $n^2/4 + \Theta_m(n)$.

Proof of Theorem 1.17. Upper bound. By Lemma A.15, a simple digraph with $\lambda_D^{\max} \le m-1$ or with $\kappa_D^{\max} \le m-1$ is $(m-1)$-budgeted. So Lemma A.18 with $C = m-1$ gives the stated bound.

Lower bound. The digraph of Construction A.10 has $\lambda^{\max} \le m-1$ (Lemma A.11). Since $\kappa \le \lambda$, also $\kappa^{\max} \le m-1$. For $n \ge m$ this gives

$$k_m^{\mathrm{dir}}(n) \;\ge\; \ell_m^{\mathrm{dir}}(n) \;\ge\; \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor \;=\; \frac{n^2}{4} + \frac{(m-2)n}{2} + O_m(1).$$

For $m \ge 3$ the term $(m-2)n/2$ is positive. So both functions are $n^2/4$ plus a term that is at least

and at most a constant times n. □

The linear coefficient is not known. The construction gives $(m-2)n/2$, and Theorem 1.17 allows $(m-1)n$. A theorem of Huang and Lyu [13, Theorem 2], with $t = m-1 \geq 2$, also gives the linear term $(m-1)n$, for $n \geq \max(t^3+4t^2+3t+4,\ 17t^2/2+30t+27)$. It applies because a feasible digraph has at most $m-1$ paths of length two between any two vertices (Lemma A.15). Theorem 1.17 holds for every n.

The next proposition is about the shape of a feasible digraph. After deleting $O(\sqrt{m}\,n^{3/2})$ arcs, all remaining arcs go from one part to the other. The parts need not be balanced. For fixed m this error is larger than the linear error of Theorem 1.17.

Proposition A.19 (Almost all arcs go one way)**.** *Let D be a simple digraph on $n \geq 2$ vertices with $\lambda_D^{\max} \leq m-1$. Then deleting at most $\sqrt{m}\,n^{3/2}$ arcs leaves D a subgraph of a one-way bipartite digraph, and*

$$|A(D)| \ \leq\ \left\lfloor \frac{n^2}{4} \right\rfloor \ +\ \sqrt{m}\,n^{3/2}.$$

Proof. By Lemma A.15, D is $(m-1)$-budgeted. We use $\min(a,b) \leq \sqrt{ab}$, the Cauchy–Schwarz inequality $\sum_x \sqrt{a_x} \leq \sqrt{n\sum_x a_x}$, and Lemma A.16:

$$\begin{aligned}\sum_{x\in V} \min\bigl(d^+(x), d^-(x)\bigr) \ &\leq\ \sum_{x\in V} \sqrt{d^+(x)\,d^-(x)} \ \leq\ \sqrt{n\sum_{x\in V} d^+(x)\,d^-(x)} \\ &\leq\ \sqrt{n\cdot(m-1)\,n(n-1)} \ =\ n\sqrt{(m-1)(n-1)} \ \leq\ \sqrt{m}\,n^{3/2}.\end{aligned}$$

Lemma A.17 now gives both statements. □

So for $m \geq 3$ both directed problems have leading term $n^2/4$ and a linear second term. The linear coefficient and the exact values are open. It is also open whether the two values are equal.

A.5 The extremal directed multigraphs

We recall the result of Chapter 1.

Theorem 1.18 (restated). *For all $n \geq 2$ and $m \geq 2$, the largest number of arcs of a directed multigraph on n vertices with $\lambda^{\max} \leq m-1$ is*

$$L_m^{\mathrm{dir}}(n) \ =\ (m-1)\,M(n), \qquad M(n) = \max\Bigl(2(n-1),\ \left\lfloor \frac{n^2}{4} \right\rfloor\Bigr).$$

Its proof of the upper bound goes as follows. Take a reachability skeleton R_1 of D with at most $M(n)$ arcs (Lemma A.13), and delete one copy of each of its arcs. By Lemma 1.21, this lowers $\lambda^{\max}$ by at least one. Repeat this with skeletons $R_2, R_3, \ldots$ of what is left. After deleting $R_1, \ldots, R_{m-1}$, we have $\lambda^{\max} = 0$, so no arc is left. So $|A(D)| = \sum_i |A(R_i)| \leq (m-1)M(n)$.

At $m = 2$ a feasible directed multigraph has no parallel arcs, since two copies of an arc are two disjoint routes. So Theorem 1.18 at $m = 2$ is Theorem 1.11.

A.5.1 The extremal multigraphs for large n

Theorem 1.18 gives the value, but not the extremal multigraphs. This subsection describes them for large n. The idea is to look at the proof of Lemma A.13 in the case of equality. Equality fixes the number of strongly connected components, then the shape of the skeleton, and then the multigraph.

Lemma A.20 (What equality forces)**.** *Let $m \geq 2$ and $n \geq 8$, so that $\lfloor n^2/4 \rfloor > 2(n-1)$ and $M(n) = \lfloor n^2/4 \rfloor$, and let D be a directed multigraph on n vertices with $\lambda_D^{\max} \leq m-1$ and $|A(D)| = (m-1)\lfloor n^2/4 \rfloor$. Then*

(1) D is acyclic*: it has no directed cycle, and*

(2) D has a reachability skeleton R from which no arc can be removed without changing who reaches whom, and which, with directions ignored, is the complete bipartite graph $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$.

Proof. As recalled above, the proof of Theorem 1.18 deletes skeletons $R_1, \ldots, R_{m-1}$, built as in Lemma A.13, and then no arc is left. So $|A(D)| = \sum_i |A(R_i)|$. Each $|A(R_i)| \leq M(n)$, and the sum is $(m-1)M(n)$. So $|A(R_i)| = M(n) = \lfloor n^2/4 \rfloor$ for every i. Let $R := R_1$.

Let q be the number of strongly connected components of D. In the proof of Lemma A.13, $|A(R)| \leq f(q) = 2(n-q) + \lfloor q^2/4 \rfloor$. So $f(q) \geq \lfloor n^2/4 \rfloor = f(n)$. We show that this forces $q = n$. Since $n \geq 8$, $f(1) = 2(n-1) < \lfloor n^2/4 \rfloor = f(n)$. The differences $f(q+1) - f(q)$ never decrease as q grows. So between 1 and n, the values of f stay on or below the straight line from $f(1)$ to $f(n)$. Since $f(1) < f(n)$, this line goes up, so it is below $f(n)$ at every $q < n$. So $f(q) < f(n)$ for $q < n$, and therefore $q = n$: every strongly connected component is a single vertex. A directed cycle would put its vertices in one component, so D has no directed cycle. This is (1).

With $q = n$, each component has one vertex, so Step 1 of Lemma A.13 chooses no arcs. So R consists of the arcs chosen for Q, and Q is R itself, with $|A(Q)| = \lfloor n^2/4 \rfloor$. With directions ignored, Q has no triangle. A graph on n vertices with no triangle and with $\lfloor n^2/4 \rfloor$ edges is $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$ (the case of equality in Mantel's theorem [18]). No arc of Q can be removed without changing reachability, by the choice of Q. This is (2). □

The next lemma uses that no arc of R can be removed.

Lemma A.21 (A minimal bipartite skeleton has no long path)**.** *Let R be as in Lemma A.20, with parts A and B. Then R contains no directed path with three arcs.*

Proof. Suppose $u \to v \to w \to x$ is such a path, with four different vertices. Every arc of R joins A and B, so the path alternates between the parts. It has three arcs, so u and x lie in different parts. The underlying graph of R is complete bipartite, so R has an arc between u and x. It cannot be $x \to u$, since that would close a directed cycle, and D is acyclic. So $u \to x$ is in R. But u also reaches x along the path, so $u \to x$ could be removed without changing reachability. This is a contradiction. □

Theorem A.22 (Extremal classification on the quadratic branch)**.** *Let $m \geq 2$ and $n \geq \max(8, 2m)$, so that $M(n) = \lfloor n^2/4 \rfloor$ and n is on the quadratic branch. If D is a directed multigraph on n vertices with $\lambda_D^{\max} \leq m-1$ and $|A(D)| = L_m^{\mathrm{dir}}(n) = (m-1)\lfloor n^2/4 \rfloor$, then*

$$D \in \left\{(m-1) \cdot B_{\lceil n/2 \rceil, \lfloor n/2 \rfloor},\ (m-1) \cdot B_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}\right\},$$

where $B_{a,b}$ has all arcs from its part of size a to its part of size b, each once, and $(m-1) \cdot B_{a,b}$ takes every arc $m-1$ times as in Lemma 1.19. These two digraphs are the same up to renaming the vertices when n is even, and reverses of each other when n is odd.

Proof. Take R, A and B from Lemma A.20, and write $\alpha = |A| \geq \beta = |B| = \lfloor n/2 \rfloor$.

Step 1: R has no directed path with two arcs. Suppose R has a path $a \to b \to a'$ with $a, a' \in A$ and $b \in B$. Suppose a had an incoming arc $b_0 \to a$. Then $b_0 \neq b$, since $b \to a \to b$ would be a directed cycle, and D has none. Also $b_0 \neq a'$, since they lie in different parts. So $b_0 \to a \to b \to a'$ would be a path with three arcs, which Lemma A.21 forbids. So a has no incoming arc. With directions ignored, R is complete bipartite, so a is joined to every $b'' \in B$. All these arcs leave a, so $a \to b''$ for every $b'' \in B$. In the same way, a' has no outgoing arc, so $b'' \to a'$ for every $b'' \in B$. The β routes $a \to b'' \to a'$ use different arcs, and they lie in D, since R is part of D. So $\lambda_D(a, a') \geq \beta = \lfloor n/2 \rfloor \geq m$, since $n \geq 2m$. This breaks the condition $\lambda_D^{\max} \leq m-1$. A path $b \to a \to b'$ with $b, b' \in B$ is excluded in the same way, with the $\alpha \geq \beta \geq m$ routes through A. This is the only step that uses $n \geq 2m$.

Step 2: all arcs of R go the same way. By Step 1, no vertex of R has both an incoming and an outgoing arc. Take $a \in A$. It is joined to every vertex of B. If all arcs at a go out, then every $b \in B$ has an incoming arc, so it has no outgoing arc. So all arcs go from A to B. If all arcs at a come in, all arcs go from B to A in the same way. From now on, call the part that the arcs leave A, and the other part B.

Step 3: D has no arc outside R. By Step 1, every path in R has one arc, so u reaches x in R only when $u \to x$ is an arc of R. By Step 2 these are exactly the pairs (u, x) with $u \in A$ and $x \in B$. If D has an arc $u \to x$, then u reaches x in D, so also in R. So $u \to x$ is an arc of R.

Step 4: every multiplicity is $m-1$. Parallel arcs are disjoint routes, so every multiplicity is at most $m-1$. By Step 3, all arcs of D go from A to B. There are $|A|\,|B| \leq \lfloor n^2/4 \rfloor$ such pairs, so $|A(D)| \leq (m-1)|A|\,|B| \leq (m-1)\lfloor n^2/4 \rfloor$. Equality holds, so every pair in $A \times B$ has multiplicity $m-1$, and $|A|\,|B| = \lfloor n^2/4 \rfloor$. The product $|A|\,|B|$ with $|A| + |B| = n$ equals $\lfloor n^2/4 \rfloor$ only when the parts have sizes $\lceil n/2 \rceil$ and $\lfloor n/2 \rfloor$. □

The hypothesis $n \geq 2m$ is used only in Step 1. Every vertex of an extremal multigraph has degree at least $(m-1)\lfloor n/2 \rfloor$. Indeed, deleting the vertex leaves a feasible multigraph on $n-1 \geq 7$ vertices, with at most $(m-1)M(n-1) = (m-1)\lfloor (n-1)^2/4 \rfloor$ arcs. So the vertex has at least $(m-1)\big(\lfloor n^2/4 \rfloor - \lfloor (n-1)^2/4 \rfloor\big) = (m-1)\lfloor n/2 \rfloor$ arcs. For even n, every vertex of $(m-1) \cdot B_{n/2,\,n/2}$ has degree exactly $(m-1)n/2$.

Part III. Hypergraphs

Menger for hypergraphs, simple hypergraphs that attain the edge bound, the vertex problem at $m = 3$, multigraphs with vertex separation, and directed hypergraphs.

A.6 The hypergraph bounds

Hypergraphs have several notions of connectivity. This thesis uses Berge paths throughout. We first need Menger's theorem for Berge paths. It also shows that the gate network of Chapter 2 computes the right value.

Theorem A.23 (Menger for multihypergraphs)**.** *Let u, v be distinct vertices of a multihypergraph $\mathcal{H}$, and count parallel copies of a hyperedge separately. The largest number of u–v Berge paths that share no hyperedge copy equals the smallest number of hyperedge copies whose removal separates u from v.*

Proof. We turn $\mathcal{H}$ into a directed multigraph N, the *gate network*, and use Menger's theorem for directed multigraphs (Theorem 1.1). Let c be the total number of hyperedge copies. Every hyperedge e with $\mu(e)$ copies gets two new nodes e^- and e^+ and $\mu(e)$ parallel arcs $e^- \to e^+$, its *gate*. Every vertex $x \in e$ gets $c+1$ parallel arcs $x \to e^-$ and $c+1$ parallel arcs $e^+ \to x$.

Paths. A Berge path $u, e_1, v_1, \dots, e_k, v$ gives the directed walk $u \to e_1^- \to e_1^+ \to v_1 \to \cdots \to v$ in N. If it uses different copies of the same hyperedge, this walk revisits the corresponding gate nodes. Conversely, a directed u–v path in N alternates between original vertices and gates. Since it repeats no gate node, choosing the hyperedge copy represented by each gate arc gives a Berge path.

Take a family of Berge paths that share no hyperedge copy. Each hyperedge e has $\mu(e)$ copies and $\mu(e)$ gate arcs, so distinct copies of e get distinct gate arcs. Each path uses at least one hyperedge copy, so the family has at most c paths. Every other arc has $c+1$ copies, so distinct paths can use distinct copies of every other arc. This gives an equally large family of arc-disjoint directed walks in N. Deleting closed portions from each walk yields directed paths with the same endpoints. This only removes arcs, so arc-disjointness is preserved. Conversely, arc-disjoint directed paths in N use distinct gate arcs and therefore distinct hyperedge copies, so the Berge paths they give share no hyperedge copy. So the largest families have the same size in N and in $\mathcal{H}$.

Cuts. Every u–v path in N passes a gate. So removing all c gate arcs separates u from v, and a smallest separating set of arcs in N has at most c arcs. So it never contains all $c+1$ copies of an arc that is not a gate arc. Removing some but not all copies of an arc stops no path, so we can put these copies back, and the set still separates u from v. So there is a smallest separating set that consists of gate arcs only. The gate arcs of e stand for the copies of e. So removing a set of gate arcs separates u from v in N exactly when removing the same hyperedge copies

separates u from v in $\mathcal{H}$. So the smallest separating sets have the same size in N and in $\mathcal{H}$.

Menger's theorem in N says that the two sizes are equal in N. So they are equal in $\mathcal{H}$. □

The same proof works for directed hyperedges and directed Berge paths, with arcs $x \to e^-$ only for the tails x of e, and arcs $e^+ \to x$ only for its heads.

Theorem 1.13 says that an r-uniform multihypergraph on n vertices with $\lambda^{\max} \le m-1$ has at most $\lfloor (m-1)(n-1)/(r-1) \rfloor$ hyperedge copies. The next theorem shows that for $r \ge 3$ and $m-1 \le \binom{n-2}{r-2}$, simple hypergraphs attain this bound. For graphs ($r=2$) this is different: there, simplicity lowers the bound from $(m-1)(n-1)$ to $\lfloor m(n-1)/2 \rfloor$.

Theorem A.24 (Simplicity is free for $r \ge 3$)**.** *Let $r \ge 3$, $n \ge r$, and $2 \le m$ with $m-1 \le \binom{n-2}{r-2}$. Then the maximum number of hyperedges in a* simple *r-uniform hypergraph on n vertices with $\lambda^{\max} \le m-1$ is exactly $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$.*

Proof. The upper bound is Theorem 1.13. For the lower bound, take a hub h and let $V' = V \setminus \{h\}$. Apply Lemma A.1 with uniformity $r-1$, with the $n-1$ vertices of V', and with $d = m-1$. The lemma needs $n-1 \ge r-1$ and $m-1 \le \binom{n-2}{r-2}$, and both hold. It gives a simple $(r-1)$-uniform hypergraph $\mathcal{G}^*$ on V' with maximum degree at most $m-1$ and $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$ hyperedges. Add h to each of its hyperedges. Different $(r-1)$-sets give different r-sets, so the result $\mathcal{H}$ is simple. Every vertex $w \ne h$ lies in at most $m-1$ hyperedges, and every route from or to w uses one of them. So removing these hyperedges separates w from every other vertex. Every pair contains a vertex other than h, so $\lambda_{\mathcal{H}}^{\max} \le m-1$. □

Remark A.25 (The condition is needed)*.* Without the condition $m-1 \le \binom{n-2}{r-2}$, simple hypergraphs may not reach the bound. The smallest example is $n = r = m = 3$. There is only one 3-set, so a simple hypergraph has at most one hyperedge, while the bound is $\lfloor 2 \cdot 2/2 \rfloor = 2$. Two copies of the one hyperedge do reach the bound. Theorem 1.16 needs the same condition with $m-1 = 2$, which is the range $2 < r < n$.

Construction A.26 (Twelve hyperedge copies on six vertices)**.** Take a hub h and five other vertices a, b, c, d, e. Take all ten triples that consist of h and two of the other vertices. Add a second copy of $\{h, d, e\}$ and one copy of $\{a, b, c\}$.

Proposition A.27 (The edge value at $(n, m, r) = (6, 6, 3)$)**.** *The multihypergraph of Construction A.26 has 12 hyperedge copies and $\lambda^{\max} \le 5$. So the largest number of hyperedge copies of a 3-uniform multihypergraph on 6 vertices with $\lambda^{\max} \le 5$ is 12, although $r-1 = 2$ does not divide $n-1 = 5$.*

Proof. Each non-hub vertex lies in four of the ten triples, one for each other non-hub vertex. The second copy of $\{h, d, e\}$ adds one to the degrees of d and e. The triple $\{a, b, c\}$ adds one to the degrees of a, b and c. So every non-hub vertex lies in exactly five hyperedge copies. Every pair of vertices contains a non-hub vertex w. Every route from or to w uses one of the five copies at w, so removing them separates the pair. So $\lambda \le 5$ for every pair. There are $10 + 1 + 1 = 12$ copies. Theorem 1.13 gives at most $\lfloor 5 \cdot 5/2 \rfloor = 12$, so 12 is the largest value. □

Since $\kappa \leq \lambda$, the same multihypergraph also gives 12 for the vertex problem. We do not know whether 12 is optimal there.

A.7 The hypergraph vertex problem

The vertex problem for hypergraphs becomes a problem about graphs (Figure A.4). Recall the *incidence graph* $I(\mathcal{H})$ of Section 1.3.1: one node per vertex, one node per hyperedge copy, and an edge when the vertex lies in the hyperedge. A Berge path is the same as a path in $I(\mathcal{H})$ between two vertex nodes. Its inner nodes are the hyperedges it uses and the vertices where it changes hyperedge. So two Berge routes share no hyperedge and no inner vertex exactly when the two paths in $I(\mathcal{H})$ are internally disjoint. So

$$\kappa_{\mathcal{H}}(u, v) \;=\; \kappa_{I(\mathcal{H})}(u, v).$$

The vertex problem asks for the largest number of hyperedge nodes, each of degree r, such that no two of the n vertex nodes are joined by m internally disjoint paths.

Proposition A.28 (General lower bound)**.** *For $r \geq 3$, $m \geq 2$ and $m - 1 \leq \binom{n-2}{r-2}$,*

$$k_m^{(r)}(n) \;\geq\; \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor,$$

attained by a simple hypergraph.

Proof. The simple hypergraph of Theorem A.24 has $\lambda^{\max} \leq m - 1$. Since $\kappa \leq \lambda$, it is also feasible for the vertex problem. □

That construction needs different $(r-1)$-sets, which explains the condition $m - 1 \leq \binom{n-2}{r-2}$. A multihypergraph can repeat a hyperedge instead. This covers other cases.

Proposition A.29 (The multihypergraph lower bound)**.** *For $r \geq 2$, $m \geq 2$ and $(r-1) \mid (n-1)$,*

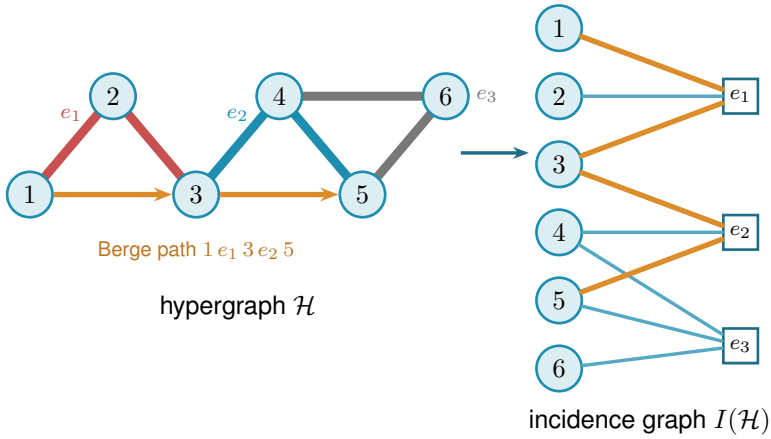
$$K_m^{(r)}(n) \;\geq\; \frac{(m-1)(n-1)}{r-1},$$

attained by the star hypertree of Figure A.4b with every hyperedge taken at multiplicity $m - 1$. At $m = 3$ this meets the upper bound of Theorem 1.16, so

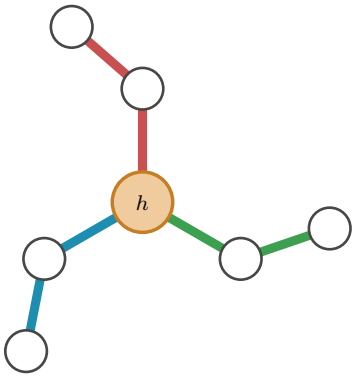
$$K_3^{(r)}(n) \;=\; \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor \qquad \textit{whenever } (r-1) \mid (n-1).$$

Proof. Take the hub h and split the other $n - 1$ vertices into groups $G_1, \ldots, G_q$ of size $r - 1$, where $q = (n-1)/(r-1)$. Give each hyperedge $\{h\} \cup G_i$ multiplicity $m-1$. This gives $(m-1)q$ hyperedges.

Now take two vertices $u \neq w$. Suppose first that both lie in one hyperedge $\{h\} \cup G_i$. At most one of them is h, so one of them lies in G_i. Every route from or to this vertex uses one of the $m-1$



(a) A 3-uniform hypergraph $\mathcal{H}$ (left) and its bipartite incidence graph $I(\mathcal{H})$ (right): circles for vertices, hollow squares for hyperedges, an edge when the vertex lies in the hyperedge. The Berge path $1\, e_1\, 3\, e_2\, 5$ is traced in orange in both pictures. In $\mathcal{H}$ it rides the red line e_1 and the blue line e_2, changing at the shared vertex 3, while the grey line e_3 lies off it. In $I(\mathcal{H})$ it is the ordinary path $1-e_1-3-e_2-5$ alternating sides.



(b) A star hypertree: every hyperedge $\{h\} \cup G_i$ shares only the hub h (orange), and the other $r-1=2$ vertices of each block sit in a single hyperedge. Its incidence graph is a tree, so $\kappa^{\max} \leq 1$.

Figure A.4. The translation the vertex problem rests on. Internally vertex-disjoint Berge paths in $\mathcal{H}$ correspond exactly to internally disjoint paths in $I(\mathcal{H})$, so $\kappa_{\mathcal{H}}(u,v) = \kappa_{I(\mathcal{H})}(u,v)$. **(top)** A hypergraph and its incidence graph with one Berge path drawn in both. **(bottom)** The star hypertree of Theorem 1.15, whose incidence graph is a tree.

copies of $\{h\} \cup G_i$, so removing these copies separates u from w. So $\kappa(u, w) \leq \lambda(u, w) \leq m-1$. Otherwise $u \in G_i$ and $w \in G_j$ with $i \neq j$. The hyperedges at u and the hyperedges at w meet only in h, so every route from u to w has h as an inner vertex. So $\kappa(u, w) \leq 1$. So the hypergraph is feasible.

At $m = 3$ there are $2(n-1)/(r-1)$ hyperedges, which equals the upper bound of Theorem 1.16. □

The upper bound at $m = 3$ follows from a bound on the cycles of a graph, proved next. For a connected multigraph G, the *cycle rank* is $\operatorname{rank}(G) = |E(G)| - |V(G)| + 1$. A tree has cycle rank 0, and each extra edge raises it by one. For the incidence graph of a hypergraph, $|E|$ counts the pairs of a vertex and a hyperedge copy that contains it. So a bound on the cycle rank is a bound on the number of hyperedges.

We use the words cut vertex and k-connected from Section A.1.1. A *separating pair* is a pair of vertices whose removal disconnects the graph. By Menger's theorem [4, Ch. 3], a graph is 3-connected exactly when every two vertices are joined by 3 internally disjoint paths.

Lemma A.30 (Incidence rank lemma)**.** *Let G be a connected loopless multigraph with a partition $V(G) = X \dot\cup Z$ such that (i) no two vertices of Z are adjacent, (ii) no edge incident to Z is parallel to another edge, (iii) every $z \in Z$ has degree at least 2, and (iv) $\kappa_G(x, x') \leq 2$ for all distinct $x, x' \in X$. Then $\operatorname{rank}(G) \leq |X| - 1$.*

Read X as the vertices of a hypergraph and Z as its hyperedges, so that G is the incidence graph. Then (i) says that two hyperedge nodes are never adjacent, (ii) that no hyperedge repeats a vertex, (iii) that every hyperedge has at least two vertices, and (iv) that the hypergraph is feasible at $m = 3$. We allow edges inside X, because Step 2 of the proof creates them.

Proof. We use induction on $|V| + |E|$. Step 1 handles cut vertices. Steps 2 and 3 remove vertices of degree two. Step 4 handles what is left: a 2-connected graph in which every degree is at least three. In each step we build one or two smaller graphs, check that they satisfy (i) to (iv) and are connected, and apply the induction hypothesis to them.

Base: $|V| \leq 2$. A vertex in Z has degree at least 2 by (iii), and by (ii) its edges go to different vertices. So it has at least two neighbours. With at most two vertices this is impossible, so Z is empty and all vertices lie in X. With one vertex, there are no edges (the graph is loopless), so $\operatorname{rank} = 0 = |X| - 1$. With two vertices, the parallel edges between them are internally disjoint routes. By (iv) there are at most two, so $\operatorname{rank} \leq 2 - 2 + 1 = 1 = |X| - 1$.

Step 1: a cut vertex. Suppose G has a cut vertex c. Split the components of $G - c$ into two non-empty groups. Let G_1 consist of c, the first group, and all edges among these vertices. Define G_2 in the same way with the second group. Every edge of G lies in exactly one of G_1, G_2, and the two share only the vertex c. So

$$\operatorname{rank}(G_1) + \operatorname{rank}(G_2) \;=\; |E(G)| - \big(|V(G)| + 1\big) + 2 \;=\; \operatorname{rank}(G).$$

Each G_i is connected and smaller than G. Conditions (i) and (ii) still hold in G_i, since it is part of G. Condition (iv) still holds, since a part of G has no more disjoint paths than G. Condition (iii) can only fail at c, when $c \in Z$ has a single neighbour in G_i. In that case delete c from G_i. This removes one vertex and one edge, so G_i stays connected, and its rank and its X-vertices do not change. By induction, $\mathrm{rank}(G_i) \leq |X \cap V(G_i)| - 1$ for $i = 1, 2$. If $c \in X$, the vertex c is counted in both G_1 and G_2, so the two X-sets together have $|X| + 1$ vertices. If $c \in Z$, they have $|X|$ vertices. Either way,

$$\mathrm{rank}(G) \;\leq\; |X \cap V(G_1)| + |X \cap V(G_2)| - 2 \;\leq\; (|X| + 1) - 2 \;=\; |X| - 1.$$

Step 2: a Z-vertex of degree two. From now on G has no cut vertex and at least three vertices, so it is 2-connected. Suppose some $z_0 \in Z$ has degree 2. By (i) its two neighbours x, x' lie in X, and by (ii) they are different. Delete z_0 and its two edges, and add a new edge xx'. Then $|E|$ and $|V|$ both drop by one, so the rank does not change, and the graph stays connected. The new edge joins two vertices of X, so $|X|$ does not change, and (i), (ii), (iii) still hold. A path through z_0 uses x, z_0, x' in a row. Replacing this by the new edge, and back, turns internally disjoint paths of one graph into internally disjoint paths of the other. So every $\kappa(x_1, x_2)$ with $x_1, x_2 \in X$ stays the same, and (iv) holds. The graph is smaller, so the induction hypothesis gives the bound.

Step 3: an X-vertex of degree two. Now Step 2 does not apply, so every Z-vertex has degree at least 3. Suppose some $x_0 \in X$ has degree 2, and delete it with its two edges. Since G is 2-connected, the result is connected. We remove two edges and one vertex, so the rank drops by one, and $|X|$ drops by one. By (ii), a neighbour of x_0 in Z loses one edge, so it keeps degree at least 2, and (iii) holds. Conditions (i) and (ii) still hold, and (iv) holds, since deleting a vertex cannot add disjoint paths. The result still has a vertex in X. Otherwise $X = \{x_0\}$, and by (i) and (ii) every z would have degree at most 1, against (iii). By induction, $\mathrm{rank}(G) - 1 \leq (|X| - 1) - 1$, that is, $\mathrm{rank}(G) \leq |X| - 1$.

Step 4: the rest. Now G is 2-connected, has at least three vertices, and every degree is at least 3.

First, G has no parallel edges. By (ii), parallel edges can only join two vertices x, x' of X. Suppose G has two parallel edges between x and x'. Since G has a third vertex and x' is not a cut vertex, x has a neighbour $w \neq x'$. Since x is not a cut vertex, $G - x$ has a path from w to x'. With x in front, this is an x–x' path with an inner vertex. With the two parallel edges this gives three internally disjoint x–x' paths, against (iv). So G is simple. A simple graph in which every degree is at least three has at least four vertices.

Suppose G is 3-connected. By Menger's theorem every two vertices are then joined by three internally disjoint paths. By (iv), X then has at most one vertex. By (i) and (ii), every $z \in Z$ then has degree at most 1, against (iii). And Z is not empty, since G has at least four vertices. So G is not 3-connected.

So G has a separating pair $\{a, b\}$. Remove it, and divide the components of $G - \{a, b\}$ into two non-empty groups, with vertex sets U and W. Each component has a neighbour of a and

a neighbour of b. Otherwise the other one of a, b would be a cut vertex. So $U \cup \{a, b\}$ contains an a–b path with all inner vertices in U, and the same holds for W. Let $s = |X \cap \{a, b\}|$, the number of a, b that lie in X. We build two smaller graphs G_U and G_W, the *pieces*. G_U consists of $U \cup \{a, b\}$ and all edges of G among these vertices, except the edge ab, together with one extra a–b connection. G_W is built in the same way. The extra connection is:

- If $ab \in E(G)$: the edge ab itself, in both pieces.
- If $ab \notin E(G)$ and $s \geq 1$: one new edge ab in each piece.
- If $s = 0$: in each piece, one new X-vertex joined to a and to b.

These cases cover everything. If $ab \in E(G)$, then by (i) at least one of a, b lies in X, so $s \geq 1$. In that case $s = 1$. Indeed, if $a, b \in X$, then the edge ab, the path through U and the path through W would give $\kappa(a, b) \geq 3$, against (iv).

Each piece is connected and satisfies (i) to (iv). We check this for G_U. It is connected, since every component in U is joined to a and b, and a and b are joined by the extra connection. (i): every added edge has an end in X. In the second case this is a or b, since $s \geq 1$. In the third case it is the new vertex. (ii): an added edge joins two vertices that had no edge in the piece, so it is not parallel to another edge. (iii): a Z-vertex in U keeps all its neighbours. If a or b lies in Z, it keeps a neighbour in U and gains the extra connection, so it has degree at least 2. (iv): take two X-vertices x, x' of G_U. If one of them is the new vertex of the third case, it has degree two, so $\kappa(x, x') \leq 2$. Otherwise both are vertices of G. Take internally disjoint x–x' paths in G_U. At most one of them uses the extra connection. Indeed, two of them cannot pass the same new vertex, since it would be a shared inner vertex. And two of them cannot both use the edge ab: both would contain a and b, so a and b would have to be the ends x and x', and then both paths would be the edge ab itself. If a path uses the extra connection, replace that part by the a–b path through W. The other paths lie in G_U, so they do not meet W. This gives the same number of internally disjoint x–x' paths in G. By (iv) for G there are at most two, so (iv) holds in G_U.

Each piece is smaller. W is not a single vertex: in a simple graph that vertex would have only a and b as neighbours, but every degree is at least three. So G_U loses the at least two vertices of W. Every vertex of W has degree at least three, and an edge inside W is counted at two vertices of W, so at least three edges of G touch W, and G_U loses them. G_U gains at most one vertex and two edges. So $|V| + |E|$ drops by at least $5 - 3 = 2$. The same holds for G_W.

Counting. Let r_U, r_W be the ranks of the pieces and X_U, X_W their X-sets. Together the pieces contain every vertex of G, with a and b counted twice, and every edge of G other than ab once.

- First case. The edge ab is counted twice, so the pieces have $|V| + 2$ vertices and $|E| + 1$ edges. So $r_U + r_W = (|E| + 1) - (|V| + 2) + 2 = \mathrm{rank}(G)$. Exactly one of a, b lies in X, so $|X_U| + |X_W| = |X| + 1$.
- Second case. Two new edges, so the pieces have $|V| + 2$ vertices and $|E| + 2$ edges, and $r_U + r_W = \mathrm{rank}(G) + 1$. The vertices of $X \cap \{a, b\}$ are counted twice, so $|X_U| + |X_W| = |X| + s$.

★ Third case. Two new vertices and four new edges, so the pieces have $|V|+4$ vertices and $|E|+4$ edges, and $r_U + r_W = \operatorname{rank}(G)+1$. The two new vertices lie in X and a, b do not, so $|X_U| + |X_W| = |X| + 2$.

By induction $r_U \le |X_U| - 1$ and $r_W \le |X_W| - 1$, so $r_U + r_W \le |X_U| + |X_W| - 2$. In the first case this gives $\operatorname{rank}(G) \le |X| - 1$. In the third case it gives $\operatorname{rank}(G) + 1 \le |X|$. In the second case it gives $\operatorname{rank}(G) + 1 \le |X| + s - 2$, so $\operatorname{rank}(G) \le |X| + s - 3 \le |X| - 1$, since $s \le 2$. □

We now prove Theorem 1.16.

Theorem 1.16 (restated). *For $r \ge 2$ and $n \ge 1$, every r-uniform multihypergraph on n vertices with $\kappa^{\max} \le 2$ has at most $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ hyperedges. For $2 < r < n$ the bound is attained by a simple hypergraph, so*

$$k_3^{(r)}(n) \;=\; \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor \;=\; \ell_3^{(r)}(n).$$

For multihypergraphs the bound is attained for every $n \ge r$, so $K_3^{(r)}(n) = \lfloor 2(n-1)/(r-1) \rfloor$ for all $n \ge r \ge 2$.

Proof of Theorem 1.16. Upper bound. Let $\mathcal{H}$ have q hyperedge copies. Apply Lemma A.30 to each component of the incidence graph $I(\mathcal{H})$, with X the vertex nodes and Z the hyperedge nodes. The conditions hold: (i) no two hyperedge nodes are adjacent; (ii) a hyperedge contains each vertex once, so no edge at a hyperedge node is repeated; (iii) hyperedge nodes have degree $r \ge 2$; (iv) $\kappa_I(u,v) = \kappa_{\mathcal{H}}(u,v) \le 2$ for vertices u, v. Every component contains a vertex node, since every hyperedge node has neighbours. Let C be the number of components. Each hyperedge copy contains r vertices, so $I(\mathcal{H})$ has rq edges. It has $n+q$ nodes. Adding $\operatorname{rank} = |E| - |V| + 1$ over the components, the ranks add up to $rq - (n+q) + C$. By the lemma, the rank of a component is at most its number of vertex nodes minus one. These bounds add up to $n - C$. So $rq - (n+q) + C \le n - C$, that is, $q(r-1) \le 2(n-1) - 2(C-1) \le 2(n-1)$. Since q is an integer, $q \le \lfloor 2(n-1)/(r-1) \rfloor$.

Lower bound, simple. For $2 < r < n$, this is Proposition A.28 at $m = 3$. It needs $2 \le \binom{n-2}{r-2}$, which holds exactly when $2 < r < n$.

Lower bound, multihypergraphs. Every simple hypergraph is also a multihypergraph, so $K_3^{(r)}(n) \ge k_3^{(r)}(n)$. For $2 < r < n$ this gives the lower bound. In the remaining cases, $r = 2$ or $r = n$, the number $r - 1$ divides $n - 1$, and Proposition A.29 gives it. □

Remark A.31 (The case $r = 2$, and where the method stops). At $r = 2$ the theorem is about multigraphs. It gives $K_3(n) = 2(n-1)$ (Corollary A.35). The proof of Lemma A.30 uses the condition $\kappa \le 2$ in several places. For example, two parallel edges and one more route already break it, and so does a 3-connected graph. The lemma is false for $\kappa \le 3$: two X-vertices joined by three parallel edges have rank two, while $|X| - 1 = 1$. For $m = 4$ a different bound would be needed, probably one that splits the graph along sets of three vertices. We do not have one.

A.8 Multigraphs with vertex separation

At $r = 2$ the hypergraph vertex problems are about multigraphs. They are not the same as the simple problems. Recall that q parallel copies of uv count as q separated routes. Write $K_m(n)$ for the largest number of edges, counted with multiplicity, of a multigraph on n vertices with $\kappa^{\max} \leq m - 1$.

The parallel edges and the longer routes do not interfere, so we can count them separately.

Lemma A.32 (Splitting the vertex measure)**.** *Let G be a multigraph with multiplicity function μ and underlying simple graph G_0. For any two vertices $u \neq v$, write $\pi(u, v)$ for the largest number of pairwise internally disjoint u–v paths of length at least two in G_0. Then*

$$\kappa_G(u, v) \;=\; \mu(u, v) \;+\; \pi(u, v).$$

Proof. A route of length one uses one copy of the edge uv and has no inner vertex. So two such routes are separated when they use different copies, and such a route is separated from every longer route.

Lower bound. Take all $\mu(u, v)$ copies of uv, and $\pi(u, v)$ internally disjoint u–v paths of length at least two in G_0. Use one copy of each edge on these paths. The inner vertices do not change, so all these routes are separated.

Upper bound. Take a family of separated u–v routes in G. At most $\mu(u, v)$ of them have length one, since they use different copies of uv. Map each longer route to its sequence of vertices, a path in G_0. Two longer routes have disjoint, non-empty sets of inner vertices. So their images are different and internally disjoint, and there are at most $\pi(u, v)$ of them. □

So a multigraph is feasible exactly when $\mu(u, v) \leq m - 1 - \pi(u, v)$ on every edge and $\pi(u, v) \leq m - 1$ on every pair.

Theorem A.33 (The multigraph vertex value at $m = 2$)**.** $K_2(n) = n - 1$*, attained exactly by the spanning trees.*

Proof. By Lemma A.32, $\mu(u, v) + \pi(u, v) \leq 1$ on every edge. If G_0 had a cycle, take an edge uv on it. Then $\mu(u, v) \geq 1$, and the rest of the cycle is a u–v path of length at least two, so $\pi(u, v) \geq 1$. So G_0 is a forest and every multiplicity is one. A forest has at most $n - 1$ edges, and a spanning tree has exactly $n - 1$. □

Remark A.34. At $m = 2$ a second copy of an edge is already a second route, so every feasible multigraph is simple. So $K_2(n) = k_2(n) = n - 1$. This is also Theorem 1.15 at $r = 2$.

The value at $m = 3$ follows from the hypergraph theorem.

Corollary A.35 (The multigraph vertex value at $m = 3$)**.** $K_3(n) = 2(n - 1)$ *for every* $n \geq 2$.

Proof. Apply Theorem 1.16 and Proposition A.29 at $r = 2$. The divisibility condition is automatic, and the doubled star attains the upper bound $2(n - 1)$. □

So at $m = 3$ the vertex value equals the edge value $L_3(n)$ of Theorem 1.8. At $m = 5$ they differ: the thickened theta below gives $K_5(4) \geq 14$, while $L_5(4) = 12$. For $m = 4$ this is open. For $m \geq 3$ there are three constructions.

- *The thickened tree.* Take a spanning tree and give every edge multiplicity $m - 1$. Tree edges have $\pi = 0$, and other pairs have $\pi = 1$, so it is feasible. It has $(m - 1)(n - 1)$ edges, the edge value of Theorem 1.8.
- *The thickened complete graph.* Give every edge of K_n multiplicity q. Here $\pi(u, v) = n - 2$: each of the other $n-2$ vertices gives a route of length two, and together they separate u from v once the edge uv is removed. So $\kappa(u, v) = q+n-2$, and it is feasible when $q \leq m+1-n$. It has $\binom{n}{2}(m + 1 - n)$ edges, which is positive only when $n \leq m$.
- *The thickened theta.* Choose two *poles*. Join each of the other $n - 2$ vertices to both poles with multiplicity $m - 2$, and join the poles with multiplicity $m + 1 - n$ if this is positive. A non-pole is joined only to the two poles, so every route from it leaves through a pole. So a pole and a non-pole have $\pi = 1$, and $\kappa = (m - 2) + 1 = m - 1$. Two non-poles are not adjacent and have $\kappa = \pi = 2$. The two poles have $\pi = n - 2$, one route through each non-pole, so $\kappa = \max(0,\, m + 1 - n) + n - 2$, which is $m - 1$ when $n \leq m + 1$. So it is feasible when $n \leq m + 1$, and it has $2(n - 2)(m - 2) + \max(0,\, m + 1 - n)$ edges.

For $n \leq m + 1$, the theta has as many edges as the thickened tree at $m = 4$, and more for $m \geq 5$ (when $n \geq 3$). For $n \geq m + 2$ the other two constructions are not feasible, so among these three the tree is best.

For $m \geq 5$, one triangle added to the thickened tree already beats it, also when $n \geq m + 2$. The next result makes this precise.

Fix a connected simple graph G_0 with $\kappa^{\max}_{G_0} \leq m-1$. By Lemma A.32, the best multigraph on G_0 takes $\mu(u, v) = m-1-\pi(u, v)$ on every edge. Its size is $(m-1)\,|E(G_0)|-\sum_{uv\in E(G_0)} \pi(u, v)$. Since $|E(G_0)| = n - 1 + \operatorname{rank}(G_0)$, it beats the thickened tree exactly when

$$\sum_{uv\in E(G_0)} \pi(u, v) \;<\; (m - 1)\operatorname{rank}(G_0).$$

Theorem A.36 (Thickened complete graphs sharing a vertex beat the thickened tree)**.** *Let $3 \leq s \leq m$ and $q = m + 1 - s \geq 1$. Let $1 \leq k \leq \lfloor (n - 1)/(s - 1) \rfloor$. Take k copies of K_s that share one vertex h and are otherwise disjoint, and give every edge multiplicity q. The remaining $n - 1 - k(s - 1)$ vertices form a path that starts at h, with every edge of multiplicity $m - 1$. This multigraph has $\kappa^{\max} \leq m - 1$, and its number of edges is*

$$(m - 1)(n - 1) \;+\; k \cdot \operatorname{gain}(s, m), \qquad \operatorname{gain}(s, m) := \frac{(s - 1)(s - 2)(m - 1 - s)}{2}.$$

So it beats the thickened tree exactly when $s \leq m - 2$. Such an s exists when $m \geq 5$.

Proof. Write G_0 for the underlying simple graph.

Feasibility. The blocks of G_0 are the copies of K_s and the path edges. As in the proof of Theorem A.6, a path between two vertices of the same block cannot leave the block, since it would pass a cut vertex twice. Take an edge uv in a copy of K_s. The other $s-2$ vertices of the copy give $s-2$ internally disjoint routes of length two. Every longer route stays in the copy, so it passes one of these $s-2$ vertices. So $\pi(u,v) = s-2$. By Lemma A.32, $\kappa_G(u,v) = q + (s-2) = m-1$. Two vertices in different blocks are separated by removing a cut vertex, so $\kappa_G(u,v) \le 1$. For an edge uv on the path, $\pi(u,v) = 0$, so $\kappa_G(u,v) = m-1$. Inside a block every two vertices are adjacent, so no other pairs need checking.

Count. Since $q \ge 1$, the underlying simple graph G_0 is connected, so $|E(G_0)| = n - 1 + \operatorname{rank}(G_0)$. Every edge uv has multiplicity $m - 1 - \pi(u,v)$, so the number of edges is $(m-1)|E(G_0)| - \sum_{uv} \pi(u,v)$. Only the k copies of K_s contain cycles or have $\pi > 0$. Each copy has rank $\binom{s}{2} - s + 1 = (s-1)(s-2)/2$, and its $\binom{s}{2}$ edges each have $\pi = s-2$. So the number of edges is

$$(m-1)\Big[n - 1 + k\tfrac{(s-1)(s-2)}{2}\Big] - k\tfrac{s(s-1)(s-2)}{2} \;=\; (m-1)(n-1) + k\,\mathrm{gain}(s,m).$$

□

At $s = 3$, $\mathrm{gain}(3,m) = m-4$. This is zero at $m = 4$ and positive for $m \ge 5$. So for every $m \ge 5$ the thickened tree is not optimal, also when $n \ge m+2$. Here is the smallest example. Take $m = 5$ and $n = 7$. Put a triangle on $\{u_1, u_2, u_3\}$ with multiplicity 3, and attach a path of four more vertices to u_1 with multiplicity 4. Two triangle vertices have three parallel edges and one route through the third vertex, so $\kappa = 4$. Every other pair is separated by a cut vertex or joined only by its parallel edges. So $\kappa^{\max} = 4 = m-1$. There are $3 \cdot 3 + 4 \cdot 4 = 25$ edges, one more than the thickened tree's 24. With k triangles the gain is k, so the gap grows linearly in n.

Sharing a vertex lets more copies fit, since each new copy uses only $s-1$ new vertices. For example, at $m = 10$, $s = 6$ and $n = 11$, two copies sharing a vertex give 150 edges. Disjoint copies would allow only one copy, with 75 edges, and 120 edges after attaching the other five vertices.

The best s is about $(m+1)/2$, where the gain per added vertex, $\mathrm{gain}(s,m)/(s-1)$, is of order m^2. So for $n/m \to \infty$ the construction gives $K_m(n) \ge \Theta(m^2)\, n$, while the thickened tree gives only $(m-1)(n-1)$.

This construction need not be optimal. The next result gives an upper bound of the same order.

Proposition A.37 (An upper bound of the same order)**.** *For all $m \ge 2$ and $n \ge 2$, the first inequality below holds, and the second holds for $n \ge m-1$:*

$$K_m(n) \;\le\; (m-1)\,k_m(n) \;\le\; (m-1)\Big[(m-1)n - \binom{m}{2}\Big].$$

Proof. Let G be feasible, with underlying simple graph G_0.

Every multiplicity is at most $m-1$. By Lemma A.32, $\kappa_G(u,v) = \mu(u,v) + \pi(u,v) \geq \mu(u,v)$, so feasibility gives $\mu(u,v) \leq m-1$ on every pair and the total is at most $(m-1)|E(G_0)|$.

G_0 *is feasible for the simple vertex problem.* Take $u \neq v$. If they are adjacent, then $\mu(u,v) \geq 1$, and the routes of G_0 between them are the edge uv and the longer routes, so $\kappa_{G_0}(u,v) = 1 + \pi(u,v) \leq \mu(u,v) + \pi(u,v) = \kappa_G(u,v) \leq m-1$. If they are not adjacent, then $\kappa_{G_0}(u,v) = \pi(u,v) = \kappa_G(u,v) \leq m-1$. So $|E(G_0)| \leq k_m(n)$, which gives the first inequality.

The second inequality is Remark A.7. □

So for $n/m \to \infty$ and then $m \to \infty$,

$$\left(\tfrac{1}{8} + o_m(1)\right) m^2\, n \;\leq\; K_m(n) \;\leq\; m^2 n.$$

This needs n large: for example $K_m(2) = m-1$. Theorem A.40 below gives a better lower constant. The first inequality, $K_m(n) \leq (m-1)k_m(n)$, says that $K_m(n)$ is at most $m-1$ times the simple vertex value. It is tight at $m = 2$ (Remark A.34).

A.8.1 The value is a block problem

The next two results reduce $K_m(n)$ to 2-connected blocks, as in Section A.1.1.

Lemma A.38 (The objective in closed form)**.** *For a simple graph* G_0 *on* n *vertices with* $\kappa_{G_0}^{\max} \leq m-1$*, the largest total multiplicity of a feasible multigraph with underlying simple graph exactly* G_0 *is*

$$W_m(G_0) \;:=\; \sum_{uv \in E(G_0)} \big(m - \kappa_{G_0}(u,v)\big),$$

and every simple graph arising as the underlying graph of a feasible multigraph satisfies $\kappa_{G_0}^{\max} \leq m-1$*. Hence*

$$K_m(n) \;=\; \max\big\{\, W_m(G_0) \;:\; G_0 \text{ simple on } n \text{ vertices},\; \kappa_{G_0}^{\max} \leq m-1 \,\big\}.$$

Proof. By Lemma A.32, a multigraph on G_0 is feasible exactly when $\mu(u,v) \leq m-1-\pi(u,v)$ on every edge, so the best choice is $\mu(u,v) = m-1-\pi(u,v)$. On an edge, $\kappa_{G_0}(u,v) = 1 + \pi(u,v)$, so this is $m - \kappa_{G_0}(u,v)$, and the total is $W_m(G_0)$. It is at least one on every edge, since $\kappa_{G_0}(u,v) \leq m-1$, so the underlying graph is really G_0. The proof of Proposition A.37 shows that the underlying graph of a feasible multigraph is feasible. So maximising W_m over feasible G_0 gives $K_m(n)$. □

The objective is a sum over edges of a quantity that depends only on the block of the edge. This gives the next result.

Theorem A.39 (The multigraph vertex problem reduces to blocks)**.** *For* $b \geq 2$ *write*

$$g_m(b) \;:=\; \max\big\{\, W_m(B) \;:\; B \text{ 2-connected or an edge},\, |V(B)| = b,\; \kappa_B^{\max} \leq m-1 \,\big\},$$

with $g_m(b) := -\infty$ when no such B exists. Then for all $m \geq 2$ and $n \geq 2$,

$$K_m(n) \;=\; \max\Big\{ \sum_i g_m(b_i) \;:\; b_i \geq 2,\; \sum_i (b_i - 1) \leq n-1 \Big\},$$

the maximum over all finite multisets of block sizes. In particular the value is determined by the finitely many numbers $g_m(2), \ldots, g_m(n)$.

Proof. This is the proof of Theorem A.6 with W_m in place of the number of edges. As there, $\kappa_{G_0}(u,v) = \kappa_B(u,v)$ for u, v in a common block B, so $m - \kappa_{G_0}(u,v)$ depends only on the block of the edge uv, and $W_m(G_0) = \sum_B W_m(B)$. The block sizes satisfy $\sum_B (|V(B)| - 1) \leq n-1$, and gluing blocks at one common vertex realises every choice of sizes, as there. With Lemma A.38 this gives the claim. □

So $K_m(n)$ can be computed from 2-connected graphs: compute W_m for each, and choose the best block sizes.

Theorem A.40 (Thickened complete bipartite blocks)**.** *Let $1 \leq s \leq t \leq m-1$. Give every edge of $K_{s,t}$ multiplicity $m - s$. The result has $s + t$ vertices, $st(m-s)$ edges and $\kappa^{\max} \leq m - 1$. Gluing k such blocks at one shared vertex gives, for $n = k(s+t-1)+1$,*

$$K_m(n) \;\geq\; \frac{st\,(m-s)}{s+t-1}\,(n-1).$$

The best choice of s and t gives the rate $\big(3 - 2\sqrt{2} + o_m(1)\big)m^2$, at $t = m-1$ and $s \approx (\sqrt{2}-1)m$. Theorem A.36 gives $\big(\frac{1}{8} + o_m(1)\big)m^2$. The new rate is larger by a factor $8(3-2\sqrt{2}) = 24 - 16\sqrt{2} \approx 1.373$.

Proof. Write S and T for the two sides, with $|S| = s$ and $|T| = t$. Every edge joins S and T.

Two vertices on the same side. Take $u \neq u'$ in S. For each $v \in T$ the route u, v, u' is a route of length two, and these t routes are internally disjoint. Every route from u first goes to a vertex of T, so removing T separates u from u'. So $\kappa(u,u') = t$. In the same way, $\kappa(v,v') = s$ for $v \neq v'$ in T.

Two vertices on different sides. Take $u \in S$ and $v \in T$, and remove the edge uv. Every other route from u first goes to a vertex of $T \setminus \{v\}$, and every route into v comes from a vertex of $S \setminus \{u\}$. So both sets separate u from v, and $\pi(u,v) \leq \min(s-1, t-1) = s-1$. Choose $s-1$ different vertices $u_i \in S \setminus \{u\}$, and $s-1$ different vertices $v_i \in T \setminus \{v\}$. This is possible because $s \leq t$. The routes u, v_i, u_i, v are internally disjoint. So $\pi(u,v) = s-1$.

Feasibility and count. By Lemma A.32, an edge uv has $\kappa = (m-s) + (s-1) = m-1$. Two vertices on the same side are not adjacent, so they have $\kappa = t$ or $\kappa = s$, and both are at most $m-1$. The block has st edges, each of multiplicity $m-s$, so $st(m-s)$ edges in total.

Gluing. If $s \geq 2$, then $K_{s,t}$ is 2-connected. By the gluing in the proof of Theorem A.39, k copies at one shared vertex are feasible, with $k\,st(m-s)$ edges on $n = k(s+t-1)+1$ vertices. If $s = 1$, then $K_{1,t}$ is a star, which is a tree, and every edge has multiplicity $m-1$. Gluing k

copies at one vertex gives a tree with every edge of multiplicity $m-1$. This is the thickened tree above, which is feasible. It has $k\,t(m-1) = k\,st(m-s)$ edges. Since $k = (n-1)/(s+t-1)$, the number of edges is $\frac{st(m-s)}{s+t-1}(n-1)$ in both cases.

The best choice. For fixed s, the factor $t/(s+t-1)$ does not decrease as t grows, since its derivative in t is $(s-1)/(s+t-1)^2 \geq 0$. So some best choice has $t = m-1$, the largest allowed value. Put $s = \alpha m$. The rate becomes

$$\frac{s\,(m-1)(m-s)}{s+m-2} \;=\; \frac{\alpha(1-\alpha)}{1+\alpha}\,m^2 \;+\; O(m).$$

The function $h(\alpha) = \alpha(1-\alpha)/(1+\alpha)$ has derivative $h'(\alpha) = (1-2\alpha-\alpha^2)/(1+\alpha)^2$. This is zero at $\alpha = \sqrt{2}-1$, positive before it and negative after it, so h is largest there. Its value there is

$$h(\sqrt{2}-1) \;=\; \frac{(\sqrt{2}-1)(2-\sqrt{2})}{\sqrt{2}} \;=\; (\sqrt{2}-1)^2 \;=\; 3-2\sqrt{2}.$$

Since $3-2\sqrt{2} \approx 0.172 > \frac{1}{8}$, this is larger than the rate of Theorem A.36, by the stated factor. □

In the best choice, one side has about $0.41m$ vertices and the other has $m-1$. So $3-2\sqrt{2}-o(1) \leq K_m(n)/(m^2 n) \leq 1$ for $m \to \infty$ and $n/m \to \infty$. Whether this ratio has a limit is open.

A.8.2 The directed problem

The splitting of κ into parallel arcs and longer routes also works for directed graphs (Lemma A.41). Corollary A.42 below gives the exact value at $m = 2$ and the leading term for every fixed m.

Lemma A.41 (Splitting the directed vertex measure)**.** *Let D be a directed multigraph with multiplicity function μ and underlying simple digraph D_0. For an ordered pair $u \neq v$, write $\pi(u,v)$ for the largest number of pairwise internally disjoint directed u–v paths of length at least two in D_0. Then $\kappa_D(u,v) = \mu(u,v) + \pi(u,v)$.*

Proof. The proof of Lemma A.32 works word for word with directed paths. A route of length one is a copy of the arc $u \to v$ and has no inner vertex. Longer routes give internally disjoint directed paths in D_0, and conversely, using one copy of each arc. Copies of the reverse arc $v \to u$ are never used.

□

Corollary A.42 (The directed multigraph vertex value)**.** *For every $n \geq 2$,*

$$K_2^{\mathrm{dir}}(n) \;=\; k_2^{\mathrm{dir}}(n) \;=\; \ell_2^{\mathrm{dir}}(n) \;=\; \max\Big(2(n-1),\; \Big\lfloor \frac{n^2}{4} \Big\rfloor\Big),$$

and for each fixed $m \geq 2$,

$$(m-1)M(n) \;\leq\; K_m^{\mathrm{dir}}(n) \;\leq\; (m-1)\Big\lfloor \frac{n^2}{4} \Big\rfloor + (m-1)^2(n-1).$$

So $K_m^{\mathrm{dir}}(n) = (m-1)n^2/4 + O_m(n)$, and the thickened bipartite digraph of Theorem 1.18 has the right leading term here too.

Proof. The case $m = 2$. Two parallel arcs would give two separated routes, so every feasible directed multigraph is simple. So $K_2^{\mathrm{dir}}(n) = k_2^{\mathrm{dir}}(n)$, and Theorems 1.11 and 1.12 give the rest.

General m. The extremal multigraphs of Theorem 1.18 have $\lambda^{\max} \le m-1$. Since $\kappa \le \lambda$, also $\kappa^{\max} \le m-1$. This gives the lower bound. A directed multigraph is a forward directed 2-uniform multihypergraph: each arc is a hyperedge with one tail and one head. So Theorem 1.22 at $r = 2$ gives the upper bound. For $n \ge 7$ we have $M(n) = \lfloor n^2/4 \rfloor$. So both bounds are $(m-1)n^2/4 + O_m(n)$. □

Construction A.43 (Thickened complete digraph, and its bouquet)**.** For $2 \le b \le m$, take the complete digraph on b vertices and give every arc multiplicity $m+1-b$. At $b = 2$ this is one bidirected edge at multiplicity $m-1$. Hanging k such blocks off one shared vertex, pairwise disjoint otherwise, uses $k(b-1)+1$ vertices, and blocks of different sizes may be mixed.

Proposition A.44 (Gluing complete digraphs)**.** *Glue blocks of Construction A.43 with sizes b_i at one common vertex, keep them disjoint otherwise, and add isolated vertices to reach n. The result is feasible, so*

$$K_m^{\mathrm{dir}}(n) \;\ge\; \max\Big\{ \sum_i b_i(b_i-1)(m+1-b_i) \;:\; 2 \le b_i \le m,\; \sum_i (b_i-1) \le n-1 \Big\}.$$

Proof. Take $u \ne v$ in one block of size b. A route that leaves the block must come back through the common vertex, which a path visits only once. So every u–v route stays in the block. A longer route inside the block has an inner vertex among the other $b-2$ vertices of the block. Separated routes have different inner vertices, so there are at most $b-2$ longer routes. The $b-2$ routes $u \to w \to v$, one for each other vertex w of the block, reach this number. So $\pi(u,v) = b-2$, and Lemma A.41 gives $\kappa(u,v) = (m+1-b) + (b-2) = m-1$. Two vertices in different blocks are separated by the common vertex, so $\kappa \le 1$. A block of size b has $b(b-1)$ ordered pairs, each with $m+1-b$ arcs. □

At $n = 3$ and $m = 6$, the thickened complete digraph has 24 arcs, more than $(m-1)M(3) = 20$. It is feasible: every route of length at least two passes through the third vertex, so $\pi = 1$ and $\kappa = 4+1 = 5 = m-1$ for every ordered pair. So the extremal multigraphs of the arc problem are not optimal for the vertex problem.

For small n the block construction is larger, and for large n the bipartite construction is larger. At $m = 3$ the best block is a bidirected edge, so the bouquet is the thickened bidirected tree, and the bipartite construction is larger from $n = 8$. At $m = 6$ the best blocks have three or four vertices. The bouquet has 84 arcs at $n = 8$, against 80, and the bipartite construction is larger from $n = 9$.

A.9 The directed hypergraph

The last four variants use directed hyperedges, in the forward, backward or general model of Section 1.3. The first result gives an upper and a lower bound that differ by a factor of four. Theorem 1.22 then closes this gap.

Proposition A.45 (Directed hypergraphs: first bounds)**.** *Let $n \geq r \geq 2$ and $m \geq 2$, and take the forward model. If every ordered pair of vertices has at most $m-1$ hyperedge-disjoint directed Berge routes, then*

$$|E(\mathcal{H})| \;\leq\; \frac{(m-1)\,n(n-1)}{r-1},$$

and, for every $1 \leq \alpha \leq n-r+1$, there is a feasible directed multihypergraph with

$$\alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor$$

hyperedges. This construction is simple when $m-1 \leq \binom{n-\alpha-1}{r-2}$. For fixed r and m and large n we can take $\alpha = \lfloor n/2 \rfloor$, so the maximum is $\Theta_{r,m}(n^2)$.

Proof. Upper bound. A hyperedge with tail u and heads $h_1, \ldots, h_{r-1}$ joins the $r-1$ ordered pairs (u, h_i). If m hyperedges joined the same pair (u, v), they would give m disjoint routes of length one. So each of the $n(n-1)$ ordered pairs is joined by at most $m-1$ hyperedges. Counting the pairs (hyperedge, joined pair) in two ways gives

$$(r-1)|E| \leq (m-1)\,n(n-1).$$

Lower bound. Split V into a set A of α tails and a set B of $\beta := n-\alpha \geq r-1$ heads. Let $k := r-1$, $d := m-1$ and $N := \left\lfloor \frac{d\beta}{k} \right\rfloor$. Number the vertices of B as $0, 1, \ldots, \beta-1$, and read the repeating sequence

$$0, 1, \ldots, \beta-1, 0, 1, \ldots$$

in consecutive blocks of k numbers. The first N blocks are the hyperedges of a k-uniform multihypergraph $\mathcal{G}^*$ on B. Since $k \leq \beta$, each block has k different vertices. The blocks use the first Nk positions of the sequence, so every vertex appears $\lfloor Nk/\beta \rfloor$ or $\lceil Nk/\beta \rceil$ times. Since $Nk \leq d\beta$, we have $Nk/\beta \leq d$, and since d is an integer, $\lceil Nk/\beta \rceil \leq d$. So every vertex has degree at most d.

Now, for every $a \in A$ and every $H \in \mathcal{G}^*$, add the hyperedge with tail a and heads H. No vertex of B is a tail, so a route can never leave B, and every route has length one. So for $a \in A$ and $w \in B$, the routes from a to w are the hyperedges with tail a whose heads contain w. There are $\deg_{\mathcal{G}^*}(w) \leq m-1$ of them. There are no routes from B, and no routes between two vertices of A, since all heads lie in B. The number of hyperedges is $\alpha N = \alpha \left\lfloor \frac{(m-1)(n-\alpha)}{r-1} \right\rfloor$. If $d \leq \binom{\beta-1}{k-1}$, we can use Lemma A.1 instead, which gives different hyperedges, so the directed hypergraph is simple. □

Construction A.46 (Bounded outgoing hyperedge count)**.** Let $n \geq r \geq 2$ and $m \geq 2$, and let $d = \min\{m-1, \binom{n-1}{r-1}\}$. Give every vertex d different outgoing forward hyperedges, with any d different sets of $r-1$ heads. This gives a simple directed hypergraph with nd hyperedges, feasible for both separations. If repeated hyperedges are allowed, give every vertex $m-1$ copies of one outgoing hyperedge, for $n(m-1)$ hyperedges in total.

Proof. Every directed route from u starts with a hyperedge whose tail is u. So removing the outgoing hyperedges of u separates u from every other vertex. There are at most $m-1$ of them, so $\lambda(u,v) \leq m-1$ for every ordered pair, and so also $\kappa(u,v) \leq m-1$. Hyperedges with different tails are different, so the hypergraph is simple when the head sets at each vertex are different. □

For small n this construction can beat the two-layer construction. For example, at $n = m = 6$ and $r = 3$ it gives 30 hyperedges, even in the simple model.

We write a directed hyperedge as (T, H), with tail set T and head set H. For $r \geq 4$ the general model has splits with both parts of size at least two. We show that the backward model is the forward model reversed, and that the general model has the same leading term as the forward model.

Lemma A.47 (Backward is the reverse of forward)**.** *Reversing every hyperedge, $(T,H) \mapsto (H,T)$, turns forward hypergraphs into backward ones and keeps the number of hyperedges. The reversed hypergraph $\mathcal{H}^R$ satisfies $\lambda_{\mathcal{H}}(u,v) = \lambda_{\mathcal{H}^R}(v,u)$ and $\kappa_{\mathcal{H}}(u,v) = \kappa_{\mathcal{H}^R}(v,u)$. So the backward model has the same extremal values as the forward model.*

Proof. Reversal turns every directed u–v Berge path into a v–u path with the same hyperedges and the same inner vertices. So it keeps both kinds of disjointness. □

Proposition A.48 (The general model obeys the forward bound)**.** *Take the general model. If every ordered pair has at most $m-1$ hyperedge-disjoint, or at most $m-1$ internally vertex-disjoint, directed Berge routes, then*

$$|E(\mathcal{H})| \;\leq\; \frac{(m-1)\,n(n-1)}{r-1}.$$

The construction of Proposition A.45 also works here, so the maximum is $\Theta_{r,m}(n^2)$.

Proof. Fix an ordered pair (u,v). Each hyperedge (T,H) with $u \in T$ and $v \in H$ gives a route $u \to v$ of length one. Different such hyperedges give separated routes, so there are at most $m-1$ of them. A hyperedge (T,H) joins exactly $|T|\,|H|$ ordered pairs. Summing over all pairs gives $\sum_{(T,H)} |T|\,|H| \leq (m-1)\,n(n-1)$. Since $|T|+|H| = r$ and both are at least one, $|T|\,|H| \geq r-1$. So $(r-1)|E| \leq (m-1)n(n-1)$. □

So both models lie between two bounds that differ by a factor of 4. This fixes the order n^2, but not the leading constant. The rest of this section finds the constant for both models. Their exact values may still differ.

The proof uses the one-step shadow digraph. Step 1 bounds the hyperedges by the shadow arcs. Step 2 shows that the shadow satisfies a two-step budget. Step 3 applies Lemma A.18.

Definition A.49 (One-step shadow)**.** Let $\mathcal{H}$ be a forward directed r-uniform hypergraph on V. Its *shadow* is the simple digraph $R(\mathcal{H})$ on V with an arc (u, v) when some hyperedge of $\mathcal{H}$ has tail u and head v. Let $\operatorname{cod}(u, v)$ be the number of such hyperedges.

We now prove the theorem of Chapter 1.

Theorem 1.22 (restated). *Let $r \geq 2$ and $m \geq 2$. Every forward directed r-uniform multihypergraph $\mathcal{H}$ on $n \geq 2$ vertices with $\kappa_{\mathcal{H}}^{\max} \leq m - 1$ satisfies*

$$|E(\mathcal{H})| \;\leq\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4} \right\rfloor \;+\; (m-1)^2(n-1).$$

Proof of Theorem 1.22. Let $R = R(\mathcal{H})$. It has no loops, since the tail of a hyperedge is not one of its heads.

Step 1: hyperedges and shadow arcs. The $\operatorname{cod}(u, v)$ hyperedges with tail u and head v give that many routes of length one, which are separated. So $\operatorname{cod}(u, v) \leq m - 1$. Every hyperedge, with tail u and heads $h_1, \ldots, h_{r-1}$, gives the $r - 1$ arcs $u \to h_i$ of R. Counting the pairs of a hyperedge and one of its arcs in two ways gives

$$(r-1)\,|E(\mathcal{H})| \;=\; \sum_{(u,v)\in A(R)} \operatorname{cod}(u, v) \;\leq\; (m-1)\,|A(R)|.$$

Proposition A.45 used the bound $|A(R)| \leq n(n-1)$ here, which loses the factor four. We now bound $|A(R)|$ better.

Step 2: the shadow is budgeted. Fix an ordered pair (u, v). Recall that a midpoint of (u, v) is a vertex $x \notin \{u, v\}$ with arcs $u \to x$ and $x \to v$ in R. Let $\mathcal{T}$ be the set of midpoints, together with v if $(u, v) \in A(R)$. So $|\mathcal{T}| = \mathbf{1}_{(u,v)\in A(R)} + p_2(u, v)$. Every $t \in \mathcal{T}$ has an arc $u \to t$ in R, so it is a head of some hyperedge with tail u. Pick hyperedges with tail u one at a time. Each time, pick one that contains an element t of $\mathcal{T}$ that is not in an earlier pick, and assign the pick to that t. Stop when every element of $\mathcal{T}$ lies in a picked hyperedge. Each new pick contains an element that the earlier picks do not contain, so the picks are different hyperedges, and they are assigned to different elements of $\mathcal{T}$. Each pick has $r - 1$ heads, so it contains at most $r - 1$ elements of $\mathcal{T}$. So the number k of picks satisfies $k(r - 1) \geq |\mathcal{T}|$.

Now turn the picks into routes. If a pick e is assigned to v, take the route $u \to v$ through e. If it is assigned to a midpoint x, take the route $u \to x \to v$: enter through e, and leave through a hyperedge with tail x and head v (one exists, since $(x, v) \in A(R)$). In the forward model every hyperedge has exactly one tail. The leaving hyperedges have different tails, namely different midpoints, so they are different. They also differ from the entering ones, which have tail u. The entering ones are different picks. So the k routes share no hyperedge. A route has at most one inner vertex, its midpoint, and different routes have different midpoints. So the routes are

separated, and $k \le \kappa_{\mathcal{H}}(u,v)$. So

$$\frac{\mathbf{1}_{(u,v)\in A(R)} + p_2(u,v)}{r-1} \;\le\; k \;\le\; \kappa_{\mathcal{H}}(u,v) \;\le\; m-1,$$

and R is $(r-1)(m-1)$-budgeted.

Step 3: conclusion. Lemma A.18 with $C = (r-1)(m-1)$ gives $|A(R)| \le \lfloor n^2/4 \rfloor + (r-1)(m-1)(n-1)$. With Step 1,

$$|E(\mathcal{H})| \;\le\; \frac{m-1}{r-1}\left[\left\lfloor \frac{n^2}{4}\right\rfloor + (r-1)(m-1)(n-1)\right] \;=\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4}\right\rfloor + (m-1)^2(n-1).$$

For the lower bound, Proposition A.45 with $\alpha = \lfloor n/2 \rfloor$ gives $(m-1)n^2/(4(r-1)) - O_{m,r}(n)$ hyperedges. This construction is feasible for both separations. For fixed $r \ge 3$ and large n it is simple. So the upper and lower bounds have the same n^2 term. □

Two remarks. For small n, the bound of Proposition A.45 is smaller, roughly until $n \approx 4(m-1)(r-1)/3$. The theorem is only better asymptotically. At $r = 2$ the shadow is the digraph itself, and the theorem gives the quadratic term of Theorem 1.18 again.

The general model needs a different Step 2. The proof above used twice that a hyperedge has one tail: leaving hyperedges with different tails are different, and they differ from the entering hyperedges. With several tails this fails. Two midpoints can leave through the same hyperedge, and for $r \ge 4$ one hyperedge can be the entrance for one target and the exit for another. Instead we take a maximal family of routes and ask why no route can be added. Each unserved target is blocked by a hyperedge that is already used, and a used hyperedge has only r vertices.

Theorem A.50 (The general orientation model shares the constant)**.** *Let $r \ge 2$ and $m \ge 2$. In the general model, every directed r-uniform multihypergraph $\mathcal{H}$ on $n \ge 2$ vertices with $\kappa^{\max}_{\mathcal{H}} \le m-1$ satisfies*

$$|E(\mathcal{H})| \;\le\; \frac{m-1}{r-1}\left\lfloor \frac{n^2}{4}\right\rfloor \;+\; \frac{(2r+1)(m-1)^2}{r-1}(n-1).$$

Since $\kappa \le \lambda$, the same bound holds when $\lambda^{\max}_{\mathcal{H}} \le m-1$. So for fixed r and m the general model has the same leading term $\big(1+o(1)\big)(m-1)n^2/\big(4(r-1)\big)$ as the forward model, under both separations.

Proof. Let R be the shadow in the general model: $(u,v) \in A(R)$ when some hyperedge has u among its tails and v among its heads. Let $\operatorname{cod}(u,v)$ be the number of such hyperedges. Tails and heads are disjoint, so R has no loops.

Step 1. As before, $\operatorname{cod}(u,v) \le m-1$. A hyperedge (T,H) gives $|T|\,|H| \ge r-1$ arcs of R. So

$$(r-1)\,|E(\mathcal{H})| \;\le \sum_{(T,H)\in E(\mathcal{H})} |T|\,|H| \;=\; \sum_{(u,v)\in A(R)} \operatorname{cod}(u,v) \;\le\; (m-1)\,|A(R)|.$$

Step 2: the shadow is budgeted. Fix an ordered pair (u,v). Let $\mathcal{T}$ be the set of midpoints counted by $p_2(u,v)$, together with v if $(u,v) \in A(R)$. Call a hyperedge e an *entrance* for $t \in \mathcal{T}$ if

u is a tail of e and t a head. Call e an *exit* for a midpoint t if t is a tail of e and v a head. Every $t \in \mathcal{T}$ has an entrance, and every midpoint has an exit. No hyperedge is both an entrance and an exit for the same t, since t cannot be both a tail and a head.

For each target we use one route. For v it is a route of length one through an entrance. For a midpoint t it is the route $u \to t \to v$ through an entrance and an exit. Build a family of such routes, for different targets and with all hyperedges different, adding routes one at a time until no route can be added. Let $\mathcal{F}$ be the result, with M routes, serving the set S of targets, and let U be the set of hyperedges it uses. Then $|U| \le 2M$. The routes share no hyperedge, and their inner vertices are different, so $M \le \kappa_{\mathcal{H}}(u, v) \le m - 1$.

Now take $t \in \mathcal{T} \setminus S$. If t had an entrance outside U and, for a midpoint, also an exit outside U, we could add a route for t. This contradicts maximality. So t is a tail or a head of some hyperedge in U. So

$$\mathcal{T} \setminus S \;\subseteq\; \bigcup_{e \in U} (T_e \cup H_e), \qquad \text{and} \qquad |\mathcal{T}| \;\le\; M + \sum_{e \in U} \bigl(|T_e| + |H_e|\bigr) \;=\; M + r\,|U|.$$

With $|U| \le 2M$ this gives $|\mathcal{T}| \le (2r+1)M \le (2r+1)(m-1)$. So R is $(2r+1)(m-1)$-budgeted.

Step 3. Lemma A.18 with $C = (2r+1)(m-1)$ gives $|A(R)| \le \lfloor n^2/4 \rfloor + (2r+1)(m-1)(n-1)$, and Step 1 gives the bound. The construction of Proposition A.45 also works in the general model, so the n^2 terms agree. □

The constant $2r + 1$ only affects the linear term.

Appendix B

The Program

Chapter 2 prints the main parts of `program/erdos915.py`. This appendix prints the remaining parts, in the order they appear in the file, and the output of one full run.

B.1 The remaining parts of the program

The file starts with a short description and the imports.

```
# Erdős Problem 915 in sixteen variants: measure connectivity, search
# for dense feasible graphs, and write a certificate that can be checked
# by hand.
#
# Run:  python erdos915.py > output.txt
# It prints one table per problem and writes every graph found, with its
# certificate, to witnesses.json.
import itertools
import json
import math
import random
import subprocess
import time
from dataclasses import dataclass
from enum import Enum
from multiprocessing import Pool

import networkx
```

Listing B.1. The description and the imports.

Two helpers list the vertex pairs that must be checked, ordered in directed models, and count the edges of a graph.

```
def vertex_pairs(problem):
    vertices = range(problem.VERTEX_COUNT)
    if problem.MODEL.IS_DIRECTED:
        return list(itertools.permutations(vertices, 2))
    return list(itertools.combinations(vertices, 2))


def edge_count(multiplicity):
    return sum(multiplicity.values())
```

Listing B.2. The vertex pairs and the edge count.

The heuristic searches keep the number of disjoint routes of every pair. The exception `OutOfTime` stops a search at its time limit.

```
def all_connectivities(problem, multiplicity):
    network = build_flow_network(problem, multiplicity)
    return {(u, v): disjoint_routes(network, u, v)
            for u, v in vertex_pairs(problem)}


class OutOfTime(Exception):
    pass
```

Listing B.3. The connectivity of every pair, and the time limit.

The generating search of Listing 2.6 reads the supports from `geng`, keeps the densest digraphs over all supports, and counts them up to renaming the vertices when it finishes.

```
def undirected_supports(vertex_count):
    """Streams the output of geng, one graph at a time. The stream counts
    as complete only if geng then exits with status zero."""
    geng = subprocess.Popen(["geng", "-q", str(vertex_count)],
                            stdout=subprocess.PIPE)
    try:
        for line in geng.stdout:
            yield networkx.from_graph6_bytes(line.strip())
        if geng.wait() != 0:
            raise RuntimeError(f"geng failed with status {geng.returncode}"
                ↪ )
    finally:
        if geng.poll() is None:      # the search stopped early
            geng.kill()
            geng.wait()


def canonical_form(vertex_count, multiplicity):
    arcs = [(e.TAILS[0], e.exit_vertices()[0], copies)
            for e, copies in multiplicity.items() if copies]
    return min(tuple(sorted((order[t], order[h], c) for t, h, c in arcs))
               for order in itertools.permutations(range(vertex_count)))


def generating_search(problem, deadline):
    best_size, extremal, finished = 0, [{}], True
    for support in undirected_supports(problem.VERTEX_COUNT):
        if time.time() > deadline:
            finished = False
            break
        size, graphs, done = best_digraphs_on_support(
            problem, sorted(support.edges()), deadline)
        finished = finished and done
        if size > best_size:
            best_size, extremal = size, []
```

```
        if size == best_size:
            extremal += graphs
    classes = None
    if finished:
        classes = len({canonical_form(problem.VERTEX_COUNT, graph)
                       for graph in extremal})
    return best_size, extremal[0], classes, finished
```

Listing B.4. Reading the supports from geng, and the generating search over all supports.

A move gives a new table, so the old one stays unchanged.

```
def apply(multiplicity, move):
    hyperedge, change = move
    changed = dict(multiplicity)
    changed[hyperedge] = changed.get(hyperedge, 0) + change
    return changed
```

Listing B.5. Making a move.

Simulated annealing, described in Chapter 2.

```
START_TEMPERATURE, COOLING_FACTOR = 2.0, 0.995


def simulated_annealing(problem, deadline, seed):
    generator = random.Random(seed)
    m = problem.THRESHOLD_M
    multiplicity, temperature = {}, START_TEMPERATURE
    connectivity = all_connectivities(problem, multiplicity)
    current = energy(0, 0, m)
    best_size, best_graph = 0, {}
    while time.time() < deadline:
        move = generator.choice(list(moves(problem, multiplicity)))
        highest = highest_after(problem, multiplicity, move,
                                connectivity)
        size = edge_count(multiplicity) + move[1]
        proposed = energy(size, highest, m)
        worse_by = proposed - current
        if worse_by <= 0 or \
                generator.random() < math.exp(-worse_by / temperature):
            multiplicity, current = apply(multiplicity, move), proposed
            connectivity = all_connectivities(problem, multiplicity)
            if highest <= m - 1 and size > best_size:
                best_size, best_graph = size, multiplicity
        temperature = max(temperature * COOLING_FACTOR, 0.01)
    return best_size, best_graph
```

Listing B.6. Simulated annealing on the same energy.

The main function runs every case in parallel, prints one table per problem, and saves every graph with its certificate.

```
def print_table(model, separation, solved, largest_m, witnesses):
    print(f"\n{model.NAME}, {separation.value} separation")
    print("n \\ m" + "".join(f"{m:>6}" for m in range(2, largest_m + 1)))
    rows = {}
    for problem, (size, exact, checked, found) in solved:
        witnesses.append({
            "model": model.NAME, "separation": separation.value,
            "n": problem.VERTEX_COUNT, "m": problem.THRESHOLD_M,
            "size": size, "search_finished": exact,
            "certificate_checked": checked, **found})
        mark = ("*" if exact else " ") if checked else "!"
        rows.setdefault(problem.VERTEX_COUNT, []).append(f"{size:>5}{mark}"
            ↪ )
    for n, cells in rows.items():
        print(f"{n:>5}" + "".join(cells), flush=True)


def main(largest_n=7, largest_m=7):
    problems = [Problem(model, separation, n, m)
                for model in ALL_MODELS() for separation in Separation
                for n in range(model.EDGE_SIZE, largest_n + 1)
                for m in range(2, largest_m + 1)]
    print("Edges of the densest feasible graph found, every one certified."
        ↪ )
    print("* the search finished, so the value is exact;",
          "! the certificate failed")
    witnesses = []
    with Pool() as pool:
        solved = zip(problems, pool.imap(solve, problems))
        for (model, separation), table in itertools.groupby(
                solved, key=lambda pair: (pair[0].MODEL, pair[0].SEPARATION
                    ↪ )):
            print_table(model, separation, table, largest_m, witnesses)
    with open("witnesses.json", "w") as file:
        file.write("[\n" + ",\n".join(json.dumps(w) for w in witnesses)
                   + "\n]\n")


if __name__ == "__main__":
    main()
```

Listing B.7. Running every case and printing the tables.

B.2 Output

Listing B.8 is the output of the program, unchanged. The command `python erdos915.py > output.txt` in the folder `program` writes it to a file, and this thesis reads that file directly, so no step lies between the program and the page. Each case had a nominal thirty-second search budget, on a computer with ten processor cores. The unfinished searches depend on the speed

of the computer, so a run on another computer can print other lower bounds. Every entry is the number of edges, arcs or hyperedges of the densest feasible graph found, and every one of these graphs passed its certificate. The same run wrote these graphs and their certificates to `program/witnesses.json`. An entry marked $*$ comes from a finished search, so it is the exact maximum. An unmarked entry is a lower bound only, and a longer run may raise it. For hypergraphs the rows start at $n = 3$. Each directed hypergraph problem has three tables, one for each version of Section 1.3. Wherever both searches finished, the tables with one tail and with one head agree, as Lemma A.47 predicts. In the general model the search finished almost only at $n = 3$. Its other entries are lower bounds, and some lie below the one-tail entries, although every one-tail hypergraph also belongs to the general model.

```
Edges of the densest feasible graph found, every one certified.
* the search finished, so the value is exact; ! the certificate failed

simple graph, edge separation
n \ m     2     3     4     5     6     7
    2     1*    1*    1*    1*    1*    1*
    3     2*    3*    3*    3*    3*    3*
    4     3*    4*    6*    6*    6*    6*
    5     4*    6*    8*   10*   10*   10*
    6     5*    7    10*   12*   15*   15*
    7     6     9    12    15    18*   21*

simple graph, vertex separation
n \ m     2     3     4     5     6     7
    2     1*    1*    1*    1*    1*    1*
    3     2*    3*    3*    3*    3*    3*
    4     3*    4*    6*    6*    6*    6*
    5     4*    6*    8*   10*   10*   10*
    6     5*    7    10*   12*   15*   15*
    7     6     9    12    15    18*   21*

multigraph, edge separation
n \ m     2     3     4     5     6     7
    2     1*    2*    3*    4*    5*    6*
    3     2*    4*    6*    8*   10*   12*
    4     3*    6*    9*   12*   15*   18*
    5     4*    8*   12    16    19    23
    6     5*   10    14    20    23    28
    7     6    12    17    23    28    35

multigraph, vertex separation
n \ m     2     3     4     5     6     7
    2     1*    2*    3*    4*    5*    6*
    3     2*    4*    6*    9*   12*   15*
    4     3*    6*    9*   14*   19*   24*
    5     4*    8*   12    19    25    33
    6     5*   10    15    24    33    42
    7     6    12    18    29    37    51

simple hypergraph, edge separation
n \ m     2     3     4     5     6     7
    3     1*    1*    1*    1*    1*    1*
    4     1*    3*    4*    4*    4*    4*
    5     2*    4*    6*    7*    8*   10*
    6     2*    5*    7    10    11    13
    7     3*    6     9    12    15    17

simple hypergraph, vertex separation
n \ m     2     3     4     5     6     7
    3     1*    1*    1*    1*    1*    1*
    4     1*    3*    4*    4*    4*    4*
    5     2*    4*    6*    7*    8*   10*
    6     2*    5*    7    10    11    13
```

```
    7    3*    6     9    12    15    17

multihypergraph, edge separation
n \ m     2     3     4     5     6     7
    3    1*    2*    3*    4*    5*    6*
    4    1*    3*    4*    6*    7*    9*
    5    2*    4*    6*    8*   10    12
    6    2*    5     7    10    12    14
    7    3*    6     9    12    15    17

multihypergraph, vertex separation
n \ m     2     3     4     5     6     7
    3    1*    2*    3*    4*    5*    6*
    4    1*    3*    4*    6*    7*    9*
    5    2*    4*    6*    8    10    12
    6    2*    5     7    10    12    14
    7    3*    6     9    12    15    17

simple digraph, edge separation
n \ m     2     3     4     5     6     7
    2    2*    2*    2*    2*    2*    2*
    3    4*    6*    6*    6*    6*    6*
    4    6*    9*   12*   12*   12*   12*
    5    8    12    16*   20*   20*   20*
    6   10    15    20    25    30*   30*
    7   12    18    24    30    36    42*

simple digraph, vertex separation
n \ m     2     3     4     5     6     7
    2    2*    2*    2*    2*    2*    2*
    3    4*    6*    6*    6*    6*    6*
    4    6*    9*   12*   12*   12*   12*
    5    8    12    16*   20*   20*   20*
    6   10    15    20    25    30*   30*
    7   12    18    24    30    36    42*

directed multigraph, edge separation
n \ m     2     3     4     5     6     7
    2    2*    4*    6*    8*   10*   12*
    3    4*    8*   12*   16*   20*   24
    4    6*   12    18    24    30    36
    5    8*   16    24    32    40    46
    6   10    20    30    37    44    48
    7   12    24    33    36    44    51

directed multigraph, vertex separation
n \ m     2     3     4     5     6     7
    2    2*    4*    6*    8*   10*   12*
    3    4*    8*   12*   18*   24*   30*
    4    6*   12    18    28    38    48
    5    8*   16    24    36    49    66
    6   10    20    30    42    60    81
    7   12    24    36    47    73    89

directed hypergraph, one tail, edge separation
n \ m     2     3     4     5     6     7
    3    3*    3*    3*    3*    3*    3*
    4    4*    8*   12*   12*   12*   12*
    5    6    10    15    20    25    30*
    6    7    12    18    24    30    36
    7    9    14    21    28    35    42

directed hypergraph, one tail, vertex separation
n \ m     2     3     4     5     6     7
    3    3*    3*    3*    3*    3*    3*
    4    4*    8*   12*   12*   12*   12*
    5    6    10    15    20    25    30*
    6    7    12    18    24    30    36
    7    9    14    21    28    35    42
```

```
directed hypergraph, one head, edge separation
n \ m     2     3     4     5     6     7
    3    3*    3*    3*    3*    3*    3*
    4    4*    8*   12*   12*   12*   12*
    5    6    10    15    20    25    30*
    6    7    12    18    24    30    36
    7    9    14    21    28    35    42

directed hypergraph, one head, vertex separation
n \ m     2     3     4     5     6     7
    3    3*    3*    3*    3*    3*    3*
    4    4*    8*   12*   12*   12*   12*
    5    6    10    15    20    25    30*
    6    7    12    18    24    30    36
    7    9    14    21    28    35    42

directed hypergraph, any split, edge separation
n \ m     2     3     4     5     6     7
    3    3*    4*    6*    6*    6*    6*
    4    4*    8    12    14    16    18
    5    6    10    14    18    21    24
    6    7    12    17    21    25    30
    7    8    13    18    24    29    35

directed hypergraph, any split, vertex separation
n \ m     2     3     4     5     6     7
    3    3*    4*    6*    6*    6*    6*
    4    4*    8    12    14    18    24*
    5    6    10    14    18    23    31
    6    7    12    17    21    27    38
    7    8    13    18    24    31    41

directed multihypergraph, one tail, edge separation
n \ m     2     3     4     5     6     7
    3    3*    6*    9*   12*   15*   18*
    4    4*    8*   12    16    20    24
    5    6    11    15    20    25    30
    6    7    13    18    24    30    36
    7    9    14    21    28    35    42

directed multihypergraph, one tail, vertex separation
n \ m     2     3     4     5     6     7
    3    3*    6*    9*   12*   15*   18*
    4    4*    8*   12    16    20    25
    5    6    11    15    21    26    31
    6    7    13    18    24    30    37
    7    9    14    21    28    35    42

directed multihypergraph, one head, edge separation
n \ m     2     3     4     5     6     7
    3    3*    6*    9*   12*   15*   18*
    4    4*    8*   12    16    20    24
    5    6    11    16    20    25    30
    6    7    13    18    24    30    36
    7    9    14    22    28    35    42

directed multihypergraph, one head, vertex separation
n \ m     2     3     4     5     6     7
    3    3*    6*    9*   12*   15*   18*
    4    4*    8*   12    16    21    24
    5    6    11    16    20    26    32
    6    7    13    19    25    30    39
    7    9    14    22    28    36    42

directed multihypergraph, any split, edge separation
n \ m     2     3     4     5     6     7
    3    3*    6*    9*   12*   15*   18*
    4    4*    8    12    15    17    20
    5    6    10    14    17    22    24
    6    7    11    15    21    24    30
```

```
    7    8    14    18    24    29    35

directed multihypergraph, any split, vertex separation
n \ m     2     3     4     5     6     7
    3    3*    6*    9*   12*   15*   18*
    4    4*    8    12    15    19    24
    5    6    10    14    18    24    30
    6    7    11    16    21    26    35
    7    8    14    18    24    33    41
```

Listing B.8. The output of the program for $2 \leq m \leq 7$ and $n \leq 7$.

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